

Master's thesis

SIMULATION OF HIGH-POWER LASER SPECTRA USING GENERATIVE MACHINE LEARNING

Bc. Anna Kapitánová

Faculty of Information Technology
Department of Applied Mathematics
Supervisor: Ing. Matěj Jech
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Student: Bc. Anna Kapitánová
Supervisor: Ing. Matěj Jech
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Instructions

High-power laser systems are a crucial part in many scientific experiments. Specifically, they are a driving force for laser wakefield electron acceleration experiments (LWFA), which function as compact particle accelerators. One such system is the L1-Allegra developed in ELI Beamlines in Dolní Břežany (<https://www.eli-beams.eu/facility/lasers/laser-1-allegra-100-mj-1-khz>).

The operation and optimization of this system is an expensive and demanding task. Furthermore, due to the nature of the experimental setup, the laser spectra cannot be measured during LWFA experimental operation. The spectra, however, has a significant impact on the laser-plasma interaction which affects the charge and energy of the accelerated electron beam. Therefore, it is a desirable diagnostic to have during experimental operation.

Laser beam optimization in the L1-Allegra system is a highly nonlinear process with a multitude of parameters. Using machine learning techniques to simulate the operation of the system could be beneficial in understanding the input-output relationships within the system. Such a model could help in understanding the complex process given its cheaper and more convenient operation compared to the real system.

The work will be conducted in collaboration with the Electron acceleration group and the L1 laser team at ELI Beamlines facility (<https://www.eli-beams.eu/>).

A successful completion of this topic includes:

- Performing a comprehensive literature review of related scientific machine learning tasks including the use of generative models for experimental physics simulations



- Preprocessing of the data obtained from scientific partners
- Designing and training a conditional generative model to simulate the aperture
- Testing and comparing multiple inference modes (samplers for diffusion-based models, multiple models if different architectures are used)

Beyond the main requirements, optional goals may be pursued:

- Exploring the possibility of including known physical laws and constraints into the training or inference to improve model performance
- Designing an optimization pipeline utilizing the trained generative model as a surrogate for the real experiments



Czech Technical University in Prague

Faculty of Information Technology

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First of all, I would like to express my sincere gratitude to my family and friends for their constant support and belief in me throughout my academic journey. A special thanks goes to my boyfriend Jakub, who has been by my side during all the challenges of my studies and whose unwavering support and motivation have been invaluable. I also would like to extend my appreciation to my thesis supervisor, Matěj Jech, for all his professional guidance and valuable advice throughout the development of this work. Thank you!

Declaration

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In Prague on May 7, 2025

Abstract

High-power laser systems, such as the L1-Allegro laser at ELI Beamlines, are essential in laser wakefield acceleration, offering compact alternatives to conventional particle accelerators. However, their operation is complex, expensive, and highly sensitive to system parameters. This thesis explores the use of diffusion-based generative models to simulate the spectral output of the L1-Allegro system, providing insights that are not accessible during real-time operation. In addition to reconstructing unmeasured spectra, the model supports the optimization of laser settings, helping to improve system performance and reduce the need for extensive experimental tuning.

Keywords LWFA, high-power lasers, diffusion models, generative ML, spectral simulation, physics informed ML, digital twin, Bayesian optimization

Abstrakt

Vysoce výkonné laserové systémy, jako je laser L1-Allegro v ELI Beamlines, hrají klíčovou roli v oblasti laser wakefield acceleration a představují kompaktní alternativu ke konvenčním částicovým urychlovačům. Jejich provoz je ovšem mimořádně nákladný a vysoce citlivý na nastavení systémových parametrů. Tato práce se zabývá využitím generativních modelů založených na difuzních modelech pro simulaci spektra laseru L1-Allegro, čímž poskytuje vhled do chování systému i v okamžicích, kdy není možné měření v reálném čase. Kromě rekonstrukce spekter model rovněž podporuje optimalizaci nastavení parametrů laseru, což napomáhá zlepšení výkonu systému a snižuje potřebu rozsáhlého experimentálního ladění.

Klíčová slova LWFA, vysoce výkonné lasery, difuzní modely, generativní strojové učení, spektrální simulace, fyzikálně informované strojové učení, digitální dvojče, Bayesovská optimalizace

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List of Abbreviations

ELI	Extreme Light Infrastructure
DDPM	Denoising Diffusion Probabilistic Model
DDIM	Denoising Diffusion Implicit Model
EDM	Elucidated Diffusion Model
MAE	Mean Absolute Error
MSE	Mean Squared Error
GAN	Generative adversarial network
VAE	Variational Autoencoders
ELBO	Evidence Lower Bound
SDE	Stochastic Differential Equation
CFG	Classifier-Free Guidance
GP	Gaussian process
DTW	Dynamic Time Warping
EM	Wasserstein Distance
PIC	Particle-In-Cell
LWFA	Laser Wakefield Accelerator
SM-LWFA	Laser Wakefield Accelerator in Self-Modulated Regime
CPA	Chirped Pulse Amplification
OPCPA	Optical Parametric Chirped Pulse amplification
OPA	Optical Parametric Amplification
ALFA	Allegra Laser For Acceleration
VHEE	Very High Energy Radio-Therapy
FID	Fréchet Inception Distance
LPI	Laser-Plasma Interaction
EMA	Exponential Moving Average

Introduction

High-power lasers play a crucial role in a wide range of scientific experiments, particularly in the field of laser wakefield acceleration (LWFA) [1], where they are used to accelerate particles over very short distances. One of the most advanced systems of this kind is the L1-Allegro laser, developed at the Extreme Light Infrastructure (ELI) Beamlines facility in Dolní Břežany [2].

Unlike traditional accelerators, such as those at the European Organization for Nuclear Research (CERN) [3] which rely on radio-frequency cavities and require facilities several kilometers long, LWFA systems operate on a fundamentally different principle. They use ultra-short, high-intensity laser pulses to create plasma waves within a gas or plasma medium. These waves generate strong electric fields that can accelerate charged particles to high energies over much shorter distances. The described method allows them to achieve extremely high acceleration gradients within a much smaller experimental area.

However, operating such systems is a complex task, highly sensitive to a variety of parameters, and very expensive. This thesis explores how machine learning, specifically diffusion models, can be used to simulate the spectral output of the L1-Allegro system. Simulating the spectral output can provide valuable insights that are otherwise difficult or impossible to obtain during real-time operation.

Diffusion models [4] are especially well-suited for this task, as they can represent the full range of possible outcomes rather than a single prediction. This feature makes them ideal for modeling complex systems like high-power lasers, where slight variations in input can produce significantly different results. This allows researchers to gain insights into the system's behavior that would otherwise remain hidden.

In addition to simulating the spectral output, the model can also be used to explore how different parameter settings affect the laser's performance. This makes searching for configurations that produce specific spectral features possible without relying entirely on time-consuming and costly experiments. This makes the model a valuable tool for optimizing system configuration and experimental planning.

Laser Plasma Acceleration

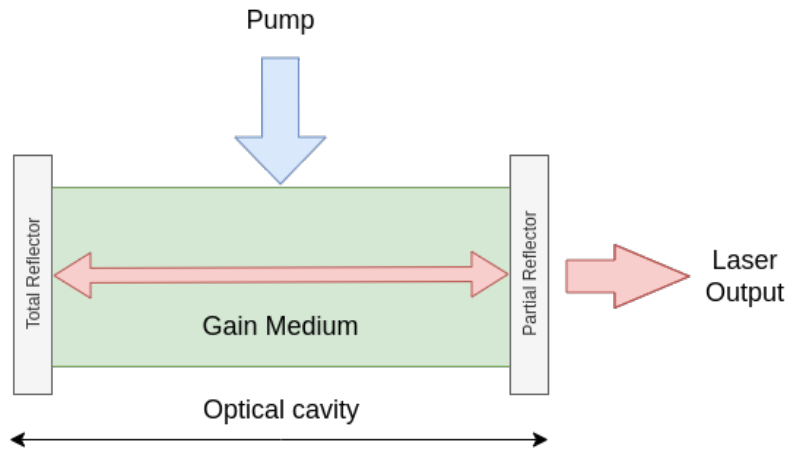
Laser stands for Light Amplification by Stimulated Emission of Radiation and is essential in many fields, such as medicine [5, 6], industry [7], or scientific research [8, 9].

The concept was introduced by Albert Einstein in 1917 in his work *On the quantum theory of radiation*, where he described the process of stimulated emission. [10] This principle later became the foundation for laser technology.

A laser is a device that produces highly directional, coherent, and intense beams of light. Unlike sunlight or other light sources, which emit photons in multiple directions, lasers are coherent sources of light. This means that the light waves produced by a laser are all in phase with each other, allowing them to remain narrow and focused over long distances. [11]

The architecture of a laser, shown in Figure 1.1, consists of three essential components:

1. Energy source (pump) that provides external energy to excite the atoms or molecules in the gain medium.
2. Gain medium (active medium) is a material where atoms absorb the energy and enter an excited state, which is necessary to achieve the stimulated emission. Stimulated emission occurs when an excited atom interacts with an incoming photon of the same energy. It releases another identical photon. These photons travel in phase, maintaining the laser's coherence and monochromaticity.
3. Optical cavity (resonator) that consists of two opposing mirrors. The mirrors allow light to bounce back and forth, amplifying the beam through repeated stimulated emission. One mirror is fully reflective, while the other is partially transparent. Once the beam reaches a sufficient intensity, it exits through the partially transparent mirror as a concentrated laser beam.



■ **Figure 1.1** Essential components of a laser.

1.1 ELI Beamlines

ELI Beamlines is a laser research facility located in Dolní Břežany. It is currently the largest research project in the history of the Czech Republic and is part of the European research initiative ELI. The facility also maintains one of the world's most powerful lasers (L4 ATON), with an output of 10 petawatts. [2]

ELI Beamlines houses four high-power lasers, producing ultra-short pulses, which last only a few femtoseconds, equivalent to 10^{-15} seconds. These extremely short pulses enable experiments that are impossible with continuous or long-pulse lasers.

A key advantage of ultra-short pulses is their ability to produce extremely high peak power. Since power is determined by energy divided by time, compressing energy into such brief pulses results in power levels reaching terawatts (TW), for the L4 laser, even petawatts (PW).

1.1.1 L1 Laser

L1 Allegra laser is one of the lasers developed at ELI beamlines. It is designed to produce high-energy, ultra-short laser pulses using a technique called Optical Parametric Chirped Pulse amplification (OPCPA), which enhances the earlier method known as Chirped Pulse Amplification (CPA). [12]

Firstly, the initial ultra-short beams, whose peak energy should be amplified, are generated by an oscillator. CPA then amplifies the beams in three main steps [13, 14]:

Stretching The pulses are stretched, meaning the pulse energy is spread in time, lowering their peak intensity. This is done to avoid the non-

linear pulse distortion or even destruction of the gain medium during the amplification process.

Amplification After the previous step, the pulses are passed through the gain medium and amplified.

Compression Finally, the amplified pulses are compressed back to their original ultra-short duration, making the peak power of the pulses higher.

OPCPA follows the same three steps as CPA, but uses a different pulse amplification method. [15, 16] It transfers energy from another laser using a nonlinear optical crystal instead of a standard laser amplifier. It is based on Optical Parametric Amplification (OPA), which uses a method called three-wave mixing. In this method, three laser beams interact inside a nonlinear crystal:

Pump Beam The strong beam, which provides the energy for amplification.

Signal Beam The weak stretched ultra-short pulse that should be amplified.

Idler beam The secondary beam generated as a byproduct of the energy transfer between the pump and signal beams.

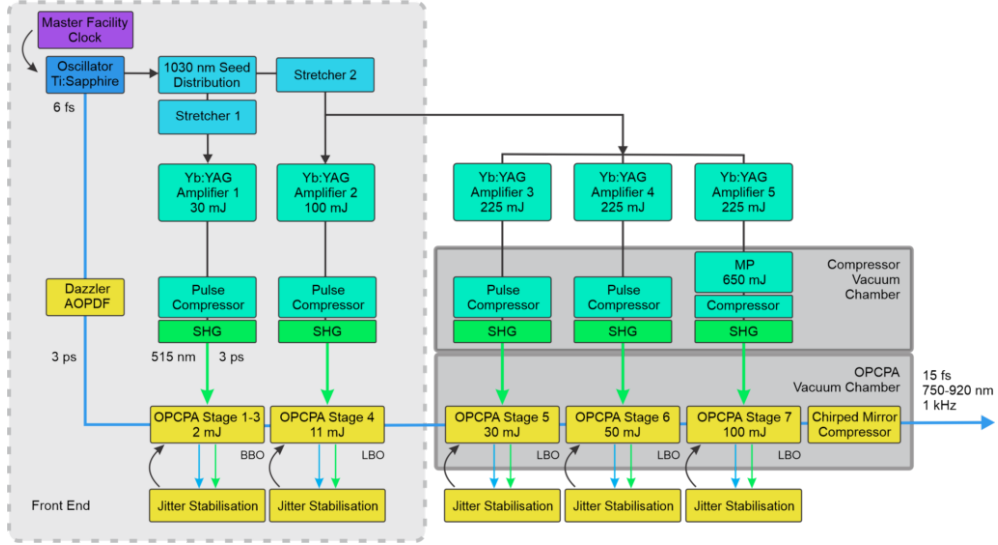
The advantage is that OPCPA enables the production of high-intensity laser pulses in various parts of the optical spectrum. These pulses are stable, highly controlled, and can reach high power levels.

In the L1 Allegra laser, the OPCPA steps occur in different parts of the system. It starts in the L1 front end, where a Ti:sapphire oscillator produces a weak 6 femtosecond long pulse. This pulse is then stretched to picosecond duration and amplified through the first three stages of the OPCPA. The pump beam is obtained from a separate Yb:YAG laser. The special JITKA timing system ensures that everything is synchronized. Next, the pulse moves to the booster stage 4, where it is even more amplified by another Yb:YAG pump laser. Afterward, the beam reaches the main amplifier, which consists of stages 5–7, designed to reach the output energy. In the end, the amplified pulse is compressed using a Chirped Mirror Compressor. The system architecture is shown in Figure 1.2.

1.1.2 ALFA Campaign

ALFA (Allegra Laser For Acceleration) is a laser-plasma electron accelerator that utilizes the high-power, high-repetition-rate (1 kHz) L1 Allegra laser to generate ultra-short electron beams (an electron beam is a collection of highly energetic electrons) lasting only a few femtoseconds, with tunable energy up to 50 MeV.

ALFA operates at relativistic intensities (5×10^{18} W/cm²), meaning the laser field is strong enough to accelerate electrons to speeds close to the speed



■ **Figure 1.2** The L1 Allegra system architecture (Source: [12]).

of light, where relativistic effects become significant. The system enables research on LWFA [17], a technique where a laser pulse creates a wave in the plasma that rapidly accelerates electrons to high energies over very short distances.

Its unique capabilities support applications in very high energy radiotherapy (VHEE), x-ray generation, ultra-fast radiobiology, radiochemistry, or studies on radiation effects in electronics. [18]

Diffusion Probabilistic Models

The diffusion probabilistic model (also known as diffusion model), first introduced in a paper *Deep Unsupervised Learning using Non-equilibrium Thermodynamics* [4] in 2015, are a powerful type of generative model inspired originally by physics. The core idea comes from the thermodynamics of gas molecules, where particles naturally diffuse from high-density regions to low-density. In physics, this process is known as an increase in entropy or the progression toward heat death.

However, from an information theory perspective, we can see it as a gradual loss of information due to an increase in noise in the system. The key idea behind the diffusion process is that if we can build a model that can learn how the information in the data is systematically lost due to increased noise, then it should also be possible to reverse this process and recover the information from the noise.

Diffusion models have been successfully used for tasks such as image generation [19], video prediction [20], or audio synthesis [21]. Their main advantage is generating high-quality and diverse samples while avoiding problems such as mode collapse, a common problem in Generative Adversarial Networks (GANs)¹ [22], where the generator produces repetitive or limited outputs instead of capturing the full data distribution [23, 24].

Another generative approach, Variational Autoencoders (VAEs)² [25], can suffer from blurry outputs due to the constraints of its surrogate loss function [26], the Evidence Lower Bound (ELBO). Unlike VAEs, diffusion models usually do not suffer from blurred outputs, as they generate data through an iterative denoising process that refines the details at each step [24].

¹GANs consist of two separate networks called a generator and a discriminator, which are trained simultaneously. The generator creates new data similar to the data in the training set while the discriminator evaluates whether the data are real or generated.

²Variational Autoencoders are generative models that learn to encode data into a latent space and then decode them back in order to reconstruct realistic samples.

2.1 One-Dimensional Diffusion Models

Although diffusion models are often associated with two- or three-dimensional data, such as images [19], where noise is added and removed across spatial dimensions, the same principles can be applied to one-dimensional data. In one-dimensional diffusion, the model operates on sequences of data. These sequences can represent various data types, such as spectral intensity distributions, waveforms, or other ordered numerical vectors.

In this setting, noise is added incrementally to the entire sequence in multiple steps, gradually transforming the structured input into noise. The model is trained to reverse this process, reconstructing the original structure step by step.

One-dimensional diffusion is computationally more efficient and faster to train and sample from compared to higher-dimensional diffusion [27, 28]. While it is naturally suited for one-dimensional data, such as spectral intensity profiles, it can also be applied to certain types of two-dimensional data that can be meaningfully reduced to one-dimensional sequences without significant loss of information [29]. Therefore, one-dimensional diffusion models are a practical choice for fast and efficient data generation.

2.2 Denoising Diffusion Probabilistic Models

Denoising Diffusion Probabilistic Models (DDPMs) were introduced by Jonathan Ho, Ajay Jain, and Pieter Abbeel in their 2020 paper, *Denoising Diffusion Probabilistic Models* [19]. Their work builds on the earlier paper *Deep Unsupervised Learning using Nonequilibrium Thermodynamics* [4]. The key innovation of DDPMs was to refine the training process and sampling efficiency, making diffusion models more usable for high-quality image generation and other generative tasks.

The diffusion model is defined as a parameterized Markov chain, which is a sequence of random variables where each state depends solely on the previous one, and the diffusion process is divided into forward and reverse processes. In the forward process, the original data are iteratively turned into noise by going through the Markov chain and introducing Gaussian noise into the data. The reverse process, also known as sampling, aims to recover information from noise by modeling reverse Markov transitions that maximize the likelihood of training data, utilizing a trained neural network model [4, 19].

2.2.1 Forward Diffusion Process

In the forward diffusion process, a Markov chain [30] is used to iteratively add Gaussian noise to the input data in T steps, creating a sequence of noisy

samples $x_1, \dots, x_T$. In the end, the original data are turned into nearly pure Gaussian noise distribution.

Step sizes are controlled by a hyperparameter variance schedule, denoted $\{\beta_t \in (0, 1)\}_{t=1}^T$. Usually, the values of β_t are very low to prevent the variance from exploding. Popular choices of schedules are linear (the schedule originally used by the authors of DDPM) or cosine schedule, which was introduced by OpenAI researchers in 2021 in a paper *Improved Denoising Diffusion Probabilistic Models* [31].

Each step of the forward diffusion process is defined as:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I) = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon$$

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}),$$

where:

- q is the forward process,
- x_t is the output of the forward process at step t ,
- x_{t-1} is the input at step t ,
- $\mathcal{N}$ is a normal distribution,
- $\sqrt{1 - \beta_t} x_{t-1}$ is the mean and $\beta_t I$ represents variance,
- $\epsilon \sim \mathcal{N}(0, I)$ is a standard Gaussian noise term that introduces randomness into the process.

We can sample x_t at any time step t in a close form using the reparameterization trick [25]. The forward step could then be written as follows:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I) = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon,$$

using notation $\alpha_t := 1 - \beta_t$, and $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$. Due to this fact, training can be done more effectively by predicting the noise at any step t without the need to go through the whole diffusion process each time (iteratively predicting the noise at every step from 0 to t). [19, 32]

2.2.2 Reverse Diffusion Process

In the reverse diffusion process (also known as sampling), the aim is to reverse the forward diffusion process to generate new realistic samples. However, because the forward process randomly adds noise, it is impossible to do the reversion $q(x_{t-1} | t)$ directly. Therefore, a learnable model p_θ is used that approximates the reverse transition $q(x_{t-1} | t)$.

The diffusion model predicts the complete noise present in the data in a given timestep t to recover the denoised data accurately. The joined distribution $p_\theta(x_{0:T})$ is called the reverse process and is defined as a Markov chain of learned Gaussian transitions starting at $p(x_T) = \mathcal{N}(x_T; 0, I)$:

$$p_\theta(x_{0:T}) := p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t),$$

$$p_\theta(x_{t-1} | x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

where $\mu_\theta(x_t, t)$ is the mean and $\Sigma_\theta(x_t, t)$ equals $\sigma_t^2 I$ which is the variance. The variance is set to be time dependent but not trainable and the authors [19] set it equal to $\beta_t I$, where β_t corresponds to β_t from the variance schedule in the forward diffusion process. The mean can be written as follows:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right)$$

where $\epsilon_\theta(x_t, t)$ is the predicted noise of the neural network. The sampling is then performed by iteratively applying the reverse transition as follows with the exception of the last iteration where the noise is not $\sqrt{\beta_t} \epsilon$ to the output:

$$x_{t-1} = \mathcal{N} \left(x_{t-1}, \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right), \sqrt{\beta_t} \epsilon \right)$$

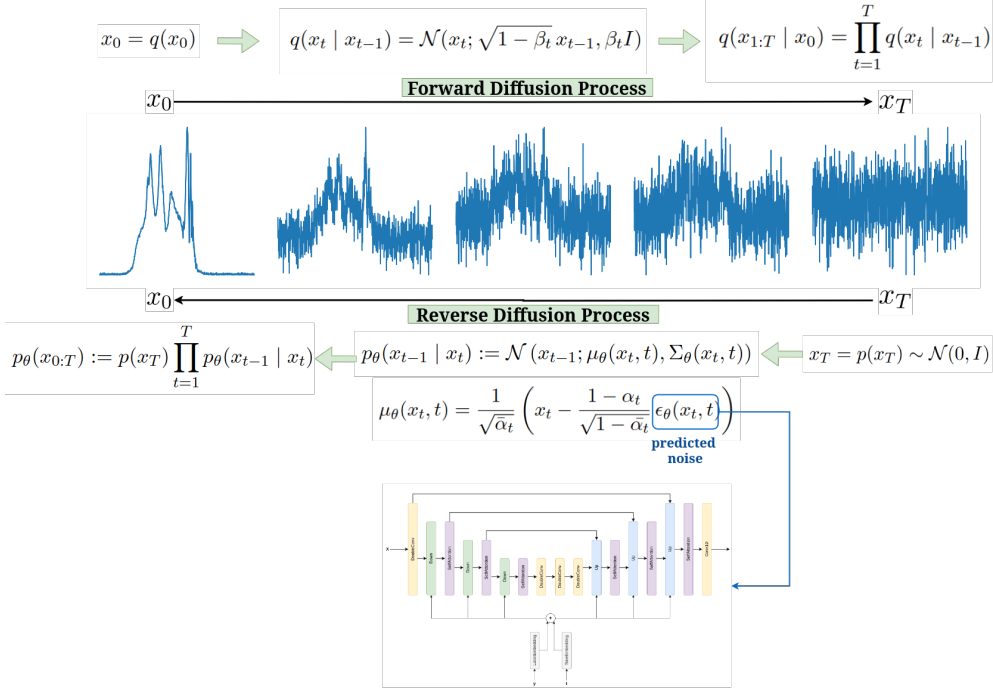
$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right) + \beta_t \epsilon$$

The whole forward and reverse diffusion process in DDPM is illustrated in Figure 2.1. [19, 32]

2.3 Denoising Diffusion Implicit Model

In 2020, researchers at Stanford University came up with a paper *Denoising Diffusion Implicit Models* [33]. DDIMs share the same training procedure as the previous DDPMs, upon which they are built. However, the new sampling process presented in the paper is more effective than the previous sampling algorithm. Moreover, it was proved that DDIMs produce high-quality samples 10-50 times faster than DDPMs [33].

The core idea is the generalization of the sampling process using a non-Markovian diffusion process, which does not follow the strict sequential structure of a Markov chain. Standard DDPMs usually require around 1000 denoising steps for high-quality sample generation because a sample at time t cannot be generated until the entire Markov chain has been generated before t . However, the new implicit generative process allows for direct transitions between distant timestamps, significantly reducing the number of required steps while maintaining the sample quality.



■ **Figure 2.1** The forward and reverse diffusion process in DDPM.

The single step in the implicit generative process of DDIM is given by the following deterministic alternative to the previously used one:

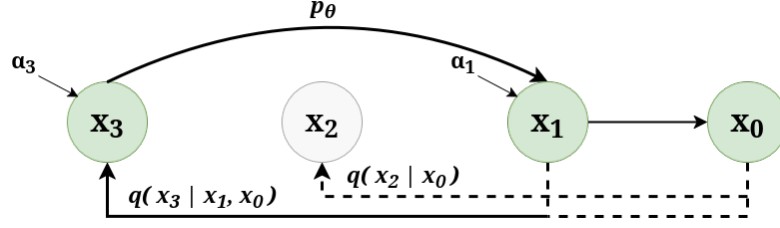
$$x_{t-1} = \underbrace{\sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t)}{\sqrt{\alpha_t}} \right)}_{\text{"predicted } x_0"} + \underbrace{\sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(x_t)}_{\text{"direction pointing to } x_t"} + \underbrace{\sigma_t \epsilon_t}_{\text{random noise}}$$

where ϵ_t is a standard Gaussian noise independent of x_t . Different choices of σ values result in different generative processes, and it is important to point out that the trained model ϵ_t is the same for all of them, therefore, it is not necessary to retrain the model. This means that trained DDPM models could be easily converted to DDIM. For the DDPM generative process, the parameter σ is set to:

$$\sigma_t = \sqrt{\frac{1 - \alpha_{t-1}}{1 - \alpha_t}} \cdot \sqrt{\frac{1 - \alpha_t}{\alpha_{t-1}}}$$

When $\sigma = 0$, the denoising process becomes completely deterministic, and the only noise added is the original noise at x_θ . Because there is no added noise, it is no longer necessary to use the Markovian chain, thus it is possible to skip steps. In Figure 2.2 there is a skip from set x_3 to x_1 , while skipping x_2 .

The authors [33] define the new diffusion process as a subsequence, τ , which consists of a selected subset of steps from the original diffusion sequence. For



■ **Figure 2.2** Graphical model for accelerated generation, where $\tau = [1, 3]$.

example, one could sample every other step in the diffusion process, resulting in a subsequence $\tau = [0, 2, 4, \dots, T - 2, T]$. Therefore, the sampling process became much faster and more efficient. [33]

2.4 Score-Based Generative Modeling through Stochastic Differential Equations

At the core of diffusion models is the idea of gradually adding noise into a signal until it becomes pure noise and then reversing this process to generate new unseen realistic samples. Researchers from Stanford University, together with researchers from Google Brain in 2021 devised an idea to describe this forward and backward process using Stochastic Differential Equations (SDEs). They described their ideas in the paper *Score-Based Generative Modeling through Stochastic Differential Equations*. [27]

The key idea is that forward diffusion processes are solutions of SDEs. In general, an SDE has the following form:

$$dx = f(x, t) dt + g(t) dW,$$

where:

- W is the standard Wiener process (Brownian motion), with dW representing infinitesimal white noise.
- $f(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a vector-valued function called the drift coefficient of $x(t)$, which controls the deterministic part of the system.
- $g(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar function known as the diffusion coefficient of $x(t)$, determining the intensity of randomness in the process.

An SDE solution is a continuous collection of random variables $\{x_t\}_{t \in [0, T]}$, which traces stochastic trajectories over time as the time index t increases from the start time 0 to the end time T [34].

In the forward process, the data distribution gradually transforms into a known prior distribution by injecting increasing amounts of noise. This

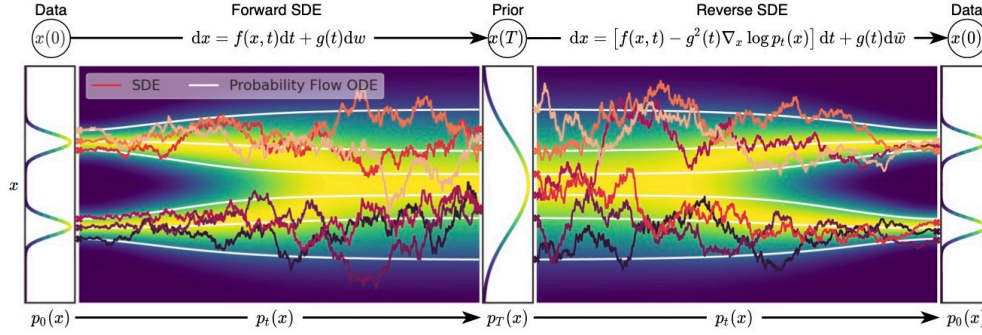


Figure 2.3 Overview of score-based generative modeling through SDEs (Source: [27]).

transition follows the path from $x(0)$, the original data, to $x(T)$, which represents pure noise. Since there are various ways to construct such a diffusion process (a solution of SDE), they diffuse into a fixed prior distribution. Visualizations of randomly selected stochastic trajectories for the forward diffusion process are shown on the left side of Figure 2.3.

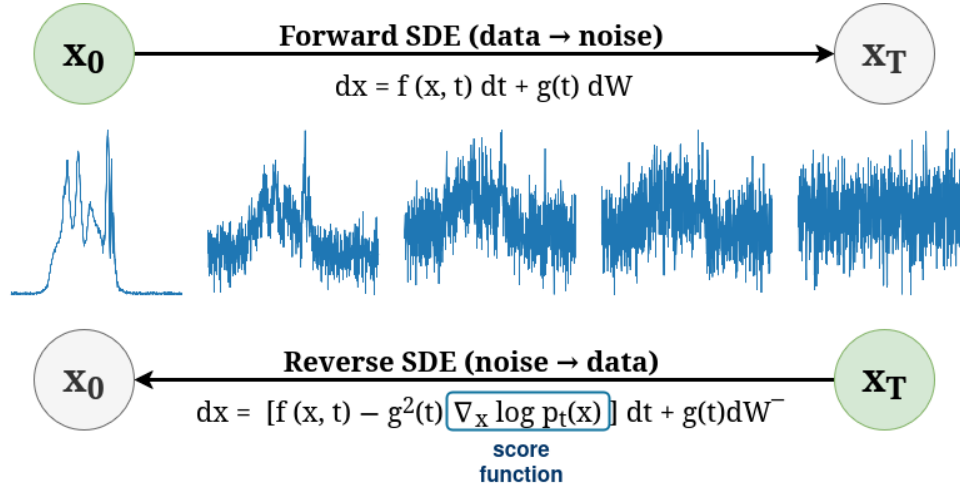
The backward process corresponding to the generative modeling entails following a random path starting at the left edge of the Figure on the right side 2.3, which would guarantee meaningful novel data at the right edge. The theory of SDEs enables us to reverse the time direction to transform the prior distribution into the data distribution, going from $x(T)$ back to $x(0)$. [35] This is possible because any SDE has a corresponding reverse SDE [36], whose close form is given by:

$$dx = [f(x, t) - g(t)^2 \nabla_x \log p_t(x)] dt + g(t) d\bar{W},$$

where $\bar{W}$ is a standard Wiener process and dt represents a negative infinitesimal time step, since the SDE needs to be solved backward in time, from T to 0. To compute the reverse SDE, it is necessary to estimate $\nabla_x \log p_t(x)$, which is the score function of $p_t(x)$. The score function is the logarithmic probability density gradient with respect to x (data), as highlighted in Figure 2.4. The approximation of the score function at different noise levels (t) is done using a trained neural network $s_\theta(x, t)$. This approach is called score-based generative modeling. [34]

2.5 Elucidating the Design Space of Diffusion Based Models

In 2022, researchers from NVIDIA published a paper *Elucidating the Design Space of Diffusion-Based Generative Models* [37]. Their work follows on from



■ **Figure 2.4** Diagram of score-based generative modeling.

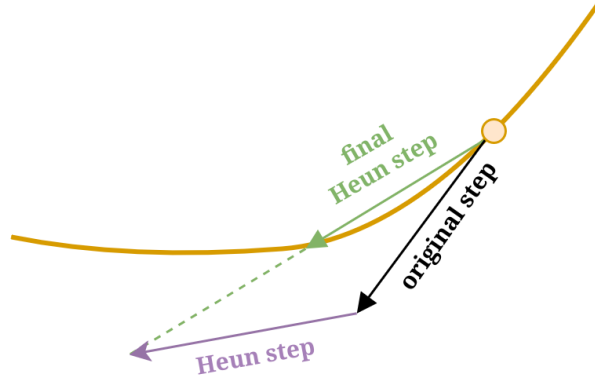
ideas in the paper *Score-Based Generative Modeling through Stochastic Differential Equations*. The work discusses several changes for both the training and sampling processes in diffusion models and shows that the design changes they made improved both the efficiency and quality of the predicted images obtained. The results were demonstrated by measuring Fréchet Inception Distance (FID)³ metric in the ImageNet-64 dataset.

2.5.1 Sampling

One of the main advantages the work introduces is the possibility of reducing the number of steps required for sampling without compromising quality. To achieve this improvement, several optimization choices were made by the authors in the sampling process [37, 35]:

1. Design an ordinary differential equation whose flow lines are as straight as possible, these lines can be easily followed. This is done using an appropriate noise schedule.
2. Do time-step discretization by identifying noise levels that do need extra stepping.
3. Adjust the step function so that it does the step in a smarter manner by using the high-order solver.

³The FID metric measures the similarity of generated images to real ones, by comparing their feature vectors. Lower values indicate better quality and diversity of the generated samples. [38]



■ **Figure 2.5** The Heun step illustrated geometrically.

Diffusion models typically add noise consistently at every step of the process. This results in curved flow lines, which correspond to the trajectories described by the probability flow ODE (as illustrated in Figure 2.3). Thus, these curved paths are difficult to approximate accurately using linear steps. The approach proposed in this work increases the noise at a steady rate over time, resulting in nearly straight flow lines.

Instead of using uniformly spaced time steps, the researchers used an adaptive strategy to optimize step size at different noise levels. At high noise levels, larger steps can be taken without a significant loss of accuracy, which reduces the number of required denoiser evaluations and improves efficiency. However, at low noise levels shorter steps should be performed because fine image details emerge rapidly and large steps may result in the loss of important information in data. The adaptive step size strategy is ensured by gradually decreasing the step sizes.

Higher-order solvers improve the accuracy of step progression in diffusion models by better following the flow lines' curvature. The researchers utilized the second-order Heun scheme. This takes a tentative step at first, then a second one, and the final step is done by retracing back halfway from the starting point. Figure 2.5 gives a visual illustration of the method. Although the method doubles the expense per iteration, it significantly reduces the number of required sampling steps while maintaining high image quality.

2.5.2 Training

The work includes several key design choices for training the denoiser. This section will explain the decision-making process behind them. [37, 35]

The authors were working with an image dataset whose pixel values were normalized in the range -1 to 1. However, the pixel value at different noise levels during the denoiser process varies in the range from -100 to 100, espe-

```

1 def calculate_loss(denoised_image, clean_image, sigma):
2     weight = (sigma^2 + sigma_data^2) / (sigma * sigma_data)^2
3     loss = weight * sum((denoised_image - clean_image)^2)
4     return loss

```

■ **Code listing 2.1** Weighted Loss Computation in EDM (Source: [37]).

cially at high noise levels. This may lead to poor final performance. Therefore, the authors approached standardization of the noisy images during denoising. Because the noise level is known, they scaled the noisy images to a standard magnitude before feeding them to the network.

Most trained diffusion models predict the noise of the noised input and then receive the denoised output by subtracting the noise from the noised input. Predicting only noise is particularly effective at low noise levels because the network receives input from nearly clear data and predicts only a small noise correction. But at high noise levels, this method amplifies errors as the large noise magnitude increases the network’s contribution.

In contrast, the authors decided to predict the expected output D_θ directly. They proposed a neural network with a σ -dependent skip connection $c_{\text{skip}}(\sigma)$ that allows it to estimate either the clean signal, the noise, or something in between. The blending weights $c_{\text{in}}(\sigma)$ and $c_{\text{out}}(\sigma)$ control how much of the input is recycled and how much the network contributes, respectively, which shows the following formula:

$$D_\theta(x; \sigma) = c_{\text{skip}}(\sigma)x + c_{\text{out}}(\sigma)F_\theta(c_{\text{in}}(\sigma)x; c_{\text{noise}}(\sigma)),$$

where F_θ is the neural network to be trained and $c_{\text{noise}}(\sigma)$ maps the noise level σ to a conditioning input for F_θ . The main idea behind using the blending weights is to achieve a minimal amplification of the network output at all noise levels, as the weights are functions of σ .

To ensure stable training across different noise levels, the authors also addressed the issue of varying gradient feedback magnitudes. Without proper scaling, the loss function produces significantly different values depending on the noise level, leading to inconsistent weight updates. This could also be understood as the application of an unintended varying learning rate. To deal with this problem, they used a statistical formula to normalize the expected loss magnitude for each noise level, ensuring uniform gradient updates. The pseudocode 2.1 shows how the loss is calculated to improve the network’s ability to generalize across different noise conditions, where σ_{data} stands for an estimated standard deviation of data.

Finally, they chose to sample noise levels from an exponential distribution controlled by parameters P_{mean} (average noise level) and P_{std} (spread of random noise levels), which were empirically tuned for robustness across different datasets and set to values -1.2 and 1.2 respectively.

The pseudocode 2.2 summarizes all the design choices made by the authors.

```

1  # Required parameters
2  P_mean = -1.2
3  P_std = 1.2
4  sigma_data = 0.5
5
6  def denoise(noisy_image, sigma):
7      c_in = 1 / sqrt(sigma^2 + sigma_data^2)
8      c_skip = sigma_data^2 / (sigma^2 + sigma_data^2)
9      c_out = (sigma * sigma_data) / sqrt(sigma^2 + sigma_data^2)
10     c_noise = log(sigma) / 2
11     return c_skip * noisy_image + c_in * net(noisy_image, c_noise)
12
13 # Loop over dataset
14 for clean_image in Dataset:
15     sigma = exp(P_mean + P_std * randn())
16     noisy_image = clean_image + sigma * noise_like(clean_image)
17     denoised_image = denoise(noisy_image, sigma)
18
19     loss = calculate_loss(denoised_image, clean_image, sigma)
20     # Perform backpropagation and optimizer update

```

■ **Code listing 2.2** Weighted EDM Denoising Algorithm (Source: [37]).

2.6 Conditioned Generation

It is possible to extend diffusion models to conditional generation, where we can add some specific constraints to consider while generating the sample. For example, if we use image data, we can specify a class label that the output image should have. Other examples may be, for instance, text descriptions or parameters of a physical model whose behavior is predicted. [32]

Unlike unconditional modeling, where samples are generated purely based on the learned data distribution, conditional diffusion models integrate extra conditioning information to direct the generation process toward specific desired outcomes. There are multiple ways to combine conditioning with diffusion models, such as concatenation-based conditioning, cross-attention, and classifier-based. [39, 40]

2.6.1 Classifier-Free Guidance (CFG)

In 2022, Jonathan Ho and Tim Salimans proposed a way to perform diffusion guidance in a paper *Classifier-Free Diffusion Guidance* [41]. The key concept lies in combining training with a conditional and an unconditional diffusion model.

The researchers used a single neural network for both approaches, changing the model's input to a null value for the unconditional model. During the training process, the unconditional input was randomly selected with probability p_{uncond} given as a model hyperparameter.

In the sampling process, they used a linear interpolation of the results obtained from conditional and unconditional approaches. The contribution of the conditional sample to the linear interpolation is determined by a weight parameter w .

It turned out that CFG improves sample diversity by balancing how much the model follows the given condition while still exploring different possible outputs, leading to more varied and realistic results.

Bayesian Optimization

Bayesian optimization is a powerful strategy for optimizing objective functions that are expensive to evaluate, such as machine learning models. It is a class of optimization methods that are concentrated on solving $\max_{x \in A} f(x)$, where f is a continuous objective function, which is usually considered as a "black box", $x \in \mathbb{R}^d$ is an input of d dimensions, and A stands for a domain of valid inputs for the objective function. [42]

In machine learning, a typical example is the hyperparameter tuning for a neural network, where x represents a set of hyperparameters, A is the possible range of these hyperparameters, and $f(x)$ is the validation accuracy or loss function that we aim to maximize or minimize. [43]

3.1 Bayesian Optimization Algorithm

Bayesian optimization contains two main components – a Bayesian statistical model that models the objective function and an acquisition function needed to decide the next direction. The algorithm consists of the following steps [44]:

1. The process begins by selecting the prior data distribution representing the initial belief about the unknown objective function. Usually, a Gaussian process (GP)¹ is used for this task.
2. Next, the initial set of observations is used to fit the model and estimate the mean and uncertainty of the objective function f .
3. The acquisition function is then used to select the next parameters x to be evaluated. It should trade off the exploration of uncertain regions against the exploitation of known high-performing areas.

¹A Gaussian Process (GP) is a distribution over an infinite set of possible function outputs, where the distribution over any finite subset of these outputs follows a multivariate Gaussian distribution. [45]

4. Afterward, the objective function is evaluated at the selected point and the result is integrated into the GP model, updating the posterior distribution.
5. Steps 3 and 4 are repeated until a predefined stopping criterion, such as a number of iterations or convergence to an optimal value, is satisfied.

Although Bayesian optimization is a powerful tool for black-box optimization, it can be time-consuming and may not always find the best solution, especially in high-dimensional spaces [46]. It works best for continuous domains with fewer than 20 dimensions, where the Gaussian process model captures the function well, and the acquisition function efficiently guides the search [42].

3.2 Commonly Used Acquisition Functions

Bayesian optimization relies on acquisition functions to guide the selection of new points to evaluate. In this section, an example of three of the most commonly used acquisition functions is presented: Lower Confidence Bound, Expected Improvement, and Probability of Improvement.

3.2.1 Lower Confidence Bound

The Lower Confidence Bound acquisition function [47] selects the next point by minimizing a combination of the predicted mean and the uncertainty (standard deviation) from the Gaussian Process model. It is defined as:

$$\text{LCB}(x) = \mu(x) - \kappa \cdot \sigma(x),$$

where $\mu(x)$ is the predicted mean at point x , $\sigma(x)$ is the predicted standard deviation, and κ is a parameter that controls the trade-off between exploration and exploitation. A higher κ encourages exploration by giving more weight to uncertain areas.

3.2.2 Expected Improvement

The Expected Improvement acquisition function selects the next point to evaluate by estimating the potential gain compared to the current best-known value. [48] It considers two criteria: how much better a point might be and how likely that improvement is. The Expected Improvement is calculated as:

$$\text{EI}(x) = \begin{cases} (\mu(x) - f_{\text{best}} - \xi) \cdot \Phi(Z) + \sigma(x) \cdot \phi(Z), & \text{if } \sigma(x) > 0 \\ 0, & \text{if } \sigma(x) = 0 \end{cases}$$

where:

$$Z = \frac{\mu(x) - f_{\text{best}} - \xi}{\sigma(x)},$$

$\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, and $\phi(\cdot)$ is its probability density function. $\xi \geq 0$ is a small constant used to encourage exploration, and $\sigma(x)$ is the predicted standard deviation.

This acquisition function tends to balance exploitation and exploration efficiently and is particularly suitable for noisy or expensive functions.

3.2.3 Probability of Improvement

The Probability of Improvement acquisition function [49] chooses points based solely on how likely they are to outperform the current best result. It is defined as the probability that a new point gives a better value than the best observed one:

$$\text{PI}(x) = \Phi \left(\frac{\mu(x) - f_{\text{best}} - \xi}{\sigma(x)} \right),$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, $\sigma(x)$ is the predicted standard deviation, and ξ is a small positive constant used to encourage exploration.

Problem Statement

Machine learning has become an essential tool for modeling complex physical systems, where traditional analytical methods struggle due to high dimensionality or nonlinear dynamics. Among various generative models, diffusion models have emerged as a state-of-the-art approach [50, 51, 52], particularly suited to handle stochastic processes and uncertainty in experimental data.

Traditional deterministic models provide a single fixed prediction, whereas diffusion models are able to generate a range of possible outcomes by systematically adding and removing noise. This ability allows them to capture uncertainty and variability in experimental conditions, making them particularly well-suited for systems where small changes in input can lead to diverse results. [53]

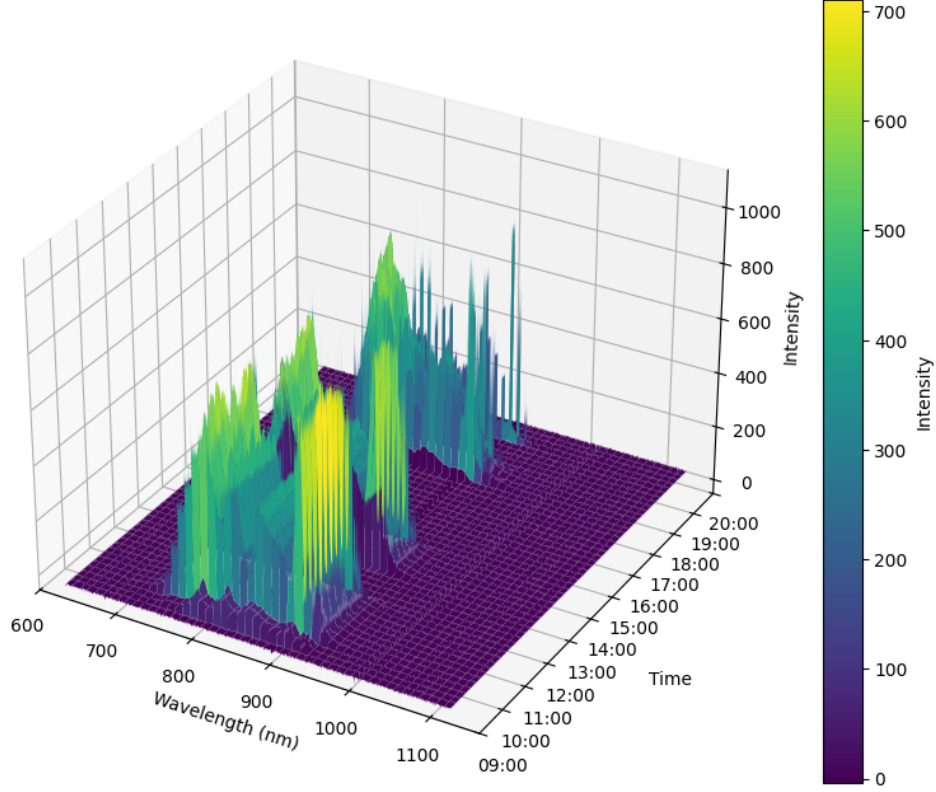
Unlike other neural network approaches that may struggle with mode collapse or biased predictions, diffusion models learn the full data distribution, ensuring diverse and realistic outputs that reflect the stochastic nature of real-world data. [54, 55]

This ability to model uncertainty and generate diverse, realistic output is particularly valuable for digital twins: virtual replicas of physical systems that simulate their behavior under varying conditions. Digital twins are widely used in engineering and science to simulate entire systems, improve performance, and optimize processes. [56, 57, 58]

4.1 Prediction of Spectrum Intensities with Diffusion models

In high-power laser experiments, many small factors, such as changes in temperature, humidity, air pressure, or even dust in the air, can influence the results. External conditions, such as weather changes or the number of people in the experimental hall, can also have an impact. Diffusion models help by learning the full range of possible laser spectra instead of just one expected

Spectrum on 03.12.2024 from 09:00 to 20:00, number of datapoints: 37775



■ **Figure 4.1** Spectrogram of spectrum intensities in time.

outcome. They can infer how the system is likely to behave under given conditions, based on previously observed patterns in the data.

This is especially useful in the context of LWFA, where direct spectral measurements from the last OPCPA stage are not available during the acceleration itself. Currently, it is technically impossible to capture accurate intensity measurements in that time. Therefore, spectral measurements are only recorded in the intervals between LWFA events when the laser power is fine-tuned to accomplish LWFA successfully. These data gaps are shown in Figure 4.1.

By conditioning the diffusion model on known system parameters, such as piezo settings, power levels, or other diagnostics, it becomes possible to predict what the spectral output might look like during those unmeasured periods. This allows researchers to reconstruct missing parts of the experimental timeline. This knowledge may lead to a more complete understanding of how the laser behaves during acceleration.

However, predicting the spectral behavior during LWFA is not the only potential application. Simulating the likely spectral response of the system under various configurations before conducting expensive laser experiments can help

save both resources and time. In addition, this predictive capability supports deeper exploration of the complex relationships between system parameters and spectral outcomes, offering insights that may otherwise remain hidden.

4.2 Prediction of Ideal Parameters

In addition to predicting the spectral output of known system parameters, another important task is determining what parameter values are needed to achieve a desired spectral property. For instance, in LWFA, certain spectral characteristics, such as the mean wavelength, the position of the maximum intensity, and the maximum intensity value itself, must fall within a specific range to ensure desired system performance. Since direct trial-and-error tuning of all system parameters is time-consuming and resource-intensive, it is helpful to have a method that can efficiently suggest suitable settings.

To solve this inverse problem, Bayesian optimization in combination with the pre-trained diffusion model might be used. The strategy is to define target values for the key spectral characteristics and allow the optimization algorithm to search for parameter configurations that are most likely to reproduce a spectrum of the desired properties. In each step, the diffusion model may be used to simulate the resulting spectrum for a proposed parameter vector.

Identifying promising configurations in advance is particularly valuable in experimental contexts, where laser time is limited and physical measurements can be costly or technically restricted.

This approach also provides insight into the relationship between the parameters of the laser system and the spectral outcomes. Over time, by observing which configurations consistently provide desired results, it becomes easier to understand the system's behavior.

Chapter 5

Related Work

Bayesian optimization is a well-established technique. In addition to its recent rise in popularity in machine learning as an optimization method for hyperparameter tuning, it is also a prominent method for optimizing complex processes in physics.

Diffusion models are a relatively new approach that has become very popular recently, and many studies show how useful they are for solving a wide range of physics-related tasks.

5.1 Applications of Bayesian Optimization

Particle acceleration using plasma-based methods [59] is a cutting-edge technique that generates high-energy particle beams over short distances. However, conducting experiments with high-power lasers is expensive and requires advanced facilities [60]. Computer simulations have a strong impact on understanding past and present plasma acceleration experiments.

For such simulations, the most popular algorithm is the Particle-In-Cell (PIC) technique since it provides the most reliable description of plasma. PIC uses groups of particles collectively representing the plasma particle dynamics, called macroparticles. The electromagnetic fields are represented on a grid. [61, 62]

Despite the effectiveness of PIC, its simulations are computationally intensive, especially when optimizing a large number of interacting parameters such as laser energy, plasma density, or acceleration length. Bayesian optimization is a powerful method for solving such problems because it efficiently explores complex parameter spaces while minimizing the number of costly simulations or experiments. [63]

One of the first demonstrations of Bayesian optimization in laser-plasma experiments was shown in a paper called *Automation and control of laser wakefield accelerators using Bayesian optimization* [64] from 2020. The authors

focused on optimizing LWFAs, which accelerate electrons to multiple giga-electron volt energies ($1 \text{ GeV} = 1 \text{ billion electron volts}$) over short distances by creating plasma waves with an ultra-short, intense laser pulse.

They implemented Bayesian optimization on a PIC simulation of a 5 TW laser system, controlling up to six key laser and plasma parameters simultaneously, including the spectral and spatial phase of the laser and the plasma density and length. The created model identified subtle optimizations in the laser pulse that would have been missed in standard manual tuning. A slight one percent change in pulse length led to an increase of 80 % in electron beam charge.

This research also demonstrated the first fully automated laser-plasma accelerator. The Bayesian optimization method provided new physical insights into the dynamics of the electron injection process, which could be used to inform the design of better future plasma accelerators.

In a paper *Multi-parameter Bayesian Optimization of Laser-Driven Ion Acceleration in particle-in-cell simulations* [65] from 2022, the researchers explored how to optimize laser-driven ion acceleration using machine learning techniques. The authors applied Bayesian optimization to guide the choice of input conditions for analyzing PIC simulated data.

A crucial component of the study was the development of BISHOP, a program that automatically generates simulation input files, analyzes results, and adjusts parameters based on Bayesian optimization.

Using Bayesian optimization significantly accelerated the search for optimal parameters compared to traditional grid-based scanning. It was four times faster for adjusting two parameters, and for adjusting four parameters, it was even 1,000 times faster. Moreover, using this optimization method resulted in discovering an unexpected optimal setting for one of the key plasma properties, the front surface density scale length, which would probably have been too expensive to find with traditional scanning.

In 2024, a paper called *Bayesian optimization of laser wakefield acceleration in the self-modulated regime (SM-LWFA) aiming to produce molybdenum-99 via photonuclear reactions* [66] was published. This study explores the use of self-modulated laser wakefield acceleration (SM-LWFA) as a cost-effective method to generate high-energy electron beams.

Traditional LWFA requires very short, ultra-intense laser pulses to produce high-quality electron beams. In contrast, SM-LWFA operates at lower laser power and repetition rates, which allows for more frequent operation, although with reduced beam quality. This method might be suitable for applications that are not sensitive to beam quality, such as the production of medical radioisotopes through photonuclear reactions. The study focused on the production of molybdenum-99, which decays into the most widely used medical radioisotope, technetium-99m.

The authors employed Bayesian optimization combined with PIC simulations to maximize the energy and charge of the accelerated electron beams.

Adjusting the gas density and laser focus improved electron energy and charge. These simulations helped predict how different gas density profiles and laser settings would influence the energy and charge of the accelerated electron beams.

5.2 Applications of Diffusion Models

Diffusion models were originally developed for image generation [4, 19, 31]. However, they have also proved their ability to be used in many other domains [67], including physics, audio synthesis, and scientific simulations. Their ability to model complex patterns makes them especially useful for tasks where traditional simulation methods are computationally expensive or impractical.

In 2021, a paper *DiffWave: A Versatile Diffusion Model for Audio Synthesis* [21] was published. The work introduced a one-dimensional diffusion model called DiffWave for high-quality waveform generation. The researcher examined different generation tasks, including neural vocoding¹ conditioned on mel spectrogram², class-conditional generation, and unconditional generation.

DiffWave is based on the DDPM framework. The architecture of the denoising network does not rely on the traditional U-Net architecture. Instead, it uses a bidirectional dilated convolution architecture motivated by WaveNet [70]. The diffusion step t is incorporated into the model using a 126-dimensional encoding vector, which captures temporal information through sinusoidal embeddings. This embedding is added to the input of each residual layer to guide the diffusion process.

The results showed that the DiffWave model outperformed previous popular models for audio synthesis: the autoregressive model based on a deep convolutional neural network WaveNet and the GAN-based WaveGAN [71]. DiffWave showed better results in both audio quality and sample diversity, especially in unconditional speech generation. These improvements were measured by several automated tests and human evaluations.

In 2024, Matěj Jech in his master thesis *Machine Learning Techniques for Laser-Plasma Acceleration Optimization* [72] focused on developing a generative model for simulating laser-plasma particle acceleration experiments using data from the ELI Beamlines ALFA campaign.

A diffusion-based architecture was used to generate images of the energy spectrum of accelerated electron beams, conditioned on experimental parameters. The approach provides a cost-effective and time-efficient alternative

¹Neural vocoders use deep learning models to convert a low-dimensional representation of an audio signal, such as spectrograms, into waveforms. Waveforms are continuous signals that represent sound vibrations over time. [68]

²A mel spectrogram is a visual representation of an audio signal's frequency content over time, mapped to the mel scale, which mimics human hearing by emphasizing lower frequencies. [69]

to costly real-physics experiments or computationally expensive PIC simulations.

Three different diffusion model architectures (DDPM, DDIM, DiffAE [73]) were tested together with two noising techniques (Gaussian noising and blurring via inverse heat dissipation [74]). The DDIM architecture with Gaussian noising demonstrated the best performance, especially in terms of the sampling speed. The evaluation of the models tested was done using the FID metric, variance difference (which evaluates the stability of the simulated beam), and MSE (to evaluate the accuracy in simulating the beam charge).

Another example of using diffusion models for the modeling of physical phenomena was presented in the 2024 paper called *Diff-PIC: Revolutionizing Particle-In-Cell Nuclear Fusion Simulation with Diffusion Models* [75]. The authors introduced a new framework called Diff-PIC, which utilizes a score-based diffusion model to efficiently generate reliable data for Laser-Plasma Interaction (LPI) in nuclear fusion research.

LPI plays a crucial role in ensuring the stability and efficiency of fusion reactions. The most popular choice for the simulation of LPI are PIC simulations. However, because of their computational intensity and time-consuming nature, the researchers decided to apply the conditional diffusion model as a computationally efficient alternative.

The authors presented the Rectified Flow Acceleration (RFA) optimization technique, based on the principles of Rectified Flow³, to accelerate data generation. RFA converts the original complex denoising trajectory (from the initial noise to the target electric field snapshot) into a direct and straight path. It is done by minimizing the squared deviations between the optimal straight path and the trajectory learned by the diffusion model.

Additionally, the paper introduced Physically-Informed Parameter Encoding (PIPE), which ensured the model accurately captured the complex relationships among the physical parameters. The experimental results showed that Diff-PIC achieves a speed increase of 16,200x compared to traditional PIC simulations while maintaining a high quality of the results.

³Rectified flow is a method for optimizing diffusion models by streamlining the denoising process, which creates a direct transition from an initial noise state to a target distribution. [76]

Evaluation Metrics

When evaluating the accuracy of predictions for one-dimensional spectral data, standard error metrics such as mean square error (MSE) or mean absolute error (MAE) may not always capture the full picture of the similarity between predicted and true data. This is because standard error metrics alone provide quantitative measures of difference.

However, they can be very sensitive to differences in alignment, where the compared data may have very similar shapes but are relatively shifted to each other. In such cases, traditional error metrics may show a big difference despite the fundamental similarity between the patterns.

To address this limitation, it is important to combine evaluation methods and thus compare both structural similarity and numerical error measurements. For this reason, two approaches to compare structural similarity will be presented.

The first one is the dynamic time warping method, which helps to compare sequences that may be misaligned by finding an optimal mapping between their points.

The second is the Wasserstein distance, a distribution-based metric that quantifies the minimal cost required to transform one distribution into another. Unlike point-wise error metrics, the Wasserstein distance takes into account the geometry of the entire distribution, making it especially useful for comparing spectra that differ in position but not in shape.

6.1 Dynamic Time Warping

Dynamic time warping (DTW) is an algorithm used to calculate the distance between two sequences. To achieve that, it is necessary to find the optimal alignment of the two sequences under certain restrictions.

It was originally developed to compare different speech patterns in speech recognition [77]. Today, DTW has found application in diverse fields such as

finance [78], biology [79], and medicine [80].

DTW [81] computes with two sequences $X := (x_1, x_2, \dots, x_N)$ of length $N \in \mathbb{N}$ and $Y := (y_1, y_2, \dots, y_M)$ of length $M \in \mathbb{N}$. Firstly, a cost matrix $C \in \mathbb{R}^{N \times M}$ is constructed. In this matrix, each element $C(i, j)$ represents the distance between points x_i and y_j . The cost is usually calculated using the Euclidean distance.

The goal is to find a warping path $p = (p_1, \dots, p_L)$, where the path length satisfies $\max(N, M) \leq L \leq N + M - 1$ and each element $p_l = (i_l, j_l)$ represents an index pair that aligns x_{i_l} and y_{j_l} . The warping path then represents a sequence of alignments that minimize the total accumulated cost $\sum_{l=1}^L C(i_l, j_l)$ while satisfying the following constraints:

Boundary condition The path starts at the beginning of both sequences $p_1 = (1, 1)$ and ends by linking the last elements $p_L = (N, M)$.

Monotonicity condition The indices must increase monotonically, which means that the warping path cannot move backward.

Step size condition The path progresses through adjacent cells in the cost matrix, allowing only the following moves:

$$(i_{l+1}, j_{l+1}) \in \{(i_l + 1, j_l), (i_l, j_l + 1), (i_l + 1, j_l + 1)\}.$$

In addition to preserving the order of assigned points, this constraint also ensures that each point in the first sequence is assigned to at least one point in the other sequence. DTW allows a point from one sequence to be linked to multiple adjacent points from the other sequence.

The DTW algorithm, which finds the optimal warping path, is based on dynamic programming, and its complexity is $\mathcal{O}(NM)$. To efficiently compute DTW, the accumulated cost matrix $D \in \mathbb{R}^{N \times M}$ is defined. Each cell in this matrix represents the minimum DTW cost needed to align the first n elements of X and the first m elements of Y :

$$D(n, m) = \text{DTW}(X(1 : n), Y(1 : m)).$$

The matrix cells $D(n, m)$ are computed recursively, each cell depends on the values of previously computed cells.

The first row and first column of D are initialized separately because they represent alignments in which one sequence is matched with only the first element of the other. For all other cells, where $n > 1$ and $m > 1$, the value $D(n, m)$ is calculated by taking the minimum value of three previously calculated cells: $D(n - 1, m - 1)$, $D(n - 1, m)$, $D(n, m - 1)$ and adding the cost of aligning x_n with y_n . The final DTW distance between the sequences X and Y is stored in the bottom right cell $\text{DTW}(X, Y) = D(N, M)$, whose value represents the minimum cost needed to align the entire sequence optimally.

6.2 Wasserstein Distance

The Wasserstein distance, also called the Earth Mover's (EM) distance, is a metric used to compare two probability distributions. It is particularly useful for finding differences in the shape and location of distributions, especially when the distributions are similar in form but shifted relative to each other. [82, 83]

Intuitively, it can be interpreted as the minimum energy cost required to transform one distribution into the other, where the cost is determined by the amount of probability mass moved and the distance it is moved. This interpretation is often illustrated by the metaphor of shifting one pile of earth into another: the goal is to find the most efficient way to move portions of the first pile so that the total amount of mass moved and the total distance transported are minimized.

Formally, given two one-dimensional probability distributions u and v , the first EM distance between them is defined as:

$$EM_1(u, v) = \inf_{\pi \in \Gamma(u, v)} \int_{\mathbb{R} \times \mathbb{R}} |x - y| d\pi(x, y),$$

where $\Gamma(u, v)$ represents the set of all valid transport plans π that define how probability mass is moved from values in u to values in v . Each transport plan specifies how much mass should be shifted from a point x in u to a point y in v . The infimum ensures that the optimal transport plan is selected, that is, the one with the least total cost.

If the cumulative distribution functions (CDFs) U and V of u and v are known, the EM distance can also be expressed as:

$$W_1(u, v) = \int_{-\infty}^{+\infty} |U(x) - V(x)| dx.$$

This alternative formulation is useful for empirical data, as it simplifies computation while giving the same result. Effectively, the area between the two cumulative distributions is measured.

Data Understanding

The data used in this work come from the final OPCPA stage of the L1 laser system at ELI Beamlines and consist of two main parts: spectrum intensities and system parameters. The spectrum intensities capture the measured signal strength across a range of wavelength points, while the system parameters describe the configuration and operating conditions of the L1 laser at the time of measurement.

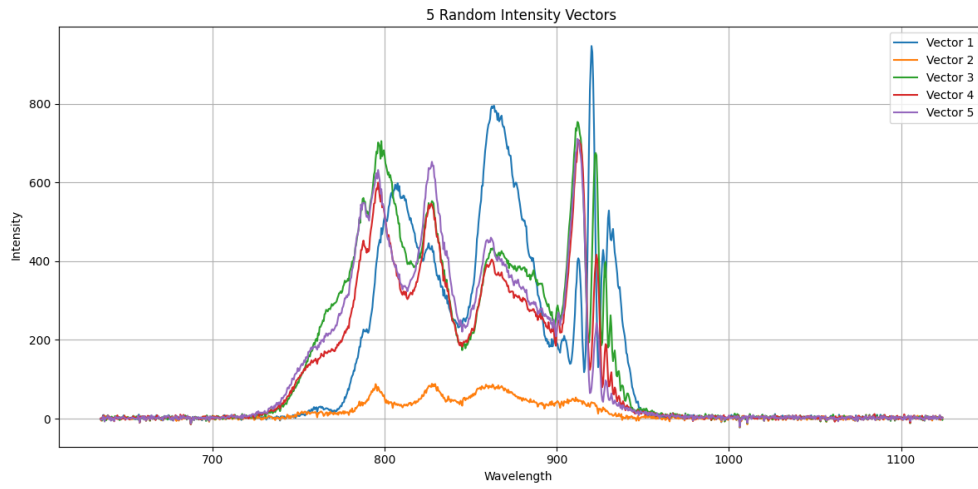
Both types of data were collected over an 18-day period between November 18 and December 11, 2024, excluding weekends.

7.1 Spectrum Intensities

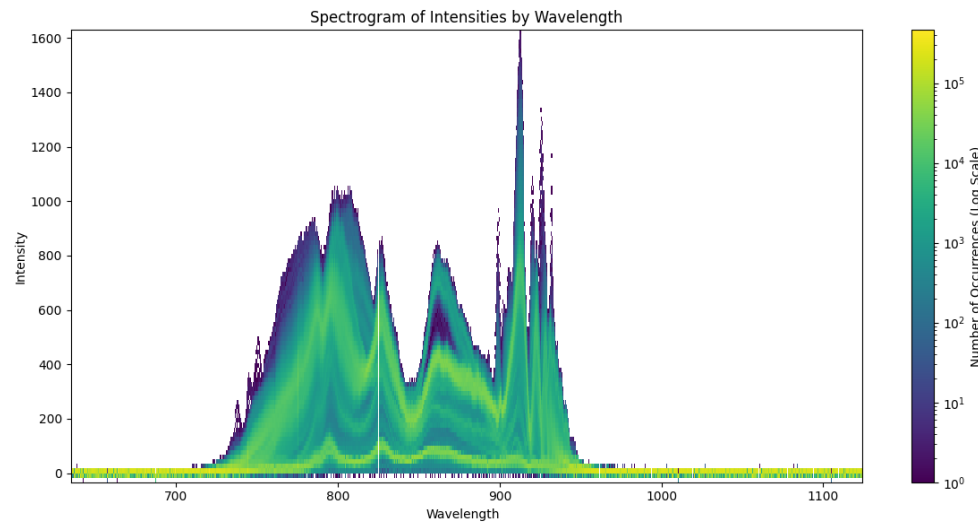
The dataset contains measured spectral intensity entries in arbitrary units, each spectrum consists of 1024 wavelength points ranging from 634 to 1124 nm. In total, the data consist of 741,568 measured entries. Figure 7.1 shows five randomly selected spectra. The intensities were recorded periodically, with an average time difference of approximately 2 seconds between consecutive measurements. The experiments were typically conducted between 7:00 and 20:00, although the exact schedule varied depending on the day.

The measured spectra can be categorized into four main groups based on their maximum intensity values:

- **600-900:** High-intensity spectra,
- **300-550:** Medium-intensity spectra,
- **80-250:** Lower-intensity spectra,
- **below 20:** These entries were filtered out as they corresponded to system states where the LWFA process is done, the acceleration chamber is open, and meaningful spectra measurements cannot be obtained. The recorded



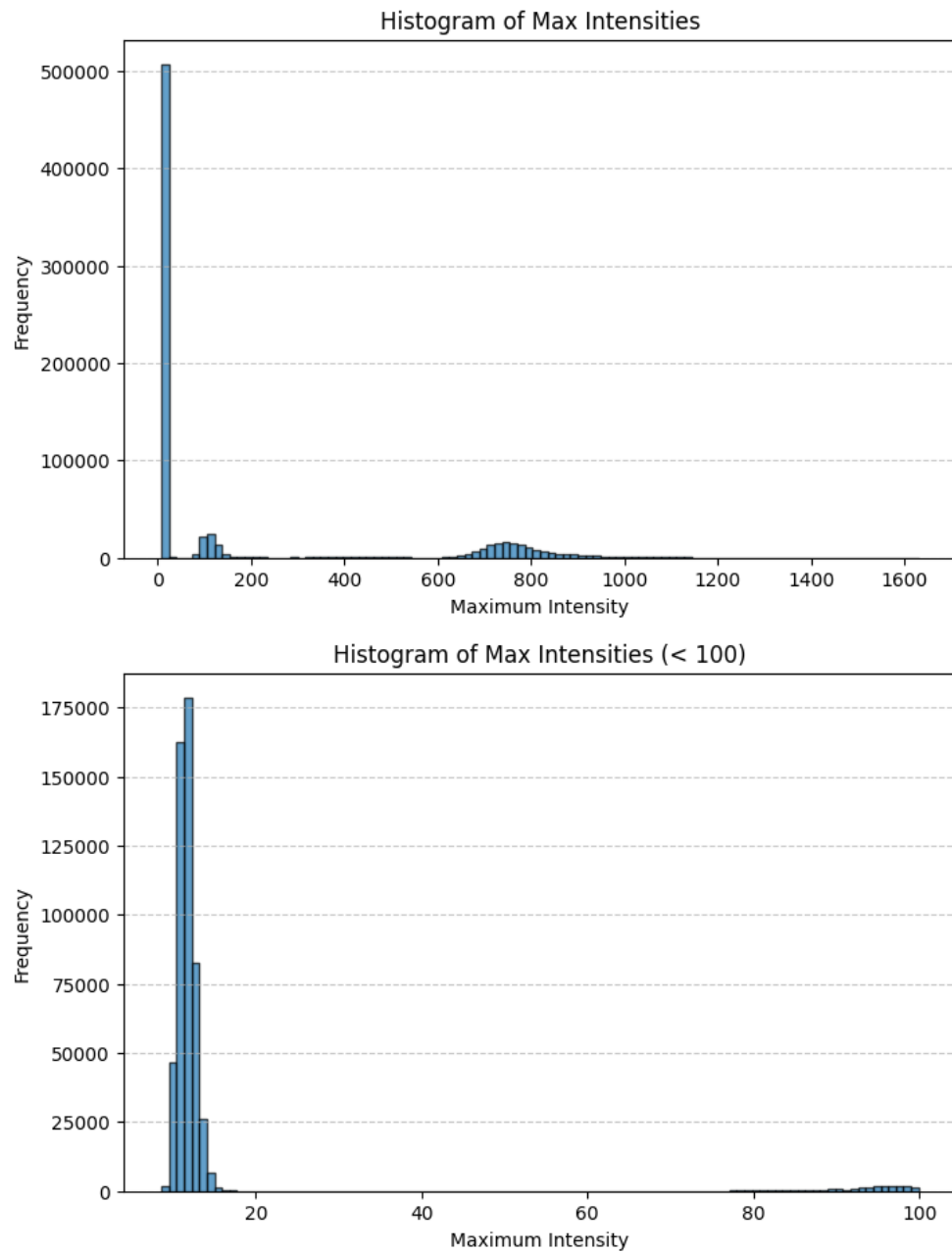
■ **Figure 7.1** Randomly selected spectrum intensities.



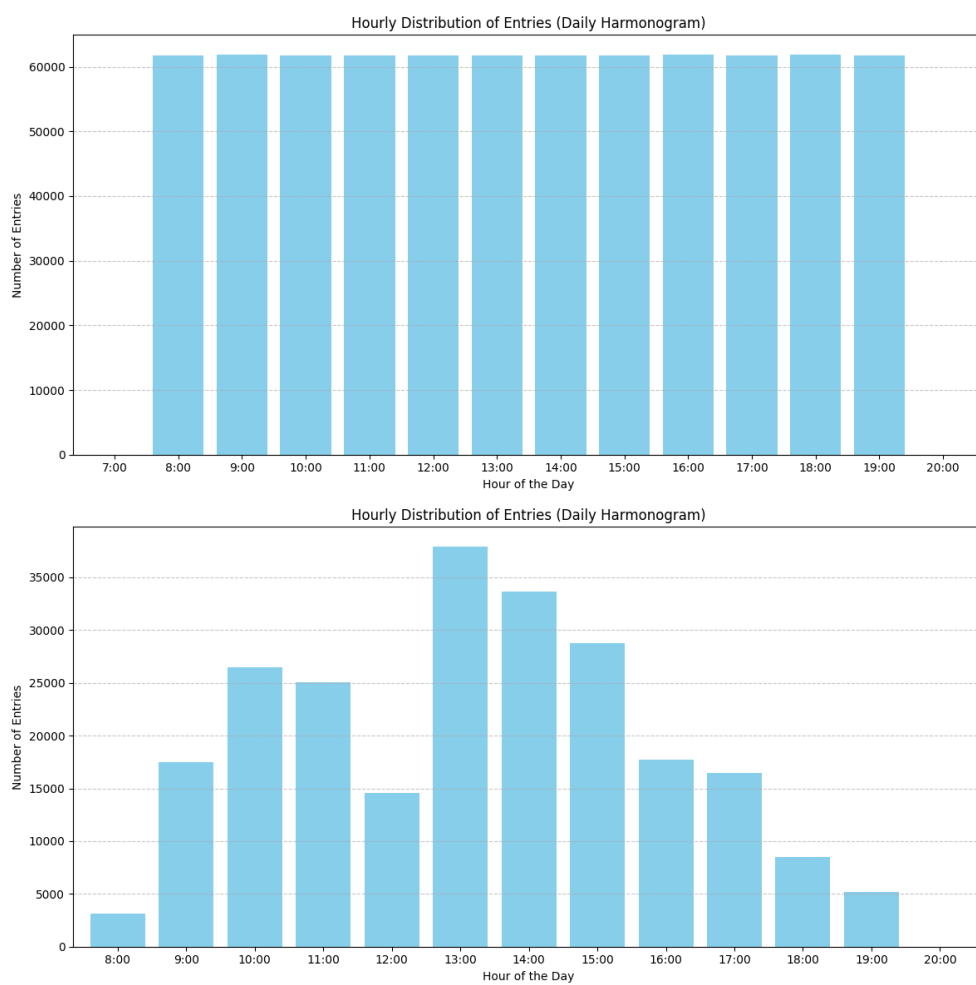
■ **Figure 7.2** Spectrogram of spectrum intensities by wavelength.

values in this range are not exactly zero due to minor measurement inaccuracies and deviations.

Figure 7.2 shows the spectrogram of the intensities measured at different wavelengths in the filtered data. Figure 7.3 shows the main groups of data entries based on maximum intensity values. The hourly distribution of all entries, as well as the filtered entries with a maximum intensity above 20, is presented in Figure 7.4.



■ **Figure 7.3** Histograms of maximal spectrum intensity values: (Top) full range, (Bottom) values smaller than 100.



■ **Figure 7.4** Hourly histogram of entries: (Top) all data, (Bottom) filtered data.

7.2 Parameters

The parameters provide insights into different components of the L1 laser, particularly its piezos and power monitoring. In total, the dataset contains 4,400,768 parameter entries, with the mean difference between consecutive data points being 0.16 seconds. Unlike spectrum intensities, which are measured systematically at regular intervals, parameters are recorded only when they change. The dataset contains 16 parameters:

- **DIRA1 Cavity Piezos:**
 - L1-PL3-PZM52:Setpoint
 - L1-PL3-PZM53:Setpoint
- **DIRA1 External Piezos:**
 - L1-PL3-PZM56:Setpoint
 - L1-PL3-PZM57:Setpoint
 - L1-PL3-PZM58:Setpoint
 - L1-PL3-PZM59:Setpoint
- **DIRA3 Cavity Piezos:**
 - L1-PL5-DIRA3:Piezo1_X_RB
 - L1-PL5-DIRA3:Piezo1_Y_RB
 - L1-PL5-DIRA3:Piezo3_X_RB
 - L1-PL5-DIRA3:Piezo3_Y_RB
- **DIRA3 External Piezos**
 - L1-PL4-PZM66:Setpoint
 - L1-PL4-PZM67:Setpoint
 - L1-PL4-PZM68:Setpoint
 - L1-PL4-PZM69:Setpoint
- **Other**
 - L1-INJ-PM70:VAL_CAL - powermeter providing actual energy values in mJ
 - L1-OPA3-5_2-PM98:VAL_CAL - Leak laser measurement on a photodiode (uncalibrated powermeter, dependent on power and spectrum changes)

Section A.1 in the Appendix summarizes the statistical summary. Section A.2 in the Appendix further illustrates how the system parameters evolve over time for each recorded day.

Data Preprocessing

Several preprocessing steps were applied before using the data for modeling to ensure data quality and consistency. These steps included removing problematic data, handling missing values, engineering new features, mapping parameter records to intensity measurements, and normalizing the data.

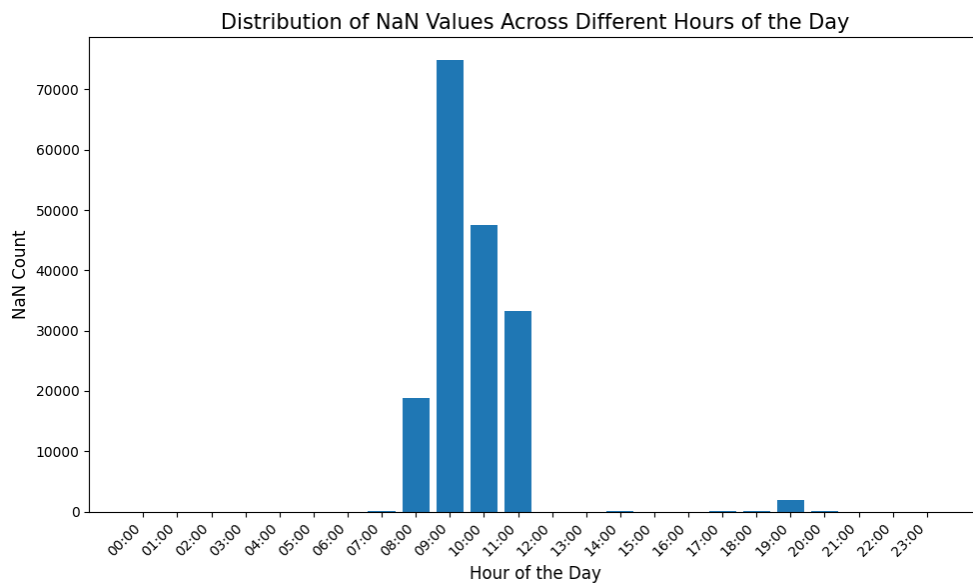
8.1 Data Cleaning

After data analysis and discussion with the ALFA experimental team, it was decided to remove the first three days of the measurements. These days, the laser system exhibited signs of instability, which could not be determined from the provided set of parameters. The final dataset consists of data from 15 days between November 21, 2024, and December 11, 2024, excluding weekends.

During data inspection, certain parameters represented as *#N/A* and *Disconnected*. These missing values predominantly appeared in the morning hours when the laser system was initializing or in the evening around 19:00 when the system was likely to shut down, as shown in Figure 8.1. Since the system was not fully operational during these periods, all records that contained missing values were removed.

To ensure consistency in the data, only measurements recorded between 7:00 and 20:00, which correspond to the standard operating hours of the laser, were included. Although some data points appeared outside this time range, further analysis revealed that these observations were sparse and inconsistent, often appearing as isolated data points recorded at random times, such as in the middle of the night.

The parameter behavior analysis showed that four DIRA3 Cavity Piezos parameters (L1-PL5-DIRA3:Piezo1_X_RB, L1-PL5-DIRA3:Piezo1_Y_RB, L1-PL5-DIRA3:Piezo3_X_RB, and L1-PL5-DIRA3:Piezo3_Y_RB) exhibit problematic dynamics. These parameters remained unchanged for extended periods, often staying constant throughout the majority of the day. In some cases, such



■ **Figure 8.1** Distribution of NaN values across different hours of the day.

as from November 25 to December 4, they retain the same value (0) without any change. When changes did occur, they were infrequent and characterized by sudden jumps to a different value, only to revert back immediately. Notably, these changes predominantly occurred in the morning, aligning with the time the laser was initializing. Given this limited variability, these parameters were removed from the set of parameters used.

8.2 Feature Engineering

The powermeter parameter `L1-INJ-PM70:VAL_CAL` represents the laser energy values. However, the power measurements are delayed by changes in laser output. Consequently, measurements taken too soon after a change may not accurately represent the true energy levels.

To address this issue, a new parameter `L1-INJ-PM70:VAL_CAL_diff` was created. This parameter is calculated by taking the difference between the value of `L1-INJ-PM70:VAL_CAL` at two specific points in time: the value that occurred 10 time steps before the current value and the value that occurred 10 time steps after the current value. The parameter provides an approximation of the rate of change in the powermeter values over time.

Additionally, records where the time gap between these two points exceeded 30 seconds were filtered out to ensure meaningful differences. This ensures that only reliable differences are calculated, filtering out problematic cases where the 10-step window could extend into different measurement blocks or even across different days.

8.3 Mapping Parameter Records to Intensity Measurements

The intensity and parameter records were aligned by assigning each intensity measurement to the closest preceding parameter record. The average time difference between an intensity record and its corresponding parameter record was 0.11 seconds. In total, the final dataset contained 204,529 records and 13 parameters:

- L1-PL3-PZM52
- L1-PL3-PZM53
- L1-PL3-PZM56
- L1-PL3-PZM57
- L1-PL3-PZM58
- L1-PL3-PZM59
- L1-PL4-PZM66
- L1-PL4-PZM67
- L1-PL4-PZM68
- L1-PL4-PZM69
- L1-INJ-PM70:VAL_CAL
- L1-OPA3-5_2-PM98:VAL_CAL
- L1-INJ-PM70:VAL_CAL_diff

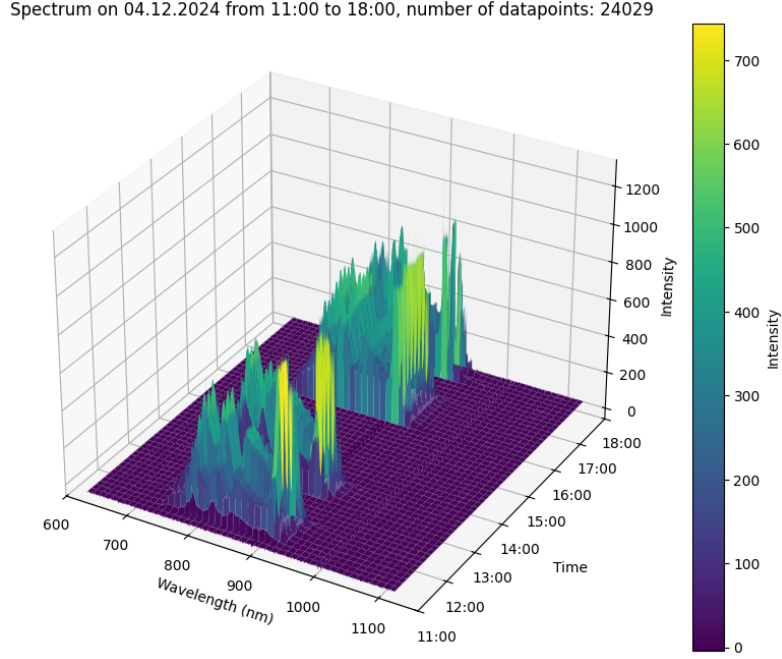
8.4 Data Splitting

The laser intensity measurements are recorded at very short intervals. As a result, adjacent spectra are very similar. If the dataset were randomly split into training, validation, and test sets, highly similar spectra would end up in all sets. This could lead to overfitting, as the model would effectively be tested on data that are nearly identical to what was already seen during training.

Intensities are measured in time blocks, with gaps between them because of the LWFA process, during which intensity data are not recorded. Instead of randomly splitting the data, whole blocks of intensity measurements were used for each set to avoid overfitting. The blocks used for test and validation sets were chosen from two randomly selected days: December 4th (Figure 8.2) and December 9th (Figure 8.3).

More specifically for the test set, 17,811 data points from the following time periods were selected:

- December 4th, from 11:15 until 13:25.
- December 9th, from 13:00 until 17:46.



■ **Figure 8.2** 3D spectrogram of spectra on December 4th.

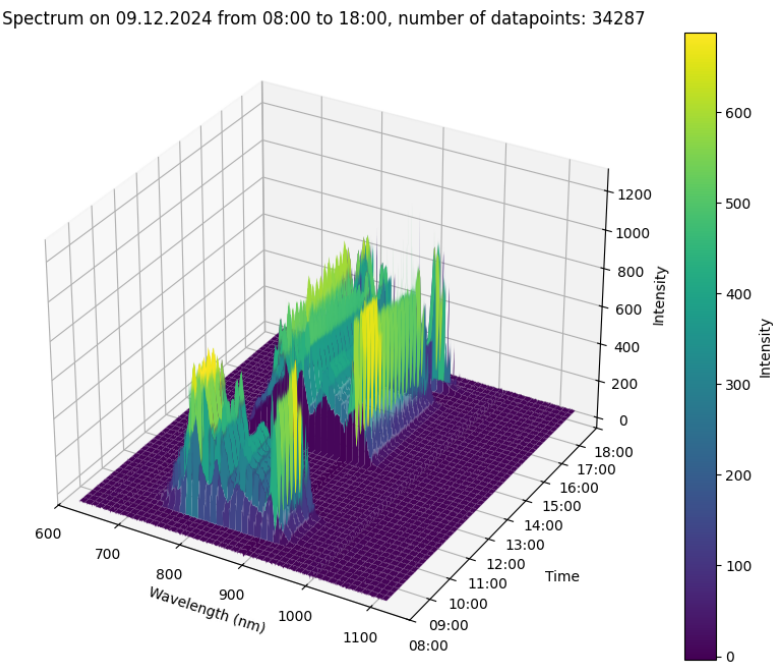
For the validation set, data from the following time periods were used:

- December 4th, from 15:38 until 17:59,
- December 9th, from 8:49 until 10:12.

In total, the validation set contained 9,757 data points. However, since the validation set would be too large, which would slow down the evaluation process, a random subset of 2,500 data points was selected from the original validation set. The train set contained the remaining 13 days, which consisted of 176,961 data points.

8.5 Data Normalization

To maintain consistency, Min-Max normalization was applied to scale the intensity values to the range -1 to 1. This range is commonly used in diffusion



■ **Figure 8.3** 3D spectrogram of spectra on December 9th.

models, where the denoising process assumes that the input data are within a bounded interval.

Only the training set was analyzed to find outlier values and upper and lower bounds for normalization. The test and validation set remained unchanged and the values were kept in the original ranges. The validation and test sets were not normalized. All predictions are later denormalized back to the original intensity scale during evaluation. This approach preserves the real-world distribution in the evaluation datasets while still benefiting from normalized training during model fitting.

Table 8.1 presents the statistical summary of the intensity values of the spectra. The first column provides statistics for the entire intensity vector, while the second and third columns focus on the minimum and maximum intensity values within each intensity vector. The minimum recorded intensity is -33.7, while the maximum reaches 1629.49.

■ **Table 8.1** Statistical summary of spectra intensity values in training set

Metric	Whole Intensity Vector	Min Value	Max Value
Mean	108.31	-20.48	611.26
Median	7.17	-20.7	730.80
Min	-33.7	-33.7	20.69
Max	1629.49	-11.7	1629.49

The intensity measurement is not entirely accurate. This can be seen especially at the edges of the spectra, where the actual intensity values shall be zero, but the measurements fluctuate between approximately -30 and 30. To prevent these inaccuracies, the lower bound was set to 30, which means that all intensity values below this threshold were set to 30. For the test and validation sets, only negative values were clipped to 0. Since the test and validation set were not normalized, a higher lower bound than 0 would shift the starting point of the spectrum values at the edges to 30, even though the true values there should be close to zero.

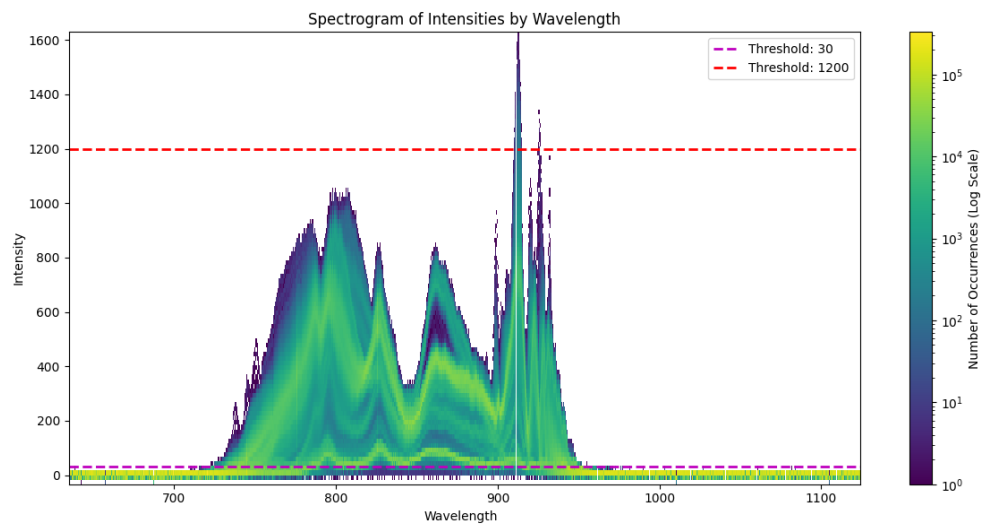
Outliers were analyzed by identifying the 99th percentile of maximum values within each intensity vector, which was calculated as 1,198.2. Based on this, the upper bound was set at 1,200, and all values above were set to 1,200. The test and validation sets were kept unchanged.

The Min-Max normalization was then applied using these bounds, transforming the values into the range -1 to 1:

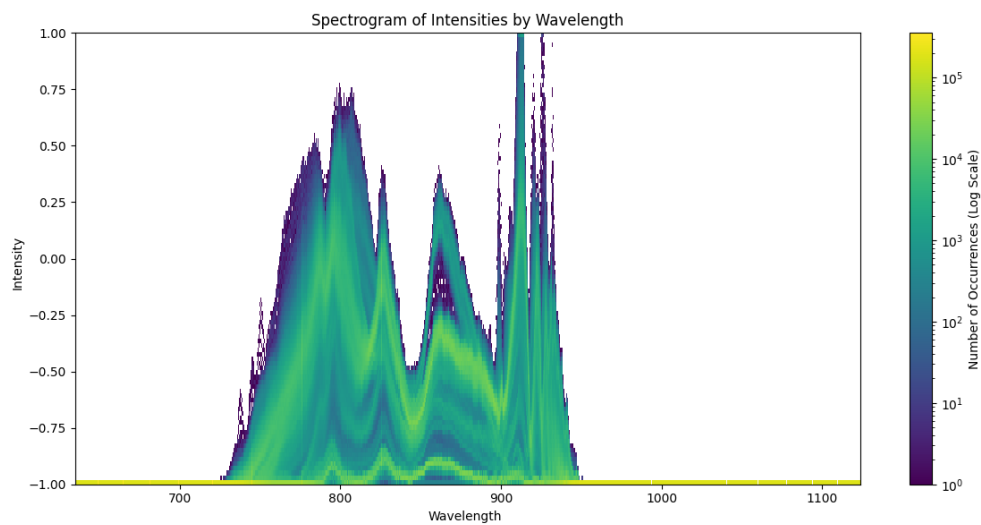
$$X_{norm} = -1 + (X - 30) \times \frac{2}{1200 - 30}$$

Records whose spectrum consisted entirely of -1 were removed (records that originally consisted of only values less than or equal to 30). A total of 7 records were removed from the training set. After this filtering, the training set contains 176,954 data points, the test set contains 17,811 data points, and the validation set contains 2,500 data points.

Figure 8.4 shows the distribution of intensity values at different wavelengths together with the upper and lower bounds highlighted before normalization. The distribution of normalized intensity values is shown in Figure 8.5.



■ **Figure 8.4** Spectrogram of intensities by wavelength with highlighted lower and upper bounds.



■ **Figure 8.5** Spectrogram of intensities by wavelength after normalization.

Architecture

This chapter describes two main architectural components used in this work: the architecture of the generative diffusion models used and the architecture of the Bayesian optimization framework. Together, these components form a system capable of generating laser spectra based on experimental conditions and finding ideal input parameters to achieve desired spectral outputs.

The first part of the chapter focuses on the architecture of the diffusion model, which serves as the core generative mechanism. It is responsible for simulating realistic laser spectra based on given system parameter settings. This model is trained using a noise-based denoising process and incorporates a U-Net backbone architecture adapted for one-dimensional spectral data.

The second part presents the Bayesian optimization framework, which is used to search for optimal system settings that generate spectra that match a given target. This framework integrates the trained diffusion model into its optimization loop and guides the search process by minimizing a custom loss function based on the properties of the spectra.

9.1 Diffusion Model

The backbone of the architecture is a neural network, which acts as a denoiser that estimates the noise added to the data during the forward diffusion process at each time step t . This neural network is implemented using a U-Net architecture [84]. The U-Net architecture consists of three main components: an encoder (downsampling part), a bottleneck, and a decoder (upsampling part), with skip connections linking the corresponding encoder and decoder layers.

The network takes the noise data x_t , the diffusion step t as input, and optionally a conditional vector y in case of conditioned generation. The diffusion step t is encoded by positional encoding, which transforms the scalar t into a high-dimensional representation using sinusoidal embedding. The con-

ditional vector, which contains the system parameters settings, is embedded through a linear transformation and activation function to match the time embedding size. Afterward, the transformed t and y are combined by summation. The combined embedding is then injected into the network's downsampling and upsampling blocks. This ensures that the model stays informed about both the diffusion step and the conditional vector during the feature extraction and reconstruction process.

The encoder consists of three downsampling blocks, each reducing the input length by a factor of 2 while increasing the number of feature channels. Each downsampling block contains the following:

- A one-dimensional max pooling layer to downsample the input.
- Two double convolutional blocks, each containing:
 - one-dimensional convolutional layers with a kernel size of 3,
 - group normalizations,
 - GELU activation function [85].
- Dropout layer to prevent overfitting.
- Self-attention [86] applied after the second downsampling block to capture long-range dependencies in the data.

The bottleneck acts as a latent feature processing block at the lowest resolution. It consists of three double convolutional layers.

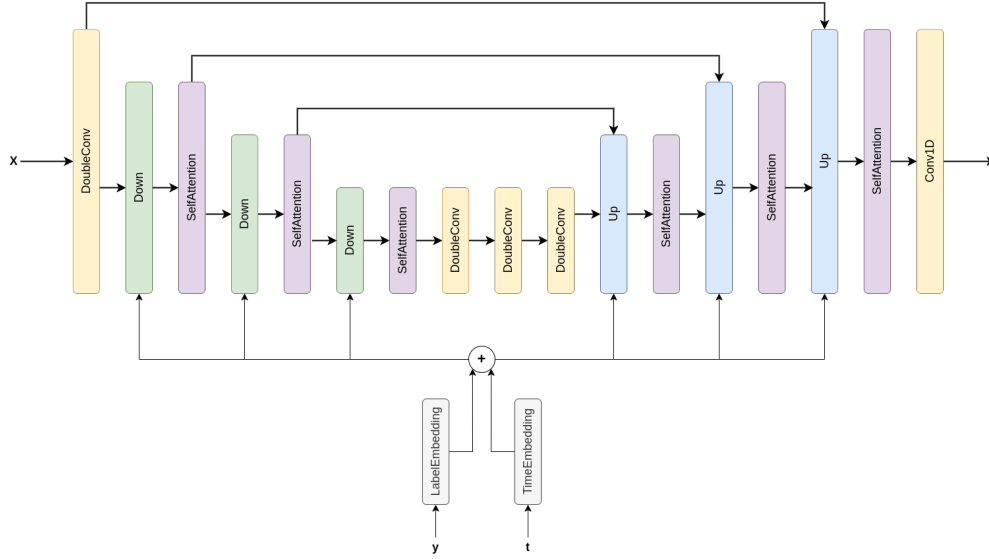
The decoder reconstructs the input to its original resolution. Downsample blocks merge the input data from the previous layer with information from the corresponding encoder layers through skip connections. It consists of three upsampling blocks:

- A one-dimensional upsampling layer using linear interpolation to increase the sequence length.
- Concatenation with the skip connections' data.
- Two double convolutional layers.
- Self-attention applied at the first two upsampling blocks.

Figure 9.1 visualizes the complete U-Net architecture, the individual block architectures are shown in Figure 9.2.

9.1.1 Training Loop

During training, the model predicts noise for different diffusion steps. Each iteration selects random timesteps, and the corresponding noisy input is generated. This ensures that the model generalizes well across all noise levels.



■ **Figure 9.1** U-Net architecture of diffusion model.

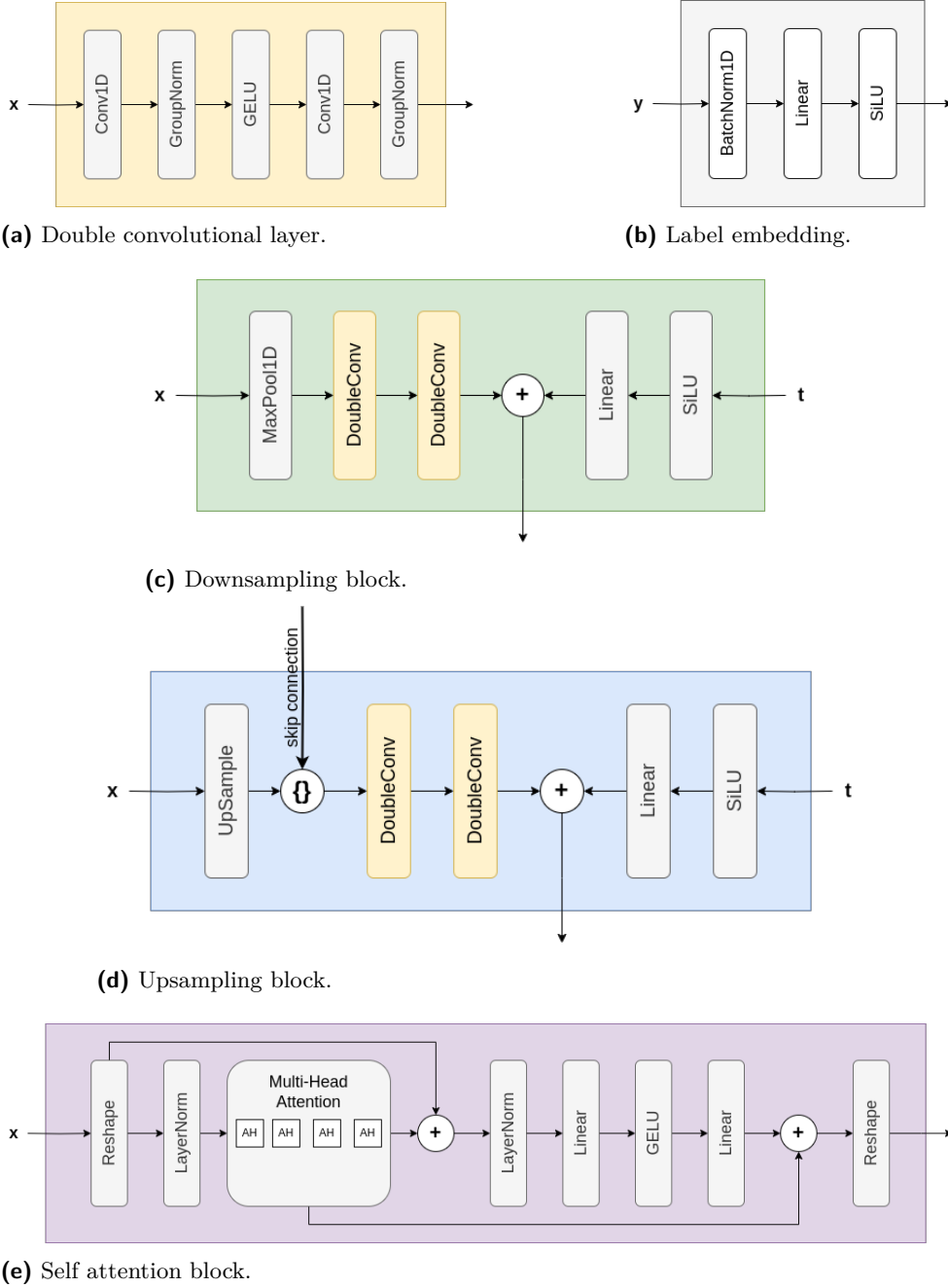
In DDIM, a discrete noise schedule is used: noise levels (betas) are pre-defined and fixed throughout the training. The model then learns to predict the noise at specific timesteps sampled during training. EDM, instead of using a fixed noise schedule, continuously samples noise levels from an exponential distribution controlled by parameters P_{mean} (average noise level) and P_{std} (spread of random noise levels).

The model was trained on 1024-length spectral intensity vectors, optionally conditioned on the system parameters. The dataset was loaded in mini-batches for efficient training. The batch size was set to 16 to balance scalability and hardware constraints.

Conditioning vectors were randomly dropped with a probability controlled by a hyperparameter guidance scale to enable classifier-free guidance.

The loss function used in DDIM is the standard MSE loss between the predicted noise and the true noise added to the input. EDM introduces a weighted loss function in which the weight depends on the noise level, which ensures stable training at different noise levels, as shown in Algorithm 2.1.

Optimization was performed using the AdamW optimizer [87], while a cosine annealing learning rate scheduler [88] gradually reduced the learning rate over time. To improve stability and performance during inference, an Exponential Moving Average (EMA) [89] model copy was maintained.



■ **Figure 9.2** Architecture of individual blocks of U-Net.

9.1.2 Sampling Algorithm

In the sampling process, the model generates new spectral intensity vectors. The sampling strategy depends on the type of diffusion model used.

DDIM follows a deterministic sampling approach, where the model denoises an initial Gaussian noise tensor in a fixed number of steps. The process follows a predefined noise schedule, and at each step, the model predicts the original signal while progressively removing noise. Unlike DDPM, which requires hundreds of diffusion steps to generate high-quality samples, DDIM accelerates the process by skipping intermediate steps, making the process significantly faster while maintaining sample quality (see Section 2.3).

The `section_counts` parameter determines how many timesteps are used during the denoising process. Instead of running through all diffusion steps, a reduced and evenly spaced subset is used. Fewer steps lead to faster sampling but may reduce output quality, while more steps can improve reconstruction at the cost of longer inference time. The DDIM implementation is adapted for one-dimensional diffusion from the official diffusion autoencoder repository [90].

EDM introduces a more flexible sampling approach with continuous noise levels instead of fixed timesteps, noise levels are sampled from an exponential distribution. A key feature of EDM is its two-step denoising process, which consists of an Euler step for an initial noise reduction, followed by a second-order correction to refine the estimate and improve stability (see Section 2.5 for further details). The EDM model is implemented based on the official code released by NVIDIA and adapted for `ssh -L 8000:localhost:8888 dgx one-dimensional diffusion` [91].

Both methods use classifier-free guidance by generating unconditional and conditional samples. The final sample is created as a linear interpolation between the results of these two approaches.

9.2 Bayesian Optimization

Bayesian optimization framework aims to find the optimal combination of laser system parameters that results in a generated spectrum closely matching a target spectrum in key characteristics. Specifically, the optimization seeks to minimize the differences in the mean wavelength, the wavelength corresponding to the maximum intensity, and the maximum intensity value.

The optimization process relies on a pre-trained conditional EDM model. The diffusion model receives a vector of system parameter settings and generates a corresponding spectral intensity vector.

For each optimization step, the current set of optimized parameters is combined with fixed parameters to construct a complete conditional vector. This vector is passed to the diffusion model to generate a synthetic spectrum. The generated spectrum is compared with the target using a custom loss function.

9.2.1 Objective Function

The objective function measures the difference between the generated and target spectra in three aspects:

- The mean wavelength, calculated as the intensity-weighted average wavelength:

$$\lambda_{\text{mean}} = \frac{\sum_{\lambda} I(\lambda) \cdot \lambda}{\sum_{\lambda} I(\lambda)},$$

where $I(\lambda)$ denotes the intensity at wavelength λ .

- The wavelength at which the spectrum reaches maximum intensity.
- The value of the maximum intensity itself.

The loss is computed as the sum of the absolute differences between the generated and the target values for these three metrics. This approach ensures that the optimization process focuses not only on aligning the overall spectral energy distribution but also on reproducing the spectral peak position and strength, which are critical for successful LWFA operation.

The loss function is formally defined as:

$$\begin{aligned} \text{Loss} = & |\lambda_{\text{mean, gen}} - \lambda_{\text{mean, target}}| \\ & + |\lambda_{\text{max, gen}} - \lambda_{\text{max, target}}| \\ & + |I_{\text{max, gen}} - I_{\text{max, target}}|, \end{aligned}$$

where λ_{mean} is the mean wavelength, λ_{max} is the wavelength at maximum intensity, and I_{max} is the maximum intensity value.

9.2.2 Parameter Search Space

In total, 13 laser system parameters were used as a conditional vector in the diffusion model. However, optimizing all of them would not only significantly increase the computational cost but also reduce the efficiency of the optimization process due to the high dimensionality of the search space.

To address this issue, the framework allows the user to manually select which subset of parameters should be optimized and which should remain fixed. The fixed parameters are set by the user and kept constant during optimization. This selective optimization is especially useful in experimental contexts, where scientists often have domain knowledge or prior experience that informs the correct setting for some parameters, while others remain uncertain or require fine-tuning.

Each parameter, from the subset of optimized parameters, is treated as a continuous variable using the `Real` class from the `scikit-optimize` library. The search space is constructed dynamically based on the chosen parameters and their respective value ranges. The ranges were selected as the minimum and maximum values of a specific parameter in a train data set, as Table 9.1 shows.

■ **Table 9.1** Parameter bounds used for Bayesian optimization.

Parameter	Lower Bound	Upper Bound
L1-PL3-PZM52_Setpoint	-3.0000	3.0000
L1-PL3-PZM53_Setpoint	-3.0000	3.0000
L1-PL3-PZM56_Setpoint	-1.5910	0.6349
L1-PL3-PZM57_Setpoint	-3.5833	-0.0177
L1-PL3-PZM58_Setpoint	-3.0000	1.0000
L1-PL3-PZM59_Setpoint	-3.0000	1.0000
L1-PL4-PZM66_Setpoint	-3.7500	3.1082
L1-PL4-PZM67_Setpoint	-3.7500	0.2802
L1-PL4-PZM68_Setpoint	-3.7500	3.7500
L1-PL4-PZM69_Setpoint	-1.9848	3.7500
L1-INJ-PM70:VAL_CAL	-0.3500	41.8900
L1-OPA3-5_2-PM98:VAL_CAL	5.6200	66.2500
L1-INJ-PM70:VAL_CAL_diff	-33.7200	1.0800

9.2.3 Optimization Process

Bayesian optimization is performed using the `gp_minimize` [92] function from the library `scikit-optimize`, which uses a Gaussian process as a surrogate model to approximate the objective function. In each iteration, the acquisition function suggests a new vector of candidate parameters.

As an acquisition function, the `gp_hedge` [92] strategy was selected. This method dynamically chooses between three commonly used acquisition functions: Lower Confidence Bound [47], Expected Improvement [48], and Probability of Improvement [49] (see Section 3.2). Each of these proposes its own candidate point in every iteration. Among them, the most promising candidate is selected and evaluated.

After each evaluation, the surrogate model is updated with the new observation. The algorithm also updates the internal weights (also called gains) for each acquisition function based on the performance of its suggestion. Initially, all weights are zero, allowing equal probability for selection. As optimization progresses, the strategy adapts by favoring acquisition functions that consistently lead to improvements.

Experiments and Results

This chapter presents the experimental evaluation of the proposed diffusion-based generative models for predicting laser spectrum intensities. The aim is to compare different model configurations and assess their ability to generate spectra that closely match real measurements.

Two sampling strategies were tested: the DDIM and the EDM, both conditioned on system parameters. A range of hyperparameters was explored, including the number of diffusion noise steps, the CFG scale, and, for DDIM, the number of denoising sections.

To evaluate the generated spectra, three complementary metrics were used, which capture different aspects of similarity between predicted and true spectra.

MSE A standard metric that measures the average of squared differences between predicted and actual intensity values. It gives a point-by-point comparison and is particularly sensitive to large deviations.

EM Distance This metric measures how different two distributions are by calculating the minimal "effort" required to transform one distribution into the other. It is especially helpful in comparing the overall shape of spectra, even if they are slightly shifted. More about this metric is explained in the theory section on EM Distance (see Section 6.2).

DTW A metric which compares two sequences by allowing for shifts and stretching in time or position. This is useful for spectral data, where two spectra might look similar but not align perfectly. The details of DTW can be found in the dedicated section (see Section 6.1).

The models were trained using the training set for 40 epochs. The training was performed using an NVIDIA A100 GPU. The generated spectra were evaluated using the selected metrics. Based on these evaluations, the most

promising models were selected for further testing of sampling hyperparameters. Only the best-performing model and sampling configurations were ultimately evaluated on the test set.

10.1 DDIM

To evaluate the performance of the DDIM for spectrum generation, a series of experiments were conducted, exploring how different configurations affect the quality of the output. These experiments were divided between modifications applied during the training phase and those applied during the sampling phase.

In order to determine which training and sampling settings lead to the most accurate and physically realistic spectral predictions during training, several models were trained independently using different dropout rates. In the sampling phase, each trained model was evaluated on validation data with different combinations of CFG scales and section counts.

10.1.1 Dropout Rate Analysis

To explore the effect of dropout rate choices during training, three DDIM models were trained using dropout rates of 0 %, 10 %, and 20 %. These models were evaluated in the validation set.

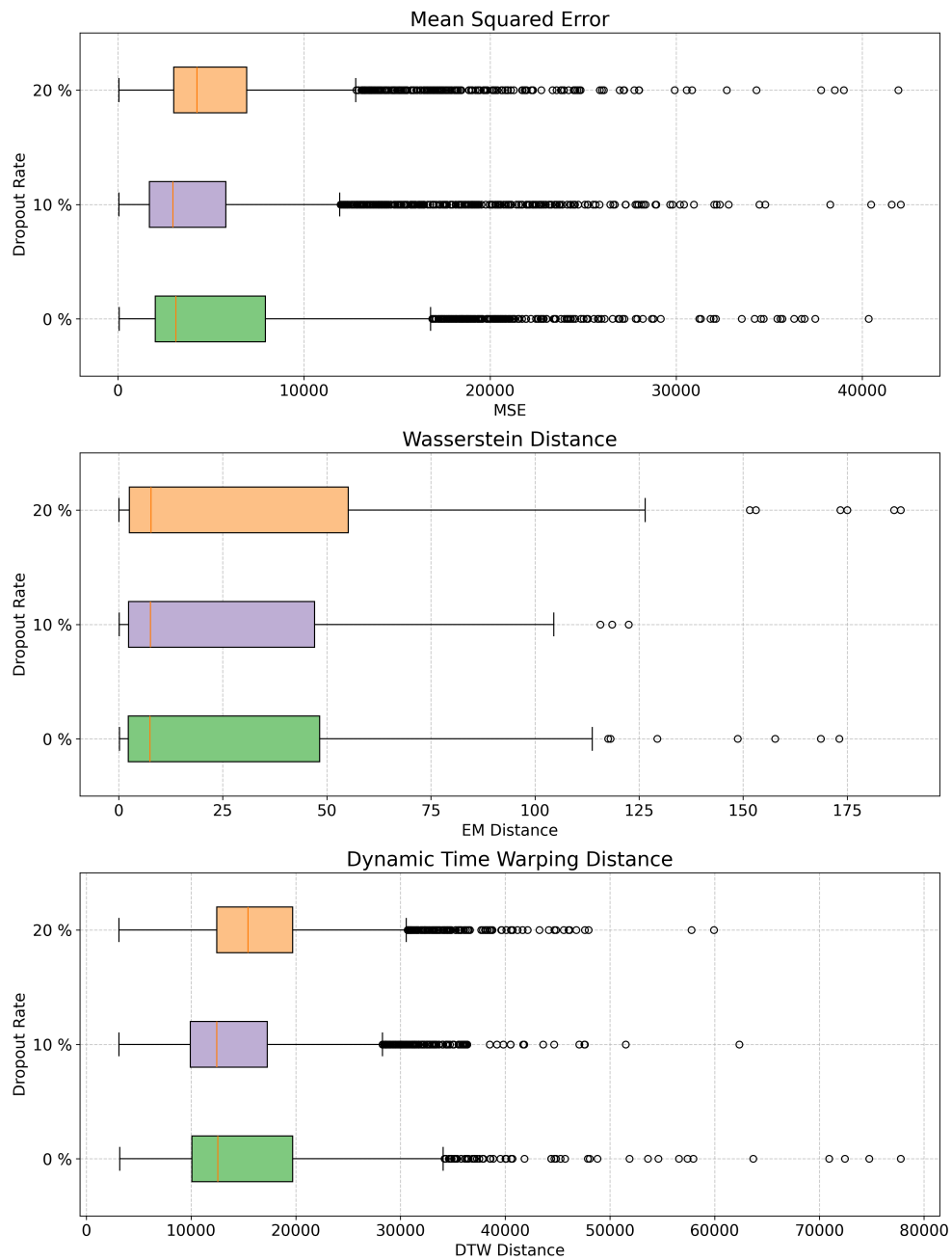
As shown in Figure 10.1 and summarized in Table 10.1, the model trained with a dropout rate of 10 % consistently demonstrated the best overall performance in all three evaluation metrics.

The model trained without dropout (0 %) showed the highest errors in most metrics. Although the 20 % dropout model showed some improvement over the 0 % model in terms of MSE, it underperformed both the EM and the DTW metrics.

■ **Table 10.1** Comparison of evaluation metrics (MSE, EM Distance, and DTW) at different dropout rates.

Parameter	MSE		EM		DTW	
Dropout Rate	Mean	Median	Mean	Median	Mean	Median
0 %	5948.08	3120.31	26.76	7.53	15516.51	12557.61
10 %	5106.82	2964.04	25.35	7.68	14719.36	12448.86
20 %	5844.75	4254.37	31.69	7.81	16996.50	15435.27

Overall, a dropout rate of 10 % offered the best trade-off between regularization and performance. It was therefore selected for all subsequent DDIM experiments.



■ **Figure 10.1** Comparison of different dropout choices for DDIM.

10.1.2 CFG Scale Analysis

Several configurations were tested during the sampling phase to investigate the influence of the classifier-free guidance (CFG) scale on the performance of the DDIM model. This included a baseline case with no CFG, along with CFG scales from 1 to 4. In this context, the no CFG condition refers to sampling using only the conditional branch of the model, meaning that the output is generated entirely based on the provided system parameters.

As shown in Table 10.2, using the conditional model alone (no CFG) produced strong baseline performance on all three metrics. The model achieved a mean MSE of 4603.23 and a median of 3152.92, a EM distance mean of 24.89, and DTW scores of 14,808.39 (mean) and 13,023.87 (median). These values suggest that the model, even without explicit CFG, is able to generate well-aligned and realistic spectra.

■ **Table 10.2** Comparison of evaluation metrics (MSE, EM distance, and DTW distance) at different classifier-free guidance scales.

Parameter	MSE		EM		DTW	
CFG Scale	Mean	Median	Mean	Median	Mean	Median
no cfg	4603.23	3152.92	24.89	7.94	14808.39	13023.87
1	4474.06	3107.86	24.87	7.93	14547.54	13039.41
2	4769.09	3002.10	24.92	7.80	14511.25	12694.77
3	5106.82	2964.04	25.35	7.68	14719.36	12448.86
4	4752.67	3349.28	25.86	7.93	15089.30	13278.90

Among the tested configurations, the CFG scale 2 delivered the best overall trade-off. It achieved the lowest median MSE (3002.10) and strong performance in DTW (mean: 14,511.25; median: 12,694.77). Although its mean MSE was higher than scale 1, it showed better median scores in both MSE and DTW distance. CFG scale 3 had the lowest median DTW (12,448.86) and a slightly better median MSE (2964.04) but a higher overall MSE and EM distance.

However, CFG scale 4 showed signs of over-guidance, resulting in a higher mean MSE (4752.67) and the worst DTW score of all configurations. Its increased median MSE (3349.28) further confirmed the reduced performance stability.

Taking into account all three metrics, CFG scale 2 was selected for further use due to its reliable and balanced results in terms of both point-wise accuracy and distributional similarity.

10.1.3 Section Counts Analysis

To explore the effect of sampling density on the performance of the DDIM model, three values of `section_counts` were tested during inference: 20, 40,

and 60. This parameter determines how many denoising steps are taken during sampling, where a lower value generally results in faster generation.

Interestingly, the results in Table 10.3 show that the best overall performance in all metrics was achieved using only 20 steps. These results suggest that a lower section count, such as 20, not only reduces generation time but also provides slightly better results.

■ **Table 10.3** Comparison of evaluation metrics (MSE, EM Distance, and DTW) at different `section_counts` for DDIM.

Section Counts	MSE		EM		DTW	
Steps	Mean	Median	Mean	Median	Mean	Median
20	4606.29	2864.54	24.41	7.64	14291.10	12371.86
40	4769.09	3002.10	24.92	7.80	14511.25	12694.77
60	4883.90	3033.53	25.96	7.86	14780.70	12812.46

10.1.4 Best DDIM Model

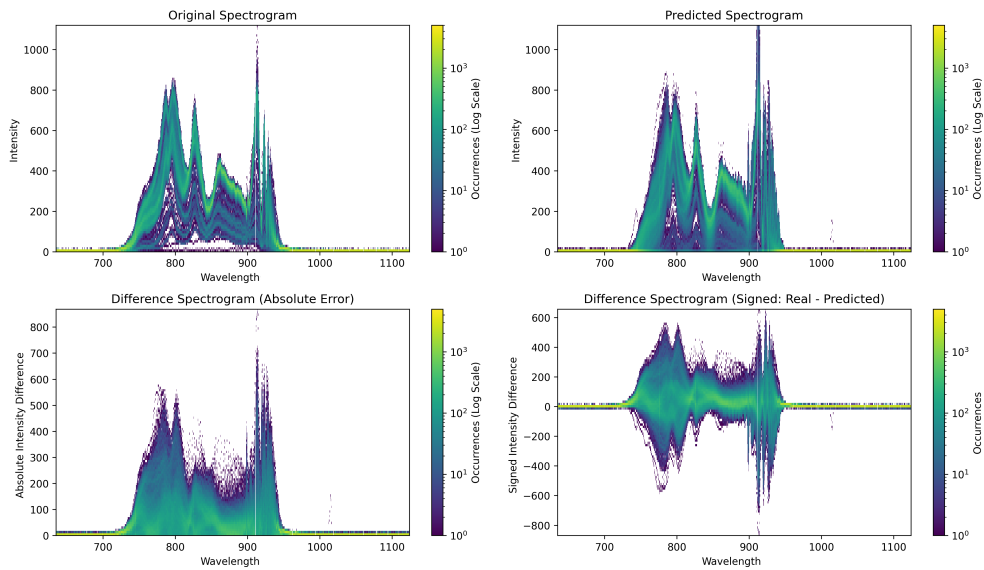
Based on the experiments conducted, the best-performing DDIM model was trained with a dropout rate of 10 % and evaluated using a CFG scale of 1 and 20 denoising sections during sampling. This configuration achieved the most consistent balance across all evaluation metrics.

Although the DDIM model demonstrates solid performance in capturing the general shape and distribution of the predicted spectral intensities, it still reveals a few limitations. The results shown in Figure 10.2 suggest that the model tends to overestimate high-intensity peaks, which are particularly visible between 900 and 950 nm, where the highest intensity peaks are measured. Additionally, at the edges of the spectrum (below 700 nm and above 1000 nm), the model occasionally generates anomalous intensity values in regions where the true spectra values are close to zero. The model also shows difficulty in accurately predicting spectra with very low overall intensity values, often underestimating their values and failing to capture their structure.

10.2 EDM

A range of experiments was conducted to evaluate the EDM model’s ability to generate realistic spectral outputs. These included tests with different dropout rates during training, as well as variations in sampling parameters such as the CFG scale and the number of noise steps.

In addition to standard classifier-free guidance, an adaptive scaling approach was also explored. Finally, the performance of EDM was compared directly with DDIM to highlight their differences. Each configuration was tested on validation data using several metrics to assess how well the generated spectra matched the real measurements.



■ **Figure 10.2** Results of the best DDIM model.

10.2.1 Dropout Rate Analysis

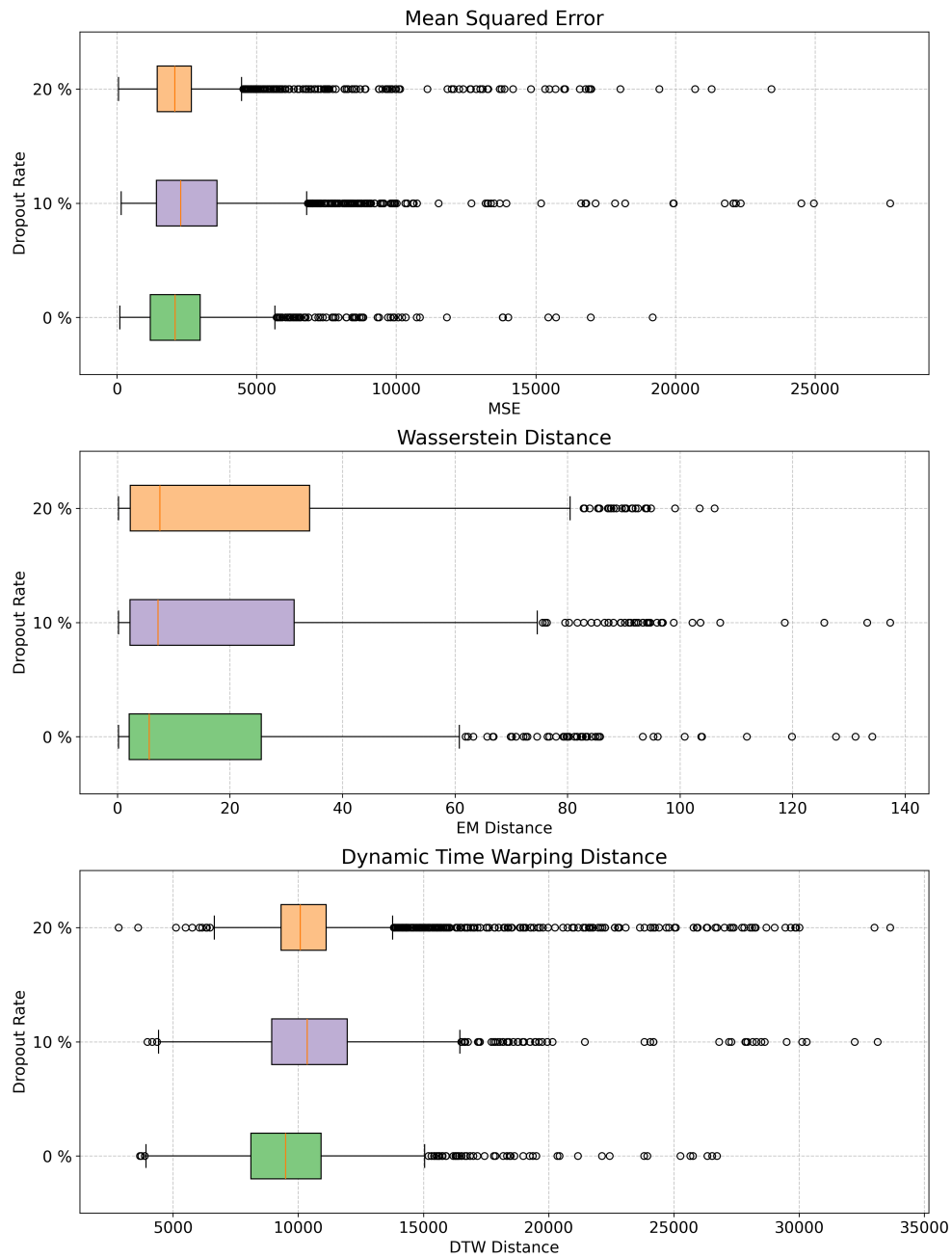
To evaluate the impact of dropout regularization on the performance of the EDM model, three separate models were trained using dropout rates of 0%, 10%, and 20%.

As shown in Table 10.4 and Figure 10.3, the model trained without any dropout (0%) achieved the best overall performance across all three evaluation metrics. It recorded the lowest mean and median MSE (2294.94 and 2083.13), the best EM distance (mean: 16.64, median: 5.65), and the lowest DTW distance (mean: 9664.63, median: 9495.38). These results suggest that in this case, adding dropout did not improve generalization, and the model trained without it was better at capturing both accurate intensities and the correct shape of the spectra.

■ **Table 10.4** Comparison of evaluation metrics (MSE, EM distance, and DTW distance) at different dropout rates.

Parameter	MSE		EM		DTW	
Dropout Rate	Mean	Median	Mean	Median	Mean	Median
0 %	2294.94	2083.13	16.64	5.65	9664.63	9495.38
10 %	2865.54	2282.02	18.91	7.24	10584.13	10354.94
20 %	2421.10	2076.42	21.11	7.53	10746.77	10080.15

In contrast, the 10% dropout model performed worst in all metrics. The 20% dropout model showed slightly better MSE and DTW results than the



■ **Figure 10.3** Comparison of dropout rates for EDM on validation data.

10% version, but its EM distance was the worst of all, pointing to a poorer alignment between predicted and true spectral distributions.

Based on these findings, the model trained with 0% dropout was selected for further use, as it provided the best and most consistent results.

10.2.2 CFG Scale Analysis

To evaluate the effect of the classifier-free guidance (CFG) scale on the performance of the EDM model, multiple values were tested during the sampling phase: no CFG, 0, and values from 1 to 10. The aim was to find the CFG scale that achieves the best trade-off between generating realistic spectral shapes and accurately reflecting the input system parameters.

In this setup, no CFG refers to using only the conditional branch of the model, which means that the output is generated solely based on the provided system parameters. In contrast, setting the CFG scale to 0 corresponds to using only the unconditional model, completely ignoring the conditioning information.

The results are presented in Table 10.5. Setting the CFG scale to 0 led to clearly worse results in all metrics. The MSE jumped to 16,313.14 (over eight times higher than the best results), and the DTW distance reached 35,122.80. This poor performance is not surprising, as the model was not guided by the actual experimental settings. Instead, it serves as a baseline to show how important conditioning is for producing accurate and meaningful spectra.

■ **Table 10.5** Comparison of evaluation metrics (MSE, EM distance, and DTW) at different classifier-free guidance scales.

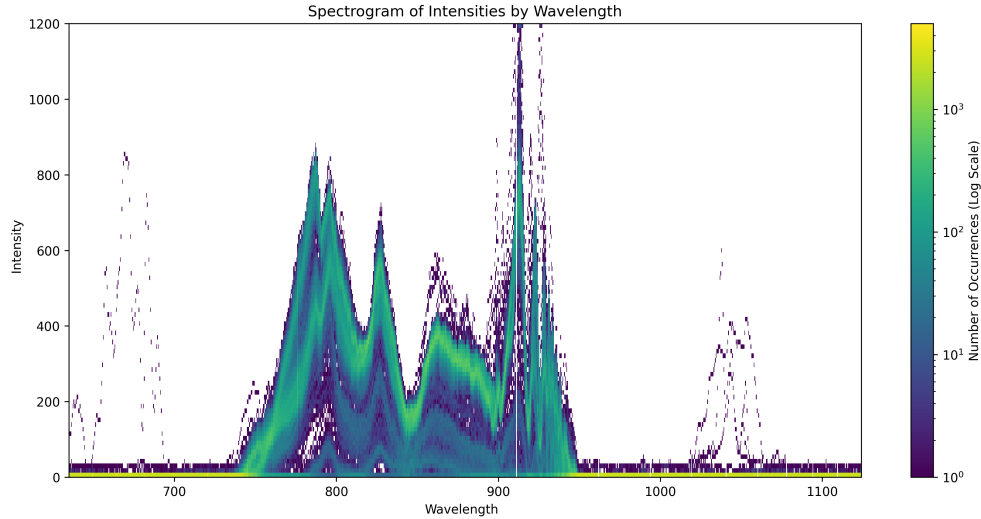
Parameter	MSE		EM		DTW	
CFG Scale	Mean	Median	Mean	Median	Mean	Median
no cfg	2424.19	2075.40	16.93	5.62	10052.26	9754.42
0	16313.14	6464.28	39.64	7.43	35122.80	16780.14
1	2380.79	2002.46	16.77	5.69	9958.80	9650.15
2	2265.93	2051.32	16.67	5.61	9658.38	9581.03
3	2294.94	2083.13	16.64	5.65	9664.63	9495.38
4	2388.41	2151.97	16.73	5.64	9806.67	9592.63
5	2424.63	2244.90	16.70	6.06	9863.19	9650.45
6	2514.22	2296.24	16.77	6.05	10020.04	9828.35
7	2609.95	2363.03	16.87	6.32	10174.11	9930.95
8	2734.82	2445.22	17.01	6.40	10383.89	10102.64
9	2816.38	2548.07	17.01	6.17	10543.76	10286.11
10	2980.24	2629.15	17.41	6.10	10852.77	10451.97

However, using only the conditional branch (no CFG) showed a strong performance with a mean MSE of 2424.19, the EM distance of 16.93 and the

DTW distance of 10,052.26. This shows that the model performs well even without using any unconditional predictions.

Among the other values, the CFG scales from 1 to 4 performed best. Figure 10.4 presents the comparison of the results. These settings offered a solid trade-off in all three metrics. In particular, CFG scale 2 produced the lowest DTW distance (mean: 9658.38) and one of the best MSE scores (mean: 2265.93, median: 2051.32) while maintaining a very low EM distance (mean: 16.67).

With higher CFG scales, performance started to degrade gradually. Higher scales like 9 and 10 resulted in an increase in the MSE and DTW values, indicating potential overguidance. When higher CFG scales (7 and above) were used, artifacts began to appear at the edges of the generated spectra. As shown in Figure 10.5, this region corresponds to areas where real spectra consistently exhibit zero or near zero intensity values.

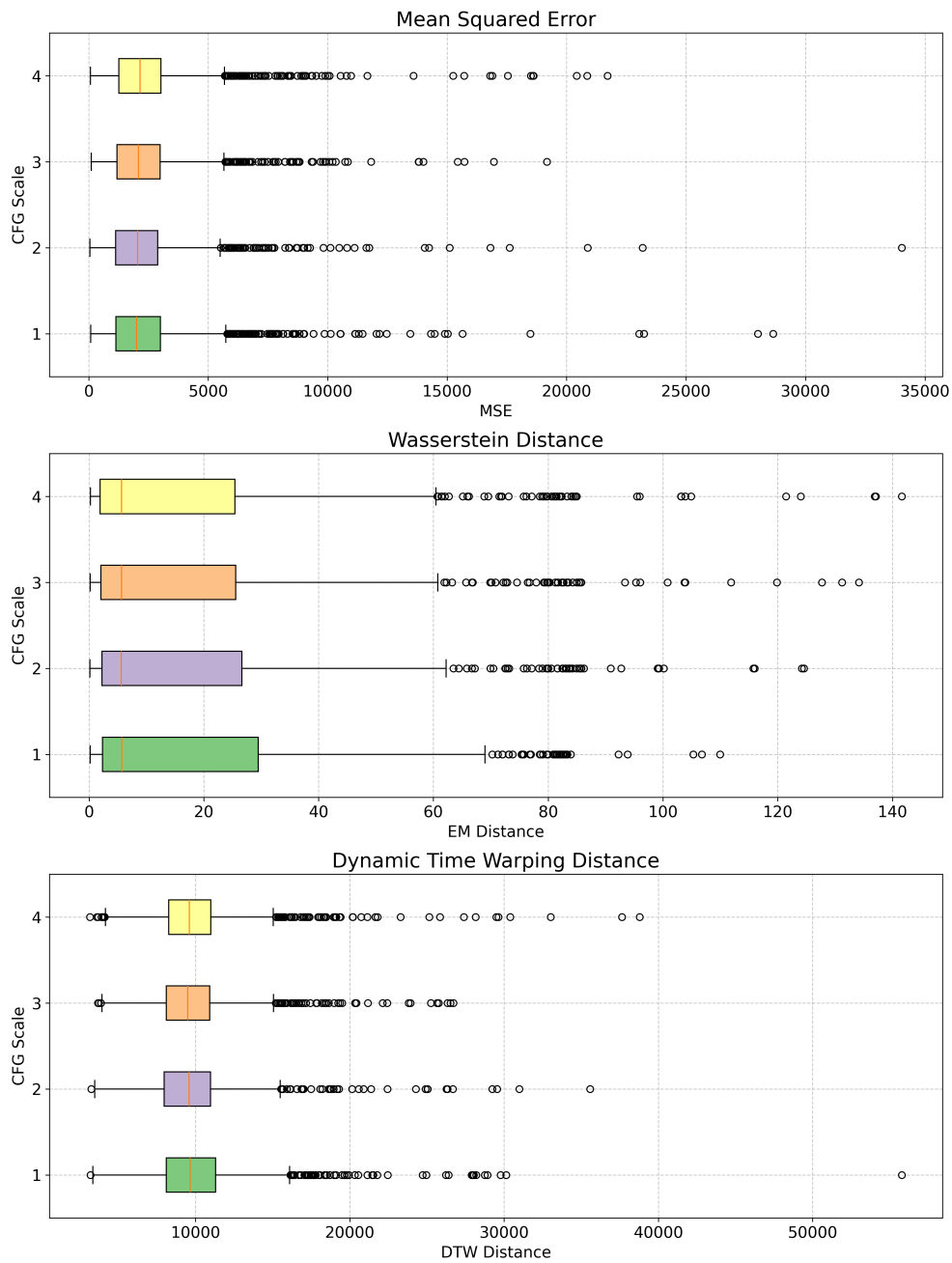


■ **Figure 10.5** Spectrogram of intensities by Wavelength for CFG scale 10.

Based on the comparison, a CFG scale of 2 was selected for further experiments.

10.2.3 Classifier-free Guidance with Adaptive Scaling

Classifier-Free Guidance [41] is a common diffusion technique that balances conditioning accuracy with output quality and diversity. Traditionally, CFG applies a fixed guidance strength across all diffusion steps. The β -CFG method [93] uses a beta distribution to adaptively scale the guidance during sampling using a beta distribution function.



■ **Figure 10.4** Spectrogram of intensities by Wavelength for best performing CFG scales.

■ **Table 10.6** Comparison of evaluation metrics with adaptive and static CFG scale.

Beta Params.	CFG Scale	MSE		EM		DTW	
		Mean	Mdn	Mean	Mdn	Mean	Mdn
a=2, b=4	4	2647.64	2276.71	16.99	5.96	9974.99	9648.34
	3	2569.39	2212.29	16.70	5.84	9915.26	9534.35
	2	2489.54	2113.84	16.53	5.81	9913.41	9566.90
a=2, b=2	4	2479.11	2237.48	17.27	6.06	9916.09	9666.79
	3	2384.90	2147.51	17.07	5.71	9762.75	9540.38
	2	2326.24	2126.86	16.85	5.53	9687.04	9545.40
None	2	2265.93	2051.32	16.67	5.61	9658.38	9581.03

The main idea is to adaptively reduce the influence of guidance at the beginning and end of the diffusion process while strengthening it in the middle steps, where the structure is typically formed.

The shape of the beta distribution is governed by the parameters a and b . When both are equal to 2.0, the distribution forms a symmetric curve centered around the midpoint. Increasing a skews the distribution toward the beginning of the diffusion process, amplifying the guidance in early steps, while increasing b shifts the emphasis toward later steps, concentrating the guidance near the end.

Table 10.6 shows that among the configurations tested, the combination of CFG scale 2 with $a = 2$, $b = 2$ achieved the best results with adaptive scaling. It reached a mean MSE of 2326.24, an EM distance median of 5.53, and a DTW median of 9545.40. These scores are close to those of the best-performing static model.

However, the fixed CFG scale of 2, without any adaptive schedule, still achieved slightly lower mean MSE (2265.93) and competitive values in both EM and DTW metrics. Although adaptive scaling can be beneficial and offers flexibility in shaping the guidance schedule, its advantage was not significant enough to justify replacing the simpler fixed CFG configuration in this case. Therefore, the static CFG scale of 2 is retained for the final model.

10.2.4 Noise Steps Analysis

To assess the impact of noise steps on model performance, experiments were conducted using 10, 20, and 30 noise steps. As shown in Table 10.7, increasing the number of steps improved the model’s accuracy in all metrics.

Using 30 noise steps delivered the most accurate predictions, achieving the lowest errors in used metrics. However, using 20 steps still produced strong results, significantly better than using only 10, while decreasing the inference time by about one-third compared to 30 steps. Nevertheless, the 30 noise steps configuration remains the most effective and was used in subsequent experiments.

■ **Table 10.7** Performance and generation time per spectrum at different noise step counts.

	MSE		EM		DTW		Time
Steps	Mean	Mdn	Mean	Mdn	Mean	Mdn	s/sample
10	2436.78	2058.77	16.54	5.94	10,221.83	9919.10	0.29
20	2312.69	2063.26	16.68	5.59	9725.68	9563.77	0.59
30	2265.93	2051.32	16.67	5.61	9658.38	9581.03	0.89

10.3 DDIM and EDM Comparison

The final evaluation metrics that compare the DDIM and EDM models in the validation data set are summarized in Table 10.8. The EDM model consistently outperforms DDIM in all metrics, as shown in Figure 10.6.

■ **Table 10.8** Comparison of evaluation metrics (MSE, EM distance, and DTW) between EDM and DDIM.

Model	MSE		EM		DTW	
	Mean	Median	Mean	Median	Mean	Median
EDM	2265.93	2051.32	16.67	5.61	9658.38	9581.03
DDIM	4606.29	2864.54	24.41	7.64	14291.10	12371.86

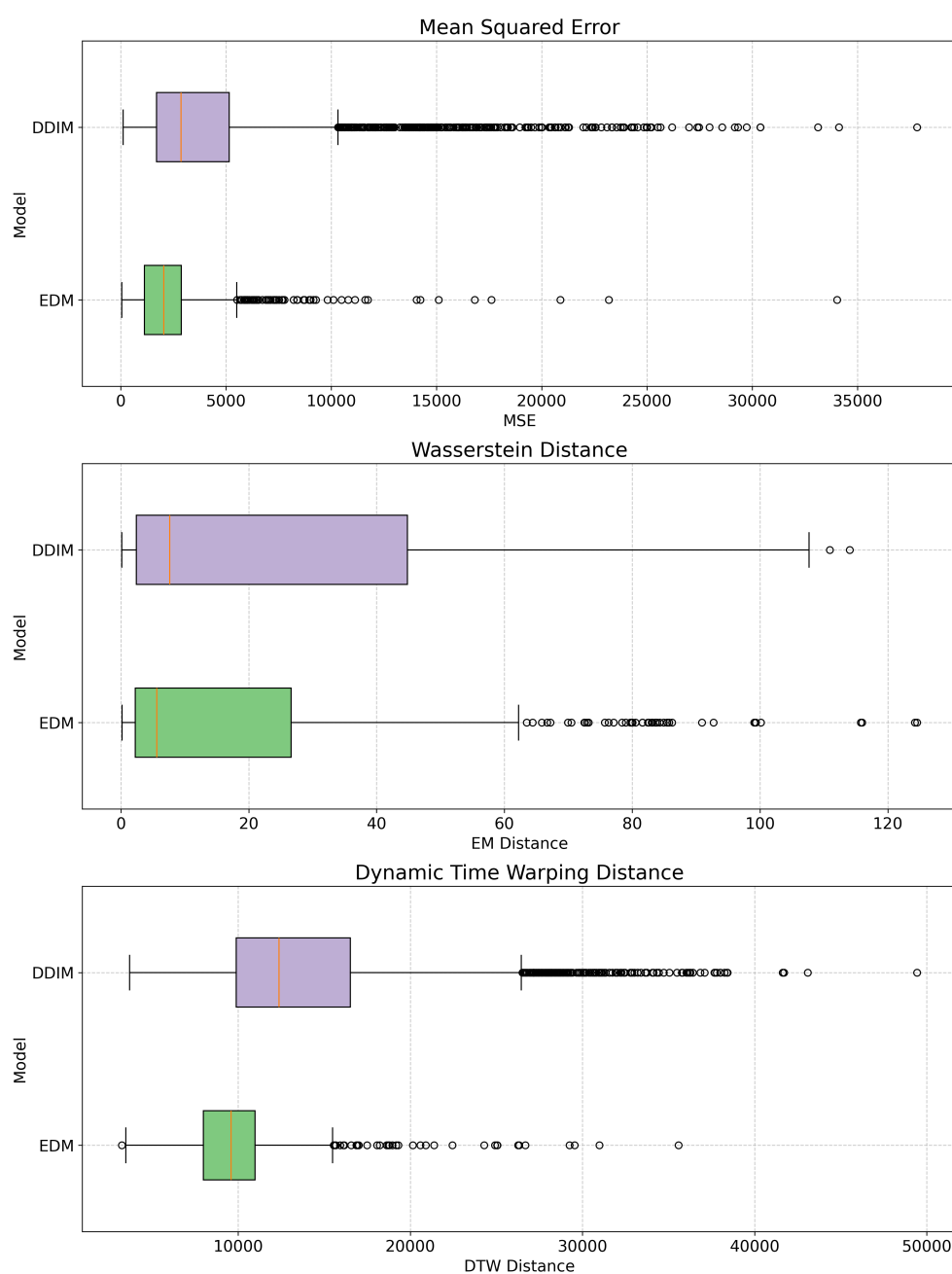
Regarding MSE, EDM shows notably better performance with a mean of 2265.93 and a median of 2051.32, compared to the higher values of DDIM (mean: 4606.29, median: 2864.54). This indicates that EDM is more accurate in point-wise spectral intensity predictions.

The fact that EDM is a better point-wise predictor is also visible in Figure 10.7. The difference spectrograms show that EDM produces more accurate and consistent results than DDIM. The DDIM model exhibits larger errors, particularly in the 780–950 nm range, where it frequently over- or underestimates peak intensities. It also introduces more unwanted fluctuations at the edges of the spectrum, where expected values are close to zero.

The EM distance, which evaluates the similarity between predicted and true spectral distributions, also favors the EDM. It achieves a lower mean (16.67) and a median (5.61), while DDIM records higher values (mean: 24.41, median: 7.64), suggesting that EDM better preserves the shape of the spectra.

Similarly, in the DTW metric, which accounts for alignment in temporal spectral patterns, EDM again leads with a lower mean of 9658.38 and a median of 9581.03. DDIM shows less favorable results (mean: 14,291.10, median: 12,371.86), pointing to greater alignment distortions.

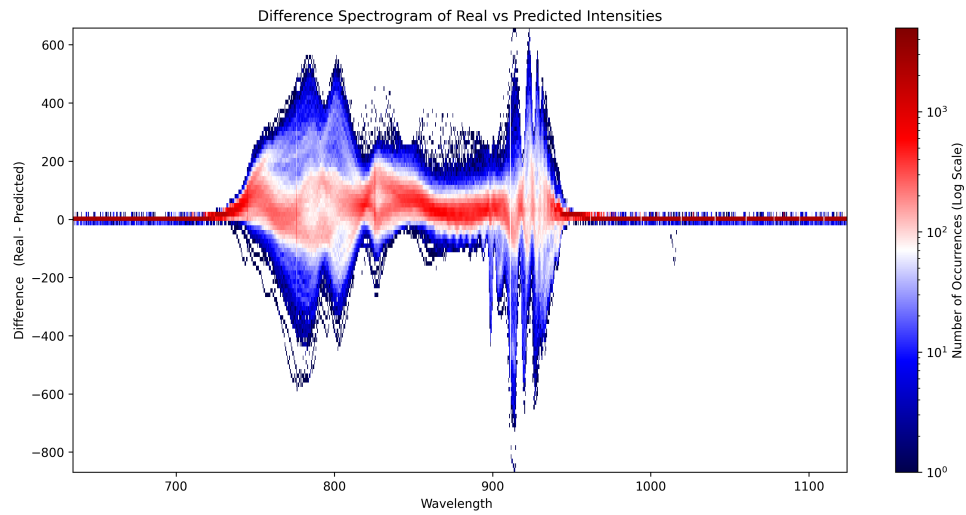
Overall, these results confirm that the EDM model generates more precise, consistent, and physically realistic spectra than DDIM. Therefore, the EDM model was chosen as the final model.



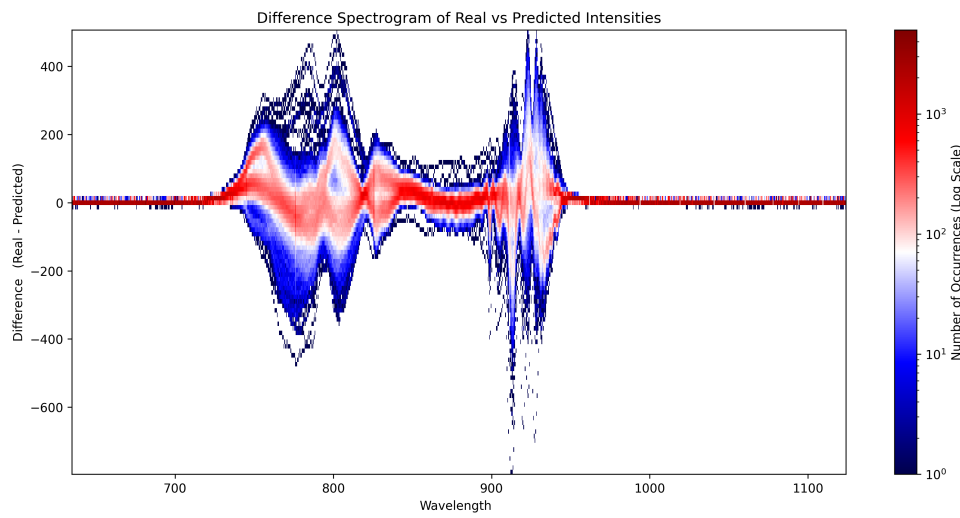
■ **Figure 10.6** Comparison of EDM and DDIM.

10.4 Final Model

The best performing EDM model, selected based on validation metrics, was further evaluated in the test data set to verify its generalizability. This model



(a) DDIM Error Spectrogram



(b) EDM Error Spectrogram

Figure 10.7 Comparison of error (real value minus predicted value) spectrograms for DDIM and EDM models.

was trained with a dropout rate of 0 %, and evaluated using 30 noise steps and a fixed CFG scale of 2. Table 10.9 summarizes the evaluation metrics in both datasets, and Figure 10.8 presents the results visually.

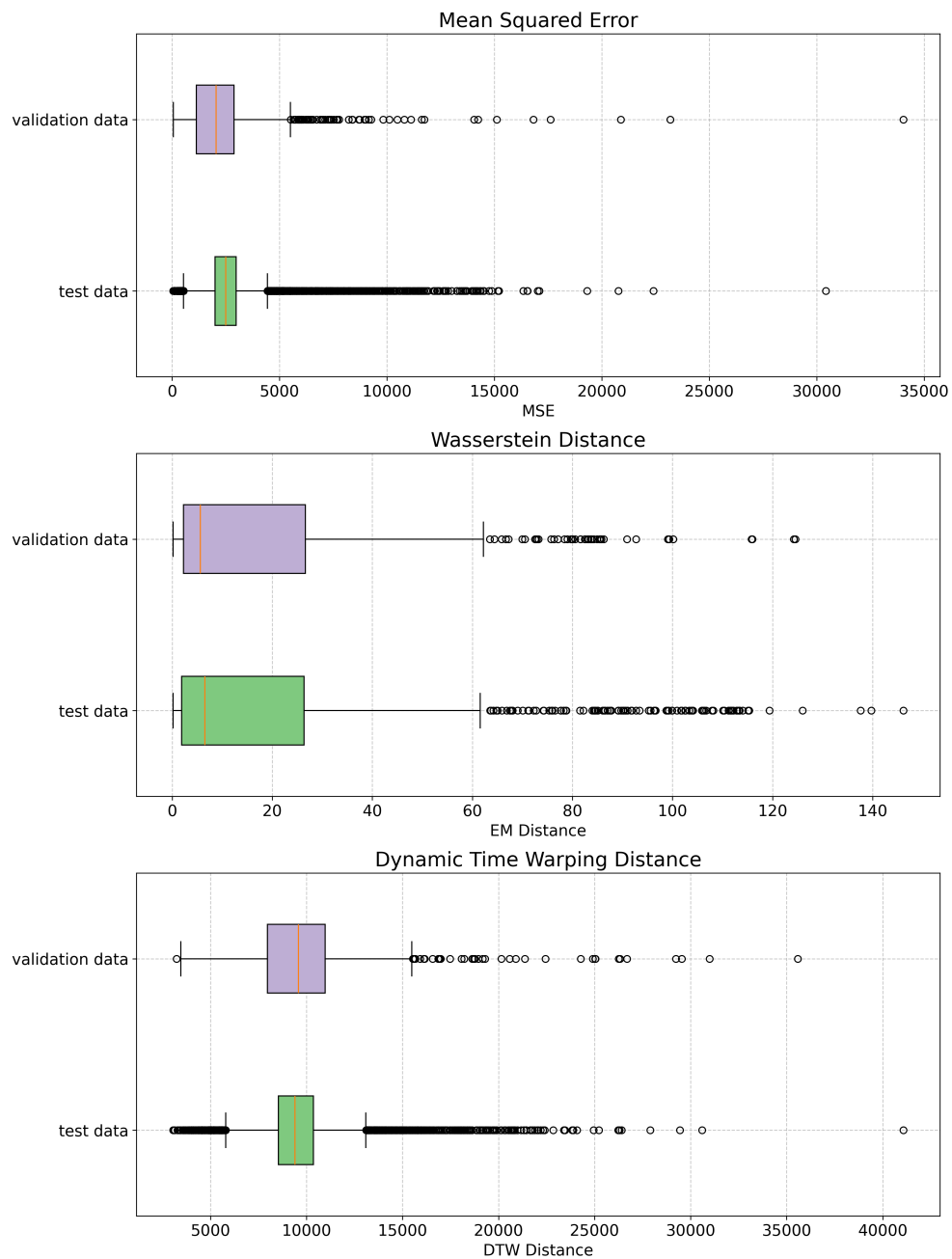
■ **Table 10.9** Evaluation metrics of the best EDM model on validation and test datasets.

Metric	Dataset	Mean	Median	Min	Max
MSE	Validation	2265.93	2051.32	54.38	34031.70
	Test	2612.67	2502.20	46.40	30428.93
DTW	Validation	9658.38	9581.03	3242.10	35577.77
	Test	9504.94	9403.86	3055.29	41077.11
EM	Validation	16.67	5.61	0.17	124.57
	Test	20.90	6.55	0.18	146.14

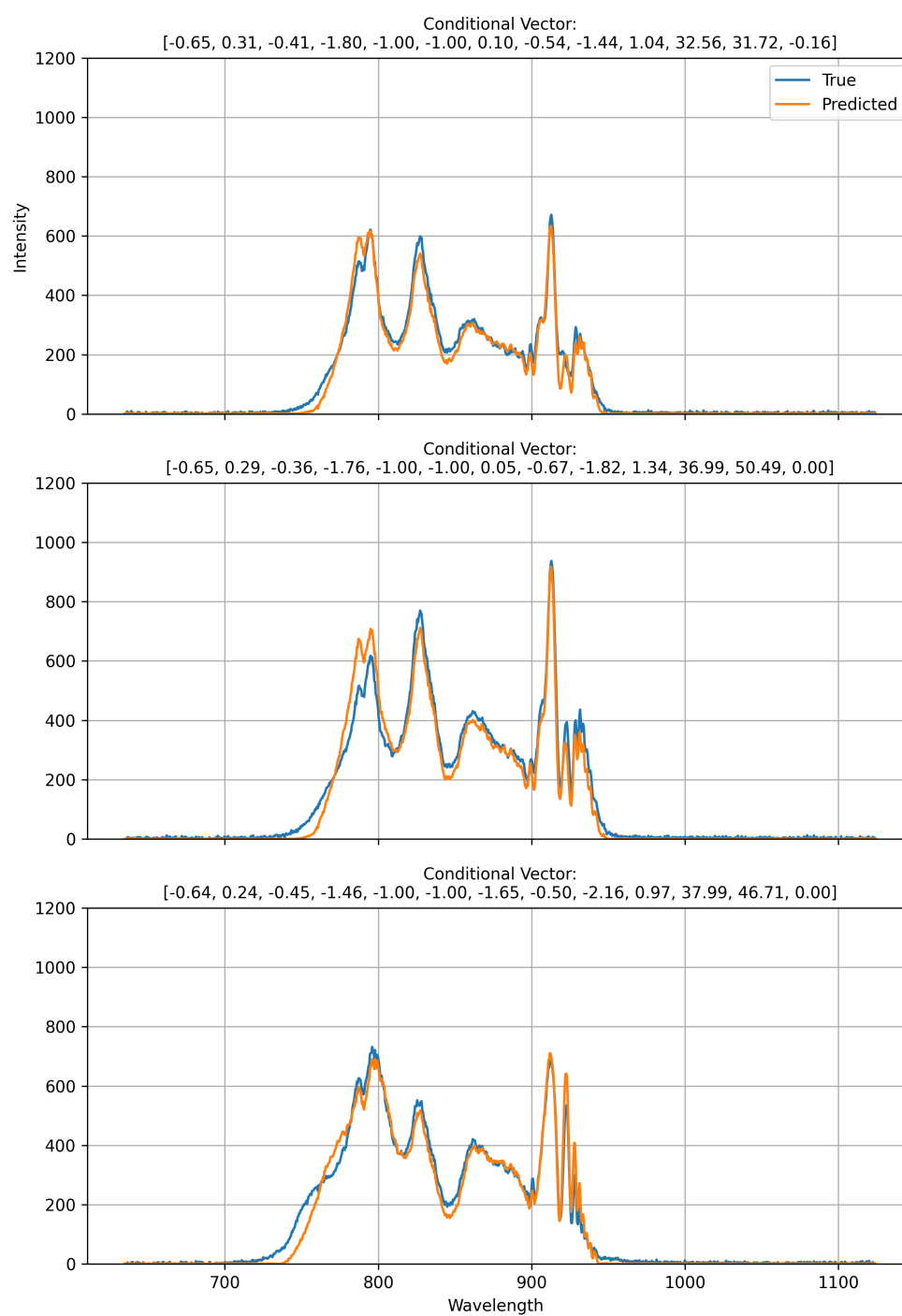
The model maintained stable performance on all three metrics. The mean MSE in the test set increased slightly from 2265.93 to 2612.67, representing a relative increase of approximately 15 %. Similarly, the DTW mean dropped marginally (from 9658.38 to 9504.94), showing that the alignment quality remained consistent or slightly improved. The EM distance showed the most noticeable increase, from a mean of 16.67 to 20.90, although the median only slightly increased from 5.61 to 6.55.

Figures 10.9 and 10.10 present six randomly selected spectra generated by the EDM model. The consistency of the predictions on varied inputs suggests a strong generalization performance.

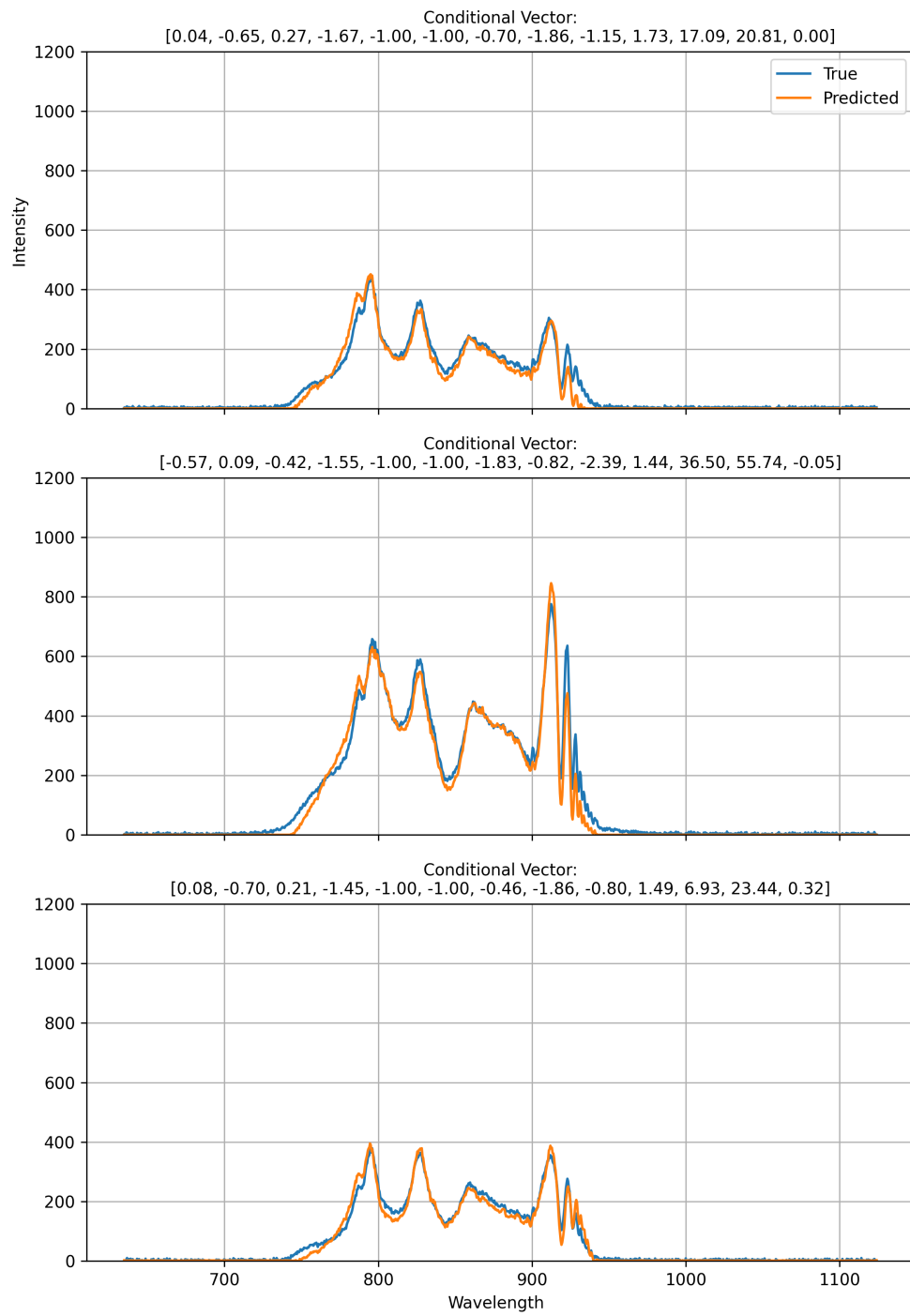
Furthermore, Figure 10.11 demonstrates the ability of the model to reconstruct spectral evolution during LWFA experiments, where direct measurement is not possible due to technical limitations.



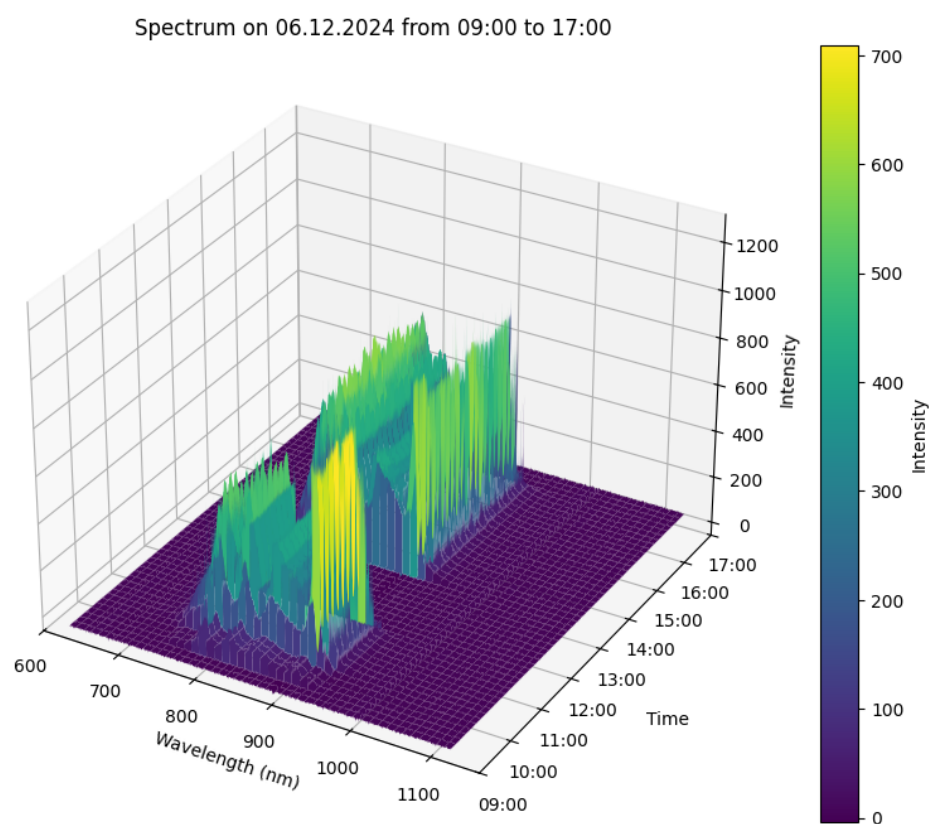
■ **Figure 10.8** Comparison of final model evaluation on test and validation data.



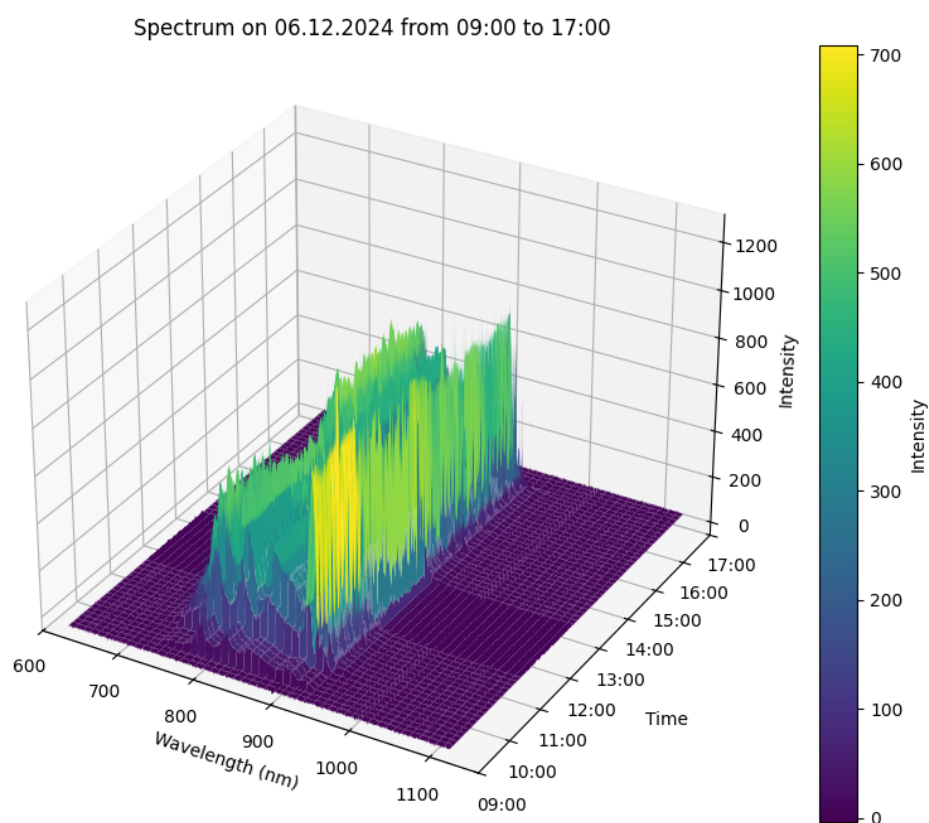
■ **Figure 10.9** Comparison of final model evaluation on test and validation data.



■ **Figure 10.10** Comparison of final model evaluation on test and validation data.



(a) Spectrogram with original data with the time gaps.



(b) Spectrogram with the predicted data in the time gaps.

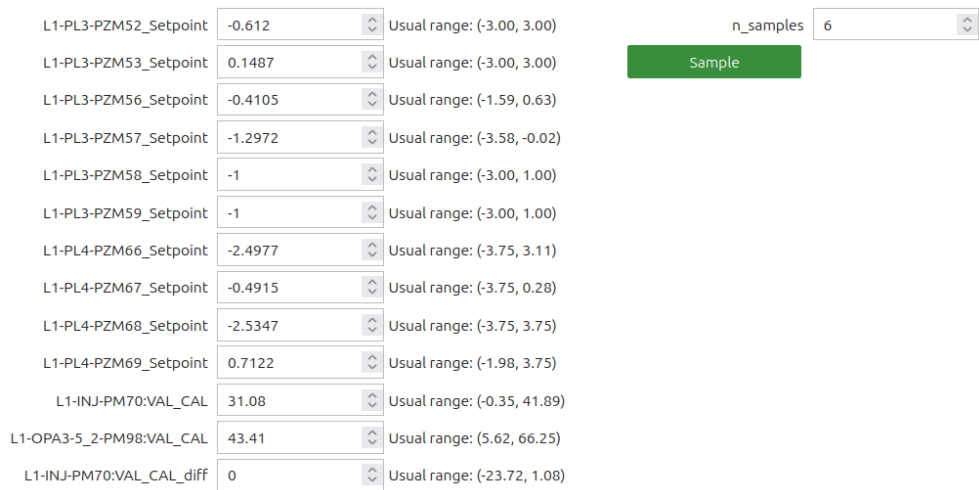
■ **Figure 10.11** Example sampling of missing data in unmeasured time gaps.

10.4.1 User Interface for Spectrum Generation

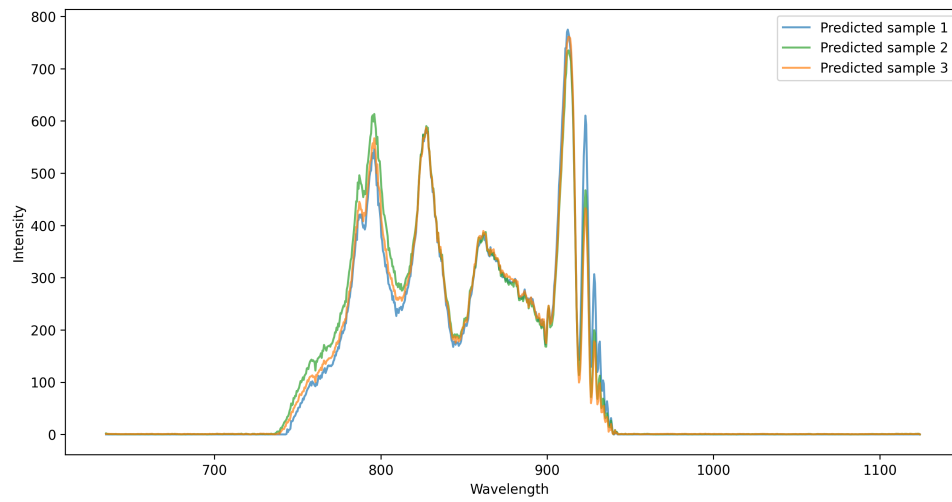
An interactive user interface was developed using `ipywidgets` within a `Jupyter Notebook` environment. This interface allows users to set parameter values manually, as shown in Figure 10.12, and visualize the resulting spectra generated by the best-performing diffusion model.

The interface provides the following functionality:

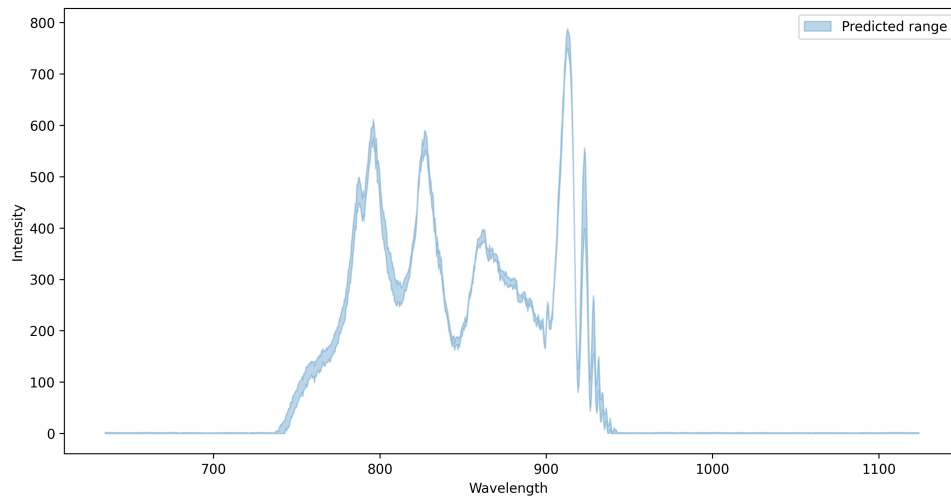
- **Parameter Input:** Users can specify a value for each system parameter. Each input is paired with a label that displays the typical operating range.
- **Sample Control:** A separate input allows the user to define how many spectral samples should be generated. This enables both single-sample inspection and multi-sample visualization of prediction variability.
- **Spectrum Generation and Visualization:** After clicking the “Sample” button, the interface constructs a conditional vector from the assigned parameter values and passes it to the diffusion model. The resulting spectral outputs are then plotted. If one to three samples are requested, they are shown as distinct lines. For larger sample counts, the visualization switches to a filled area showing the envelope (minimum to maximum) of all generated spectra. Examples of visualizations are provided in Figure 10.13.



■ **Figure 10.12** Example of user interface for generating spectra.



(a) Example visualization of three generated spectra for the same set of parameters.



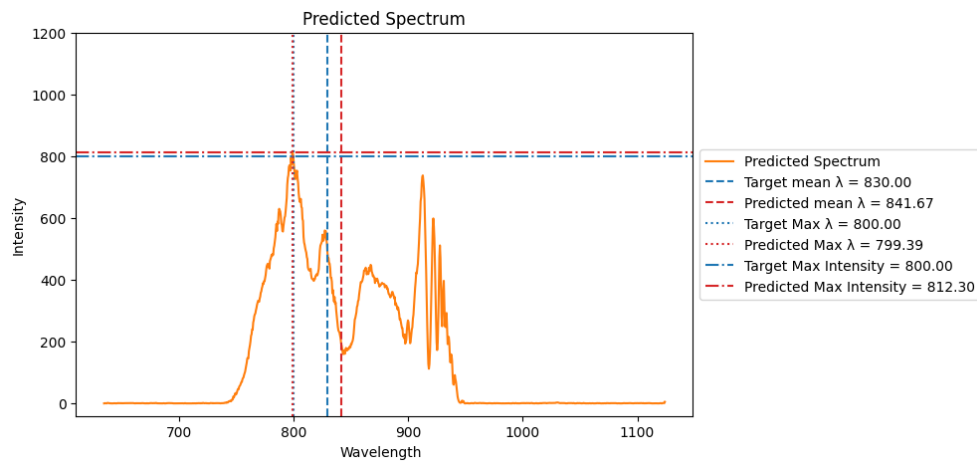
(b) Example visualization of the range between the generated spectra for number of samples bigger than three.

■ **Figure 10.13** Comparison of generated spectra visualizations.

10.5 Bayesian Optimization

Bayesian optimization allows exploring how different system parameters influence the resulting spectra. One particularly promising direction for future experiments is to investigate the relationship between the measured signals, the powermeter (L1-INJ-PM70:VAL_CAL) and the photodiode (L1-OPA3-5_2-PM98:VAL_CAL) parameters, and the other controlled parameters in the system. This relationship is not yet fully understood by ALFA researchers, and systematic optimization experiments could provide better insight into it.

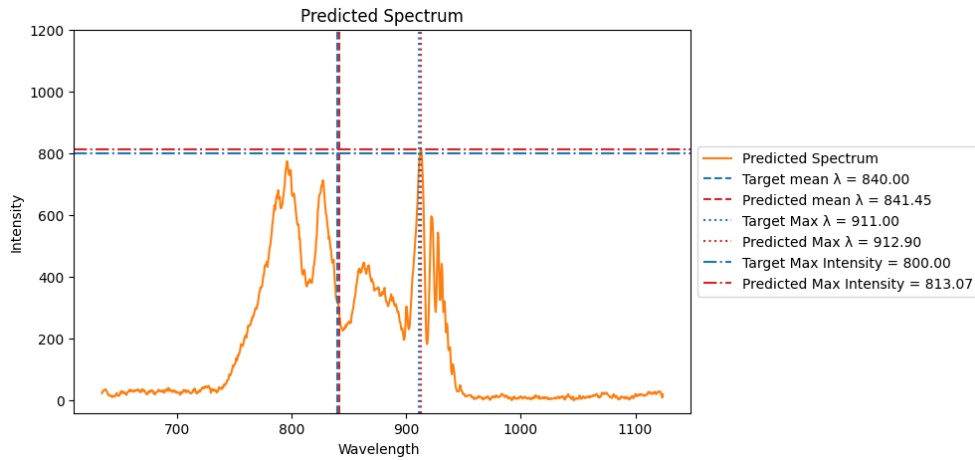
One possible experiment involves fixing the powermeter and photodiode values to those observed during an actual measurement and then optimizing a subset of controllable parameters. The objective is to identify a configuration of these parameters that, when combined with the fixed measured signals, generates a spectrum with the target spectral properties, namely the mean wavelength, the wavelength at maximum intensity, and the maximum intensity value. This setup allows for controlled and cheap exploration of how different parameter choices influence the resulting spectral shape under known energy conditions. An illustrative example is provided in Figure 10.14.



■ **Figure 10.14** Bayesian optimization experiment – fixing the powermeter and photodiode values.

Another experimental setup reverses the roles: the system parameters are kept constant, and Bayesian optimization is used to adjust the values of the powermeter and photodiode. The goal is to identify what values of these measured signals would need to be achieved in order to produce a spectrum with the specific desired properties. Figure 10.15 presents an example result from this scenario.

In more specific cases, it may be beneficial to isolate the photodiode signal for optimization. For example, one experiment could involve fixing all system



■ **Figure 10.15** Bayesian optimization experiment – fixing the system parameters.

parameters and the powermeter value and optimizing only the photodiode signal. This would help determine what photodiode response aligns best with the target spectral properties given a known system state and energy input.

In contrast, another setup could fix the parameters and the photodiode value while optimizing only the powermeter. This would allow researchers to study how energy-related changes, as captured by the powermeter, influence spectral outcomes when other conditions are held constant.

The experiments presented represent only a subset of the possible experimental directions. The flexibility of the Bayesian optimization framework allows researchers to explore a wide range of configurations and hypotheses suitable for both system analysis and in-time parameter tuning.

10.5.1 User Interface for Bayesian Optimization

To allow users to configure the optimization process, a user-friendly graphical interface was developed using `ipywidgets` in a `Jupyter Notebook` environment.

The interface supports the following functionality:

- **Parameter Selection:** Users can choose which system parameters to optimize. For each parameter, the current value can be set manually, and the interface displays the usual operating range as a reference.
- **Target Spectrum Specification:** The user can input target values for three spectral properties: mean wavelength, wavelength at maximum intensity, and the maximum intensity value. These are the key features used by the loss function during optimization.

- **Optimization Control:** Users can adjust the number of optimization steps (`n_calls`) and run the optimization. Additional controls are provided to quickly select or deselect all parameters.
- **Output and Visualization:** After optimization, the interface prints the best-found parameter values and displays a comparison between the set spectral properties and the generated spectra with its corresponding properties.

Figure 10.16 shows the user interface, Figures 10.14 and 10.15 present the example outputs.

✓ Select All

✗ Deselect All

Select parameters to optimize and set values for fixed parameters:

☐ L1-PL3-PZM52_Setpoint	-0.612	usual range: [-3.0, 3.0]
☒ L1-PL3-PZM53_Setpoint	0.1487	usual range: [-3.0, 3.0]
☐ L1-PL3-PZM56_Setpoint	-0.4105	usual range: [-1.591, 0.6349]
☒ L1-PL3-PZM57_Setpoint	-1.2972	usual range: [-3.5833, -0.0177]
☒ L1-PL3-PZM58_Setpoint	-1	usual range: [-3.0, 1.0]
☒ L1-PL3-PZM59_Setpoint	-1	usual range: [-3.0, 1.0]
☒ L1-PL4-PZM66_Setpoint	-2.4977	usual range: [-3.75, 3.1082]
☐ L1-PL4-PZM67_Setpoint	-0.4915	usual range: [-3.75, 0.2802]
☐ L1-PL4-PZM68_Setpoint	-2.5347	usual range: [-3.75, 3.75]
☐ L1-PL4-PZM69_Setpoint	0.7122	usual range: [-1.9848, 3.75]
☐ L1-INJ-PM70:VAL_CAL	31.08	usual range: [-0.35, 41.89]
☐ L1-OPA3-5_2-PM98:VAL_CAL	43.41	usual range: [5.62, 66.25]
☐ L1-INJ-PM70:VAL_CAL_diff	0	usual range: [-33.72, 1.08]

Target spectrum properties:

Target Mean λ

840

usual range: [830, 850]

Target Max λ

911

usual range: [794, 913]

Target Max Intensity

800

usual range: [600, 820]

n_calls:

10

▶ Run Optimization

■ **Figure 10.16** Example of user interface for Bayesian optimization.

Chapter 11

Future Work

This chapter proposes potential directions for extending the current work. The results achieved so far demonstrate solid predictive performance. Building on this foundation, the following ideas suggest how the quality and applicability of spectral predictions could be improved even more in future research.

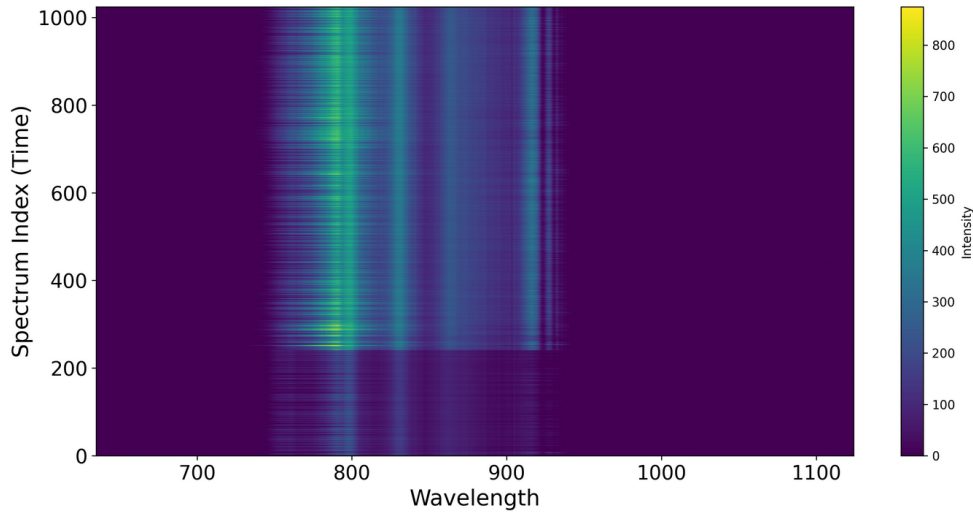
11.1 Incorporating Temporal Context

In real-world experiments, spectra are usually recorded in sequence over time, where each spectrum is influenced by previous system states.

Although the current model treats each prediction independently, it already includes inputs that partially reflect the system history. Specifically, the parameters `L1-INJ-PM70:VAL_CAL` and `L1-OPA3-5_2-PM98:VAL_CAL` are not user-controlled setpoints but measured signals. They provide feedback on the system’s output, such as laser energy, and thus carry information about the recent past. These variables already help the model indirectly to infer some time-related aspects. Building on this idea, further incorporating the temporal input could improve the robustness of the model.

There are multiple options on how to handle the temporal context. First, the model could be reformulated as a 2D diffusion process, where one axis represents the wavelength, and the other represents time. Each row in the resulting 2D array would correspond to a spectrum measured at a specific time, forming a spectrotemporal “image” with intensity values as pixel brightness, Figure 11.1 demonstrates how such input data may look like. This would allow the model to learn temporal patterns directly.

A key question in this setup is how to handle the conditional parameter vector. Several strategies can be considered: using only time windows where parameters remain constant, applying an average parameter vector across the window, using the parameters from the first spectrum as a representative, or extending the conditioning to include a distinct vector for each time step.



■ **Figure 11.1** 2D diffusion process example.

An alternative approach is to treat the model as a sequence predictor. A short history of previous spectra could be fed into the UNet as input and the model would predict the next spectrum in the sequence. This approach can help the model make more accurate predictions by capturing how the spectrum changes over time. The same options as above apply with respect to the use of system parameters during conditioning.

11.2 Physics Informed Diffusion Model

Another potential direction for future research is the development of a physics-informed diffusion model. The model could be guided toward more robust predictions by incorporating prior knowledge about the underlying physical principles that influence the system.

However, to implement such a model effectively, access to parameters that reflect key physical characteristics of the system would be essential. The current set of input variables, primarily system control values and diagnostic readings, is not suitable for this purpose.

Future work could explore the extension of conditional vectors with additional system parameters. This might include measurements or derived quantities that are more closely tied to the physical behavior of the laser or experimental setup. Further discussion and collaboration with experts from the ALFA team would be essential to identify such parameters.

11.3 Digital Twin Concept for L1-Allegra and ALFA Integration

A promising direction for future work is to integrate the spectral prediction model developed in this thesis with the existing generative model of the laser-plasma acceleration stage, developed by Matěj Jech [72] (Section 5.2), which uses diffusion models to simulate electron beam spectra based on ALFA campaign data.

Combining these efforts could enable the creation of a digital twin of the full beamline system L1-Allegra laser and the ALFA LWFA chamber. This architecture could simulate the entire process from laser input to particle acceleration. This digital twin could help plan experiments, adjust settings in real time, and predict how the system will behave.

Conclusion

This thesis explored the application of diffusion-based generative models for simulating the spectral output of the high-power L1-Allegro laser system. The primary objective was to create a digital twin of the system that is able to generate realistic spectral profiles conditioned on a set of system parameters. This involved designing and training conditional generative architectures that could learn the complex relationship between input conditions and resulting spectra.

A comprehensive literature review was conducted, focusing on recent advances in scientific machine learning and the deployment of generative models for physical simulations. This review established the methodological foundation for the implementation of a conditional diffusion framework.

The experimental data provided by the scientific partners required careful preprocessing to ensure suitability for model training. This involved cleaning corrupted entries, normalizing spectral intensities, and aligning system parameters with the corresponding measurements.

Two diffusion-based models were implemented and compared: the DDIM and the EDM.

The evaluation was carried out using three complementary metrics: MSE, EM distance, and DTW. These metrics were selected to capture different aspects of spectral similarity: MSE for point-wise accuracy, EM distance to compare the overall distributional similarity, and DTW distance, which is particularly useful for comparing sequences, which might be slightly misaligned.

The experimental results demonstrated that the EDM model consistently outperformed DDIM in all metrics. Specifically, EDM achieved a mean MSE of 2265.93 compared to 4606.29 for DDIM, while also showing a lower mean EM distance (16.67 vs. 24.41) and DTW (9658.38 vs. 14291.10), measured in the validation data set. Based on this evaluation, EDM was selected as the preferred architecture for further experiments.

The best performing EDM model, trained with 0 % dropout, 30 noise steps, and a fixed CFG scale of 2, was further evaluated in test data to assess generalization. It showed stable performance in all metrics: the mean MSE increased slightly from 2265.93 to 2612.67, the DTW remained consistent and the EM

distance increased modestly.

Two additional optional objectives were suggested as part of the broader goals of this work: the integration of physical constraints into the generative modeling process and the development of an optimization pipeline for experimental guidance.

Embedding physical knowledge into the model was not pursued, as the data provided were not suitable for this purpose. However, incorporating physically meaningful input remains a promising direction for future work.

The second optional goal, developing an optimization framework, was fulfilled using Bayesian optimization. This method allows for a structured exploration of how both measured signals and controllable parameters affect the resulting spectra.

The results show that conditional diffusion models can be valuable tools for simulating and understanding complex laser systems. The experiments confirm that the EDM architecture performs well in predicting spectral data and suggest several directions for future work. This study provides a strong basis for further research combining machine learning with experimental laser physics.



Appendix A

Parameters

This appendix provides a detailed overview of the system parameters. It includes both a statistical summary of the parameters and visualizations of their daily changes over the recorded period.

A.1 Statistical Summary of The System Parameters

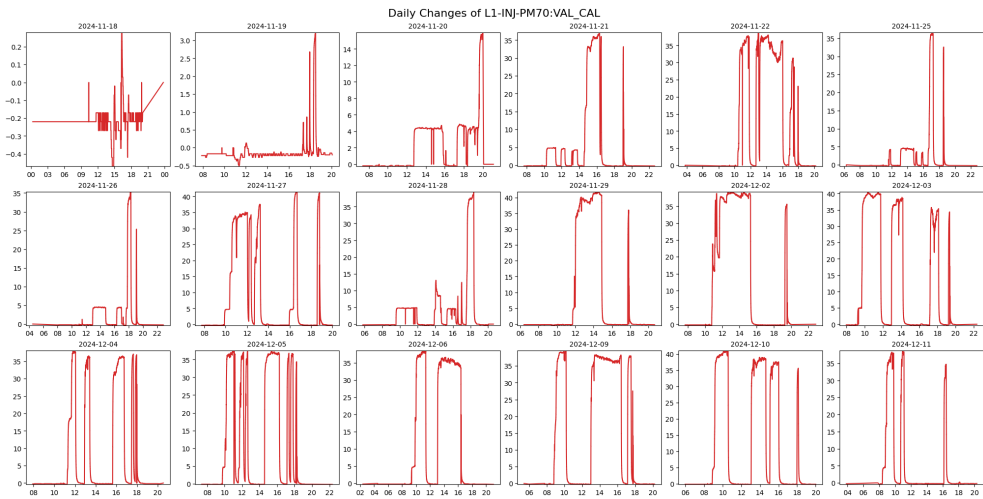
The following Table A.1 presents a statistical summary of the system parameters, including their minimum and maximum values, mean, median, and variance.

■ **Table A.1** Statistical summary of the system parameters.

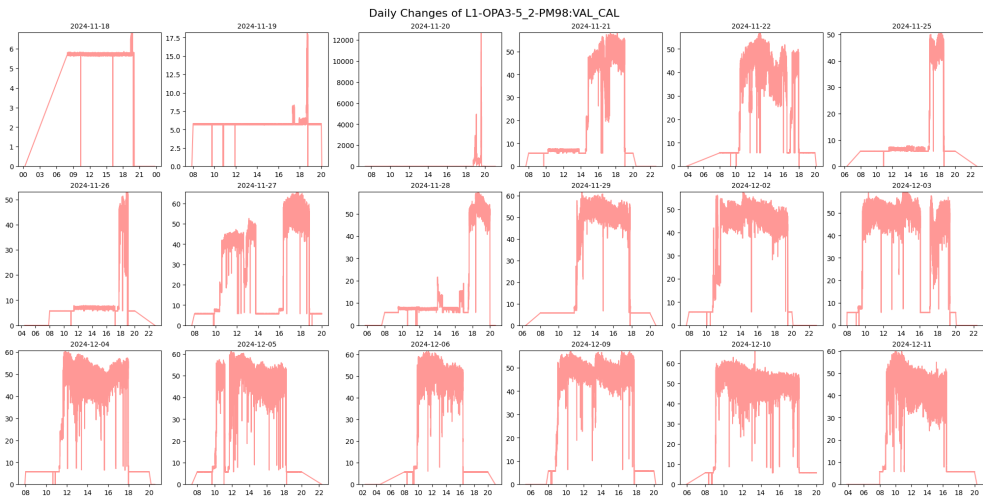
Parameter	Min	Max	Mean	Median	Variance
L1-PL3-PZM52:Setpoint	-3.00	3.00	-0.36	-0.45	0.12
L1-PL3-PZM53:Setpoint	-3.00	3.00	-0.18	-0.08	0.18
L1-PL3-PZM56:Setpoint	-1.86	0.83	-0.43	-0.43	0.17
L1-PL3-PZM57:Setpoint	-3.71	-0.02	-1.46	-1.46	0.22
L1-PL3-PZM58:Setpoint	-3.00	1.00	-1.03	-1.00	0.62
L1-PL3-PZM59:Setpoint	-3.00	1.00	-0.99	-1.00	0.46
L1-PL5-DIRA3:Piezo1_X_RB	-0.57	0.99	-0.05	-0.05	0.0005
L1-PL5-DIRA3:Piezo1_Y_RB	-0.64	0.99	0.29	0.26	0.03
L1-PL5-DIRA3:Piezo3_X_RB	-0.62	0.99	0.01	0.00	0.0004
L1-PL5-DIRA3:Piezo3_Y_RB	-1.00	0.95	-0.03	0.03	0.04
L1-PL4-PZM66:Setpoint	-3.75	3.75	-1.80	-2.20	1.82
L1-PL4-PZM67:Setpoint	-3.75	0.33	-1.20	-0.99	0.44
L1-PL4-PZM68:Setpoint	-3.75	3.75	-1.64	-2.02	3.20
L1-PL4-PZM69:Setpoint	-3.75	3.75	0.89	1.15	1.52
L1-INJ-PM70:VAL_CAL	-0.52	41.89	13.16	3.97	266.13
L1-OPA3-5_2-PM98:VAL_CAL	0.00	12706.35	39.38	45.12	12035.60

A.2 Daily Changes of Parameters

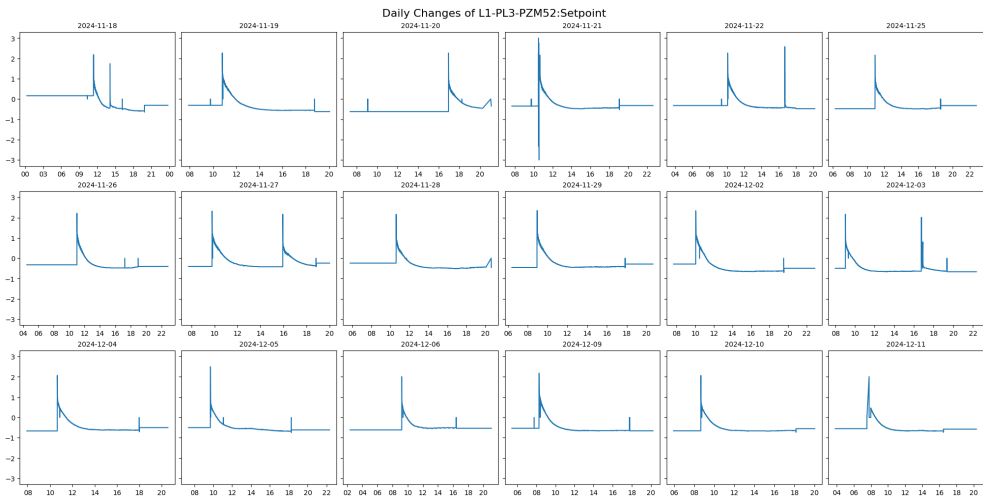
This section illustrates how the system parameters change over time, as shown in following Figures. Each Figure corresponds to a specific parameter and contains 18 subplots for each of the recorded days.



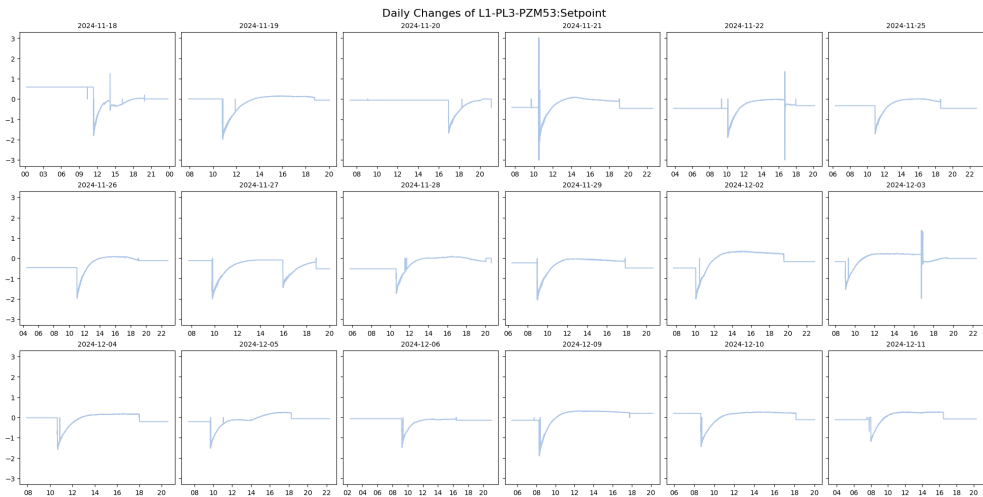
■ **Figure A.1** Daily changes of L1-INJ-PM70:VAL_CAL.



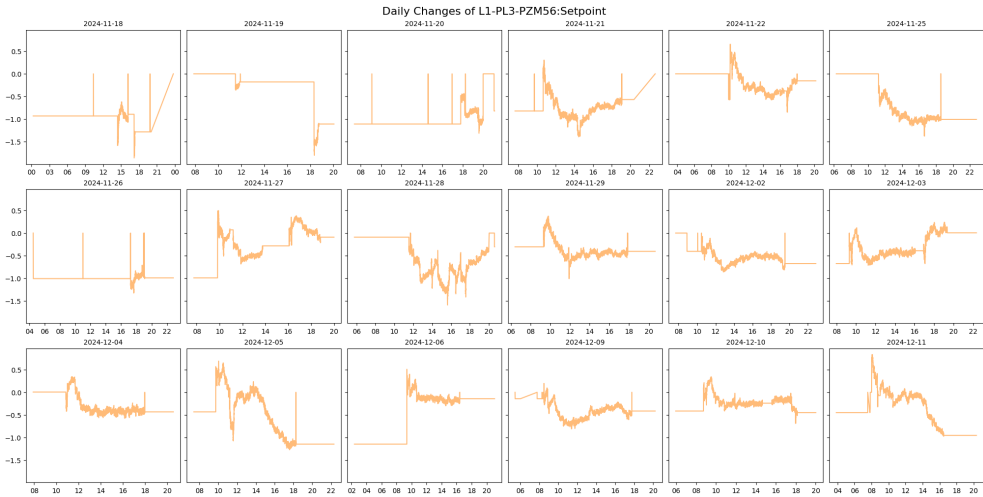
■ **Figure A.2** Daily changes of L1-OPA3-5_2-PM98:VAL_CAL.



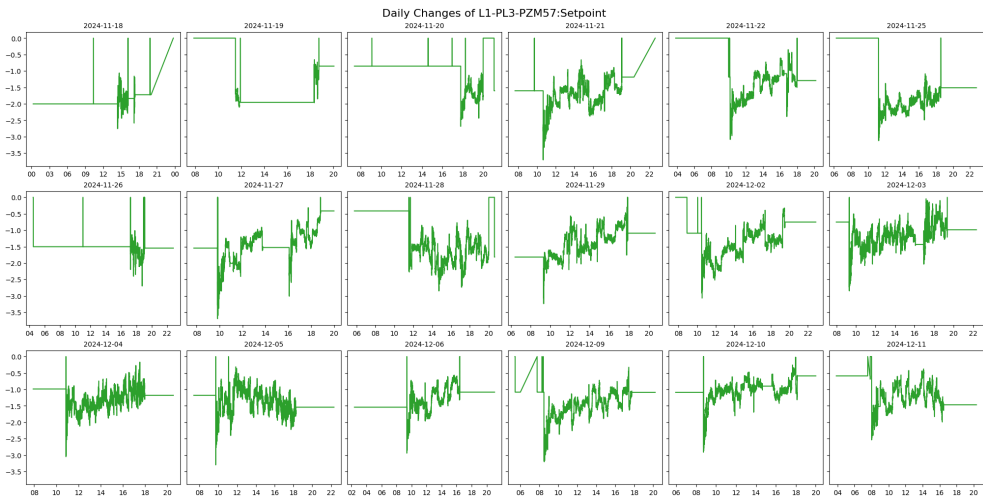
■ **Figure A.3** Daily changes of L1-PL3-PZM52:Setpoint.



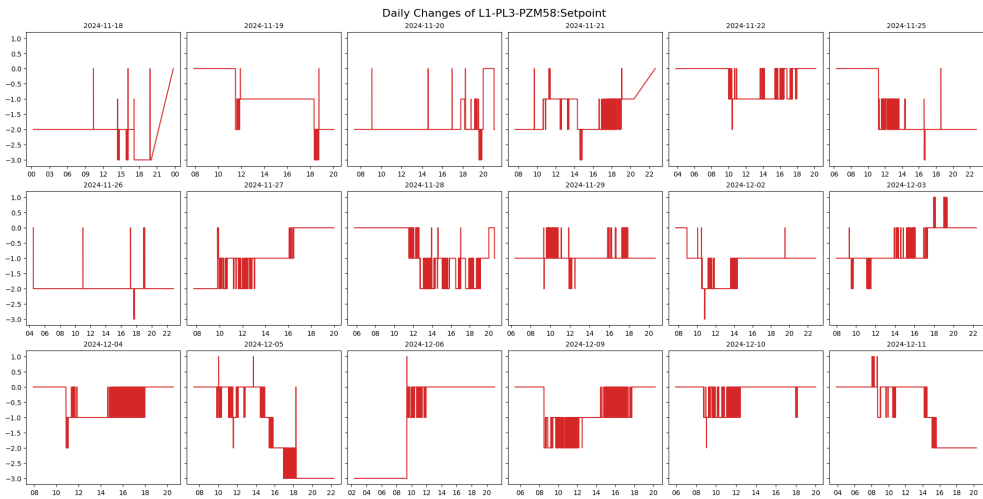
■ **Figure A.4** Daily changes of L1-PL3-PZM53:Setpoint.



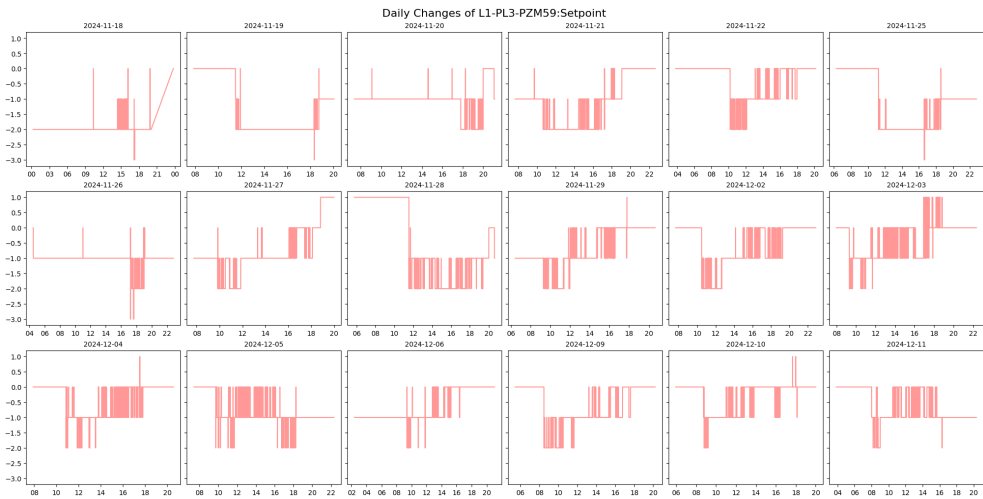
■ Figure A.5 Daily changes of L1-PL3-PZM56:Setpoint.



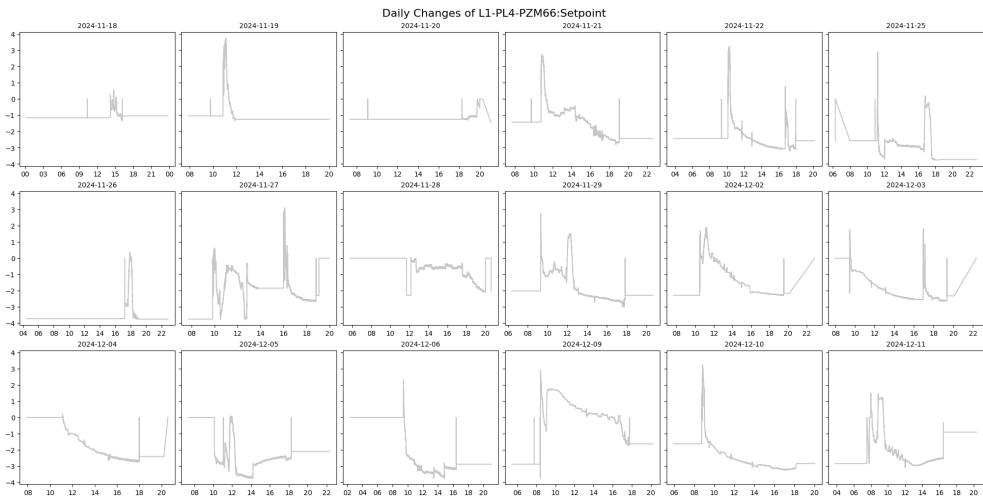
■ Figure A.6 Daily changes of L1-PL3-PZM57:Setpoint.



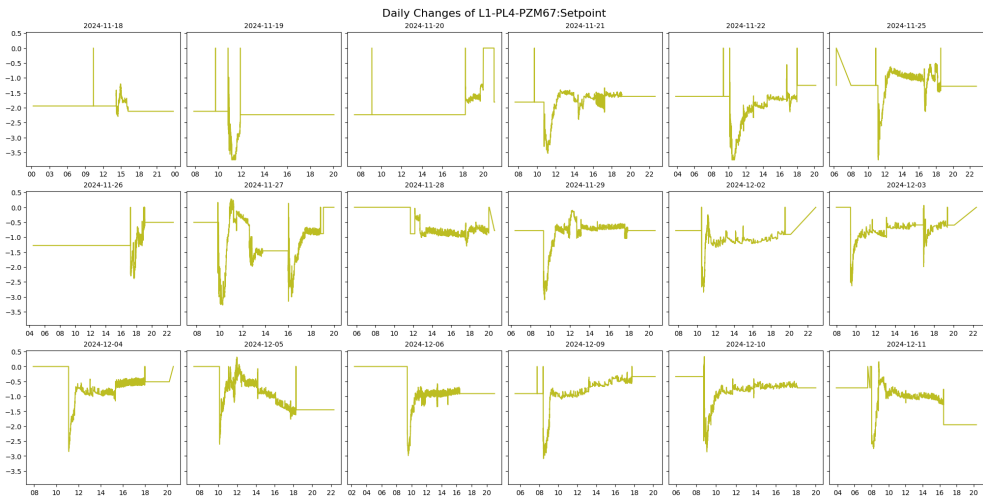
■ **Figure A.7** Daily changes of L1-PL3-PZM58:Setpoint.



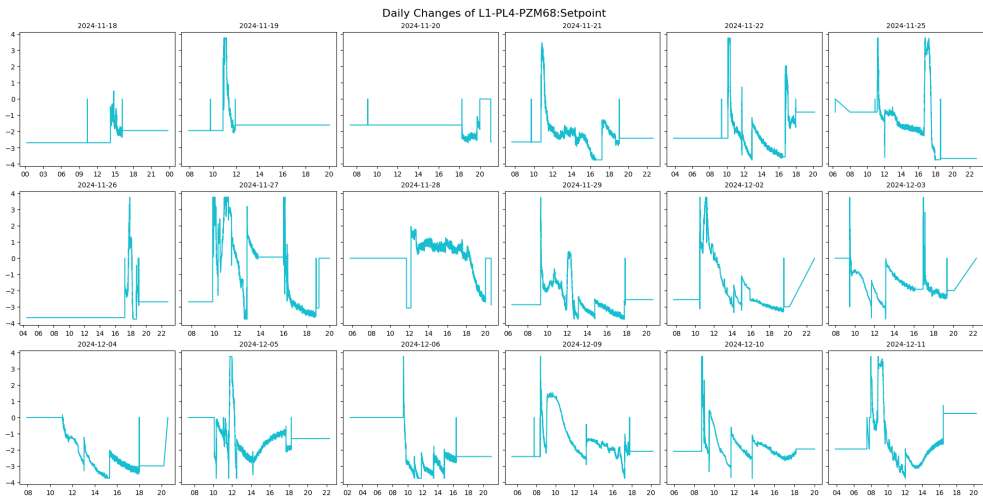
■ **Figure A.8** Daily changes of L1-PL3-PZM59:Setpoint.



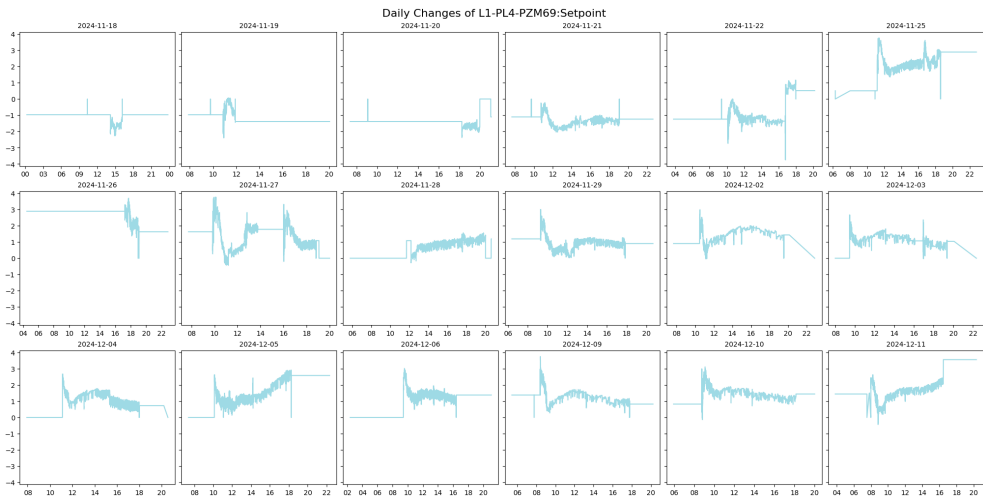
■ Figure A.9 Daily changes of L1-PL4-PZM66:Setpoint.



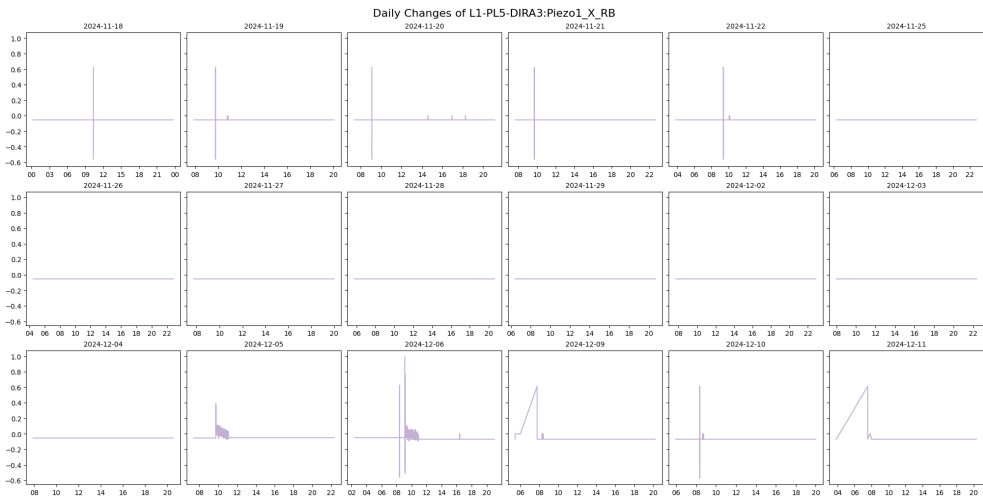
■ Figure A.10 Daily changes of L1-PL4-PZM67:Setpoint.



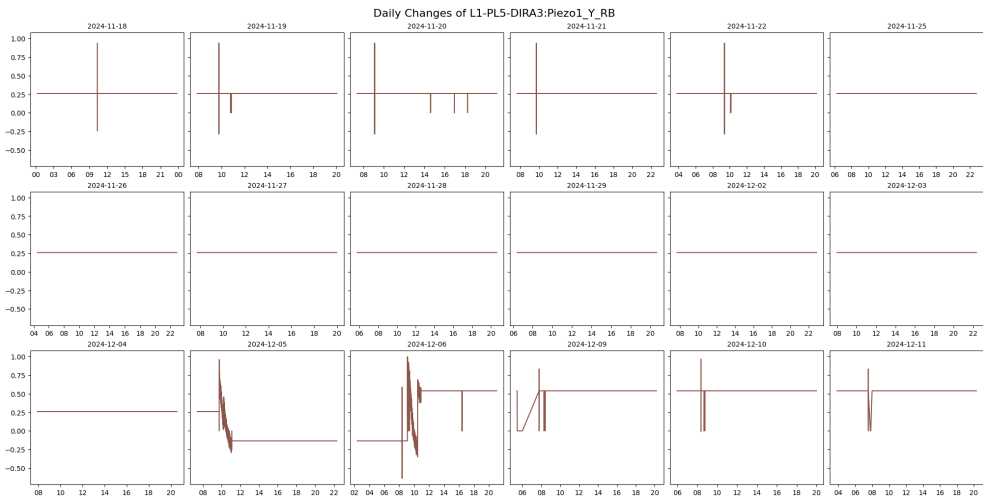
■ **Figure A.11** Daily changes of L1-PL4-PZM68:Setpoint.



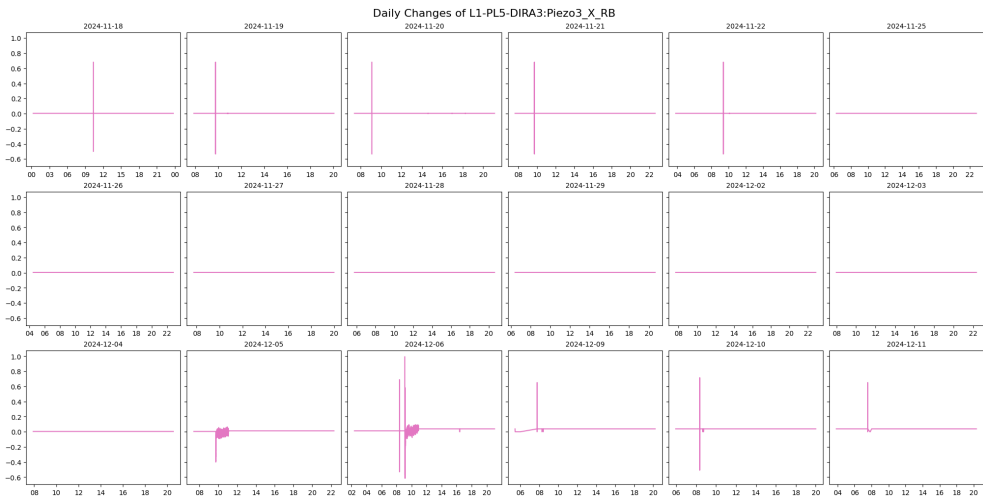
■ **Figure A.12** Daily changes of L1-PL4-PZM69:Setpoint.



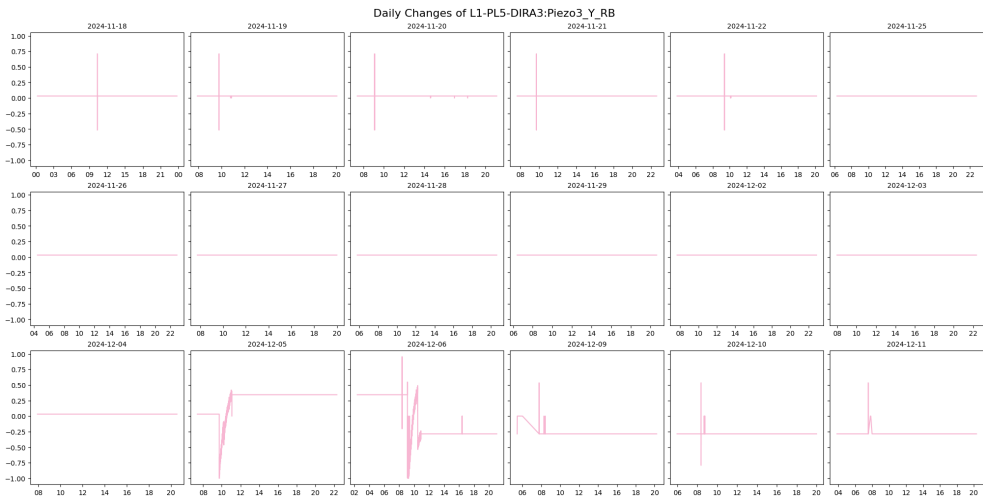
■ **Figure A.13** Daily changes of L1-PL5-DIRA3:Piezo1_X_RB.



■ **Figure A.14** Daily changes of L1-PL5-DIRA3:Piezo1_Y_RB.



■ **Figure A.15** Daily changes of L1-PL5-DIRA3:Piezo3_X_RB.



■ **Figure A.16** Daily changes of L1-PL5-DIRA3:Piezo3_Y_RB.

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Contents of the Attachment

/	
└─	readme.txt.....brief description of the contents
└─	requirements.txt.....list of Python dependencies
└─	data.....processed datasets in CSV and NPY formats
└─	preprocessing.....notebooks for data preprocessing
└─	src.....source code for models, training, and evaluation
└─	models.....saved best models for EDM and DDIM
└─	predictions.....predictions generated by the best models
└─	Bayesian_optimization_ui.ipynb .. UI for Bayesian optimization
└─	Spectrum_prediction.ipynb.....UI for spectrum generation
└─	Compare_models.ipynb.....performance of models comparison
└─	evaluate_ddim.ipynb.....evaluation for DDIM-generated samples
└─	evaluate_edm.ipynb.....evaluation for EDM-generated samples
└─	evaluate_edm_test_data.ipynb.....EDM evaluation on test data
└─	evaluate.py.....shared evaluation functions used by notebooks
└─	train_ddim.py.....training script for the DDIM model
└─	train_edm.py.....training script for the EDM model
└─	diffusion.py.....core diffusion process logic
└─	dataset.py.....dataset loading and preprocessing utilities
└─	loss.py.....EDM loss function used during training
└─	metrics.py.....metrics for model evaluation
└─	modules.py.....model architecture components
└─	utils.py.....general-purpose utility functions
└─	text.....text of the thesis
└─	thesis.pdf.....thesis in PDF format
└─	thesis.....source files of the thesis in L ^A T _E X format

The complete code is available on GitHub: https://github.com/akapitanova/master_thesis.